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► Mapping of the cotton supply chain at the community level in Pakistan

A report on selected districts in Punjab and Sindh Provinces

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► Acronyms and abbreviations

CABI	Centre for Agriculture and Bioscience International
DWCP	Decent Work Country Programme
EFP	Employer' Federation of Pakistan
EOBI	Employees' Old-Age Benefits Institution
FAO	Food and Agriculture Organization
FBR	Federal Board of Revenue
FPRW	Fundamental Principles and Rights at Work
GDP	Gross Domestic Product
GOP	Government of Pakistan
HDI	Human Development Index
ILO	International Labour Organization
ILS	International Labour Standards
INDITEX	Industria de Diseño Textil SA
KCA	Karachi Cotton Association
KPK	Khyber Pakhtunkhwa Province
NCVT	National Coordinative Variety and Traits
NGOs	Non-Governmental Organizations
NIC	National Identity Card
OP&HRD	Ministry of Overseas Pakistanis and Human Resource Development
OSH	Occupational Health and Safety
PPCC	Pakistan Central Cotton Committee
PESSI	Punjab Employees Social Security Institutions
PKR	Pakistani Rupee
PWF	Pakistan Workers' Federation
SDG	Sustainable Development Goals
TCO	Textile Commissioner's Organization
UNICEF	United Nations Children's Emergency Fund
UNDP	United Nations Development Programme
WWF	Workers Welfare Fund
WWF-Pakistan	World Wide Fund for Nature Pakistan

► Executive summary

Supply chains that originate at the roots in small farms and then wind their way up through the informal economy to the progressively more structured formal economy are messy. Their spirit and challenges defy neat flowcharts and superficial or simplistic approaches, just as they defy attempts at blueprint solutions. Such is the case with the community level cotton supply chain in Pakistan's Punjab and Sindh Provinces. Community level cotton production is the base, often remote, on which the textile value chain rests, supporting spinning, weaving, processing, printing, manufacturing (fabric or cloth, readymade garments, clothing, knitwear and cottonseed oil, among other products). In these cotton growing communities, the world's 'white gold' crop is produced, mostly through traditional agricultural practices echo of the past, but striving towards a more efficient and productive future with decent work and fundamental principles and rights at work.

Introduction

Scope of the report

Mapping of the cotton supply chain at the community level in Pakistan - A report on selected districts in Punjab and Sindh Provinces (henceforth referred to as 'the report'), seeks to better understand the status of fundamental principles and rights at work and the socio-economic context in the cotton producing communities in selected areas of Punjab and Sindh Provinces. Through evidence-based recommendations, it puts forward possibilities for improvement of the livelihoods and lives of the cotton producers at the community level.

The report presents findings of a mapping study (henceforth referenced as the "mapping exercise" or "the survey") commissioned to the Worldwide Fund for Nature Pakistan (WWF Pakistan) by the International Labour Organization (ILO). The mapping exercise was funded and implemented under two ILO projects, namely, **Promoting fundamental principles and rights at work in the cotton supply chain** (funded by the INDITEX company),¹ and **CLEAR Cotton: Eliminating child labour in the cotton, textile and garment value chains: an integrated approach**, funded by the European Union (EU).

More specifically, the scope of this report encompasses:

- The cotton producing community (tier 4), key actors involved in a range of tasks, relationships, practices, and issues (including lack of representation and collective bargaining, forced labour and compulsory labour, child labour, and discrimination in respect of employment);
- The market where seed cotton is traded;
- The ginnery, to the extent that it links with the farm and the market through the sale and purchase of seed cotton; and
- An overview of the working conditions from the perspective of fundamental principles and rights at work, of textile and garment unit workers – mainly in tiers 3, a few in tier 2 and one in tier 1.

¹ Industria de Diseño Textil SA (INDITEX), one of the largest fashion retailers, collaborates with the ILO through a global public-private partnership initiative aimed at promoting respect for fundamental principles and rights at work in the cotton sector.

Methodology

Primary data for the mapping study were collected through a total of 728 interviews and group discussions conducted in Punjab and Sindh Provinces. Of these, 300 interviews were with farmers (male), 300 with farm workers (150 male and 150 female), 16 focus group discussions and 36 interviews with stakeholders. These were conducted in five districts, namely, Vehari, Bahawalpur and Rahim Yar Khan in Punjab, and Mirpur Khas and Sanghar in Sindh. Also included in the interviews were 76 with workers in Faisalabad and Lahore (Punjab) and Karachi (Sindh).

Limitations of the mapping exercise

The report is challenged by limitations of the mapping exercise arising from (a) a reduced scope and depth in coverage of fundamental principles and rights at work, (b) data that mostly is not sex-disaggregated, nor age-disaggregated in the case of children, (c) sensitivities of data collection and access to stakeholders for discussions on issues of rights-based work in the traditional settings, particularly in remote areas, and (d) the impact of COVID-19 pandemic, particularly on mobility. Nonetheless, the findings are strong enough to result in the conclusions and recommendations provided in this report.

Socio-economic context

Pakistan's economy has faced challenges over the years but those posed by the 2020 COVID-19 pandemic, together with impacts of the country's stabilization programme, were unprecedented, plunging the growth rate to (negative) – 0.47 per cent in 2020 fiscal year. The economy started to revive and regained its pre-COVID-19 trajectory with 3.9 per cent growth in 2021 fiscal year. The Government reported broad-based recovery across sectors, underpinned by sector-specific measures. Agriculture sectoral growth of close to 2.77 per cent contributed to the overall Gross Domestic Product (GDP) growth (Pakistan FD 2021).² Continued adverse climatic conditions, including floods, drought and excessive heat, together with macroeconomic imbalances and global downturn in global economy resulted in another decline in the economic growth rate in 2022/23 – to 0.29 per cent, with agriculture sector's growth down to 1.55 per cent. Cotton production was particularly affected and registered a decline of 41 per cent. Agriculture sector's overall growth was enabled by livestock and other crops. Notwithstanding its challenges and the decline in growth rate, the agriculture sector remains critical to the country's economic and human resource growth, with its contribution of almost 23 per cent to Pakistan's Gross Domestic Product (GDP) and its total employment generation of over 37 per cent – the largest category of employment in 2022/23 (Pakistan FD 2023). Male and female agriculture employment declined in recent years, with the former at 28 per cent of total male employment; and the latter at just over 67 per cent of the total female employment.³

The compounded impact of multiple shocks - including the 2022 and previous catastrophic floods, high food and energy prices and the overall macroeconomic environment – increased poverty in Pakistan (US\$3.65/day) to an estimated over 39 per cent in 2023 – a 5 per cent increase from 2022. The year saw an additional 12.5 million people below the poverty line in Pakistan despite the increased targeted cash transfers (World Bank 2023). In addition to adverse climatic and macroeconomic conditions, Pakistan faces numerous development challenges including low and stagnant literacy rate – particularly the female literacy rate; inadequate primary and secondary education outcomes; low public expenditures on education, low-skilled agriculture labour, lack of opportunities for advancement and gender inequality. Many of these challenges are systemically embedded and impact cotton production, cotton growers and

2 Agriculture sector includes crops, livestock, forestry and fishing.

3 Data from the World Bank Data webpage, [“Employment in agriculture – Pakistan” – modeled ILO estimates](#). The agriculture sector in this data consists of activities in agriculture, hunting, forestry and fishing (consulted March 2024).

businesses. To a large extent, the human development challenges arise from and propagate in the large informal economy, that is home to cotton production. It is largely unregulated, uninspected, devoid of workers' representation and voice, and a catalyst for vulnerabilities leading to child labour, forced labour, and discrimination, and unsafe and unhealthy working environments.

Pakistan was the fourth largest cotton producer globally in 2018/19 (Pakistan MOFS&R API 2019). In 2022/23, it dropped further to seventh position, but rebounded to fifth according to current data for 2023/24, (Pakistan MOFS&R 2022, USDA 2024a and 2024b). Pakistan's global position for yield has been sliding.

The narrative of Pakistan's cotton production plays on its strength and potential, but also of missed opportunities. The health and vitality of the subsector has widespread implications for national and local economies and livelihoods. The decline in seed cotton production led to about 23 per cent decline in cotton ginning in 2022/23 (Pakistan FD 2023). In 2021, there were reports of shutdown of factories by some 60 per cent of the ginners in the previous three years. Hard-pressed cotton farmers were switching to other crops, such as wheat and sugarcane – considered to have more reliable harvest output and more support from the government (Dilawar and Mangi 2021).

The largest industrial sector in Pakistan is the **textile sector**, contributing 8 per cent to the GDP and 60 per cent to foreign exchange earnings. It accounts for some 40 per cent of the industrial workforce, employing 10 million people. The integrated cotton and textile sector has 1,050 ginneries, 430 textile mills and 350 cottonseed crushers and oil refiners (USDA and GAINS 2021). Pakistan is both an importer and exporter of cotton.

Cotton production in Pakistan

Grown mostly in Punjab and Sindh Provinces, the Federal Government is now boosting cotton production in Balochistan and Khyber Pakhtunkhwa Provinces as well. Punjab produces 68.5 per cent of the country's cotton (Samee et al./FAO 2014). There are an estimated 1.5 million cotton-growing farmers in the country and over 90 per cent of cotton is produced by small farmers cultivating less than 5 hectares of land (USDA and GAIN 2021). Over 90 per cent of cotton farmers in Punjab and 43 per cent in Sindh have less than 5 hectares of land. More farms in Punjab are below 2 hectares than Sindh, where more are 5 to 10 hectares (Pakistan MOFS&R 2019).

Low yield and low-quality cotton boll affect livelihoods of the small farmers and farmworkers and workers and employers in other industries and processes along the supply chain. Cotton yield and quality are impacted by numerous factors. **Human infrastructure** – seldom highlighted in discussions and policies on cotton yield – impacts all aspects of cotton production.⁴ Through its strengths and weaknesses, it is critical to the quality and quantity of the yield and consequently to livelihoods and the quality or vulnerability of the cotton producing communities.

Pakistan's vulnerable employment in 2022 was at 56 per cent of its total employment. The same year, 74 per cent of female employment and 51 per cent of male employment was estimated as vulnerable.⁵ The co-existence of vulnerable employment, working poor, and gender inequality in the cotton supply chain at the community level make cotton production highly challenging for women and puts children at risk of child labour. Women and girls bear an overwhelming burden of manual labour in cotton production, participating in all phases, but their role in ownership, management, marketing and selling is limited. Their access to finance, training and land ownership is systemically and traditionally difficult, thereby inhibiting their advancement.

4 In this report, the concept of 'human infrastructure' encompasses the farmers, farm workers and their families, together with those who provide farm inputs, credit, training and other services.

5 Data from The World Bank Data webpage, "[Vulnerable employment – Pakistan](#)" – modeled ILO estimates (consulted March 2024).

Child labour persists in farming in Pakistan, as in many countries globally. With little information and many misconceptions, it is difficult for families and others to grasp or acknowledge the hazardous nature of children's work that is detrimental to their health and often illegal.

The problem of child labour and hazardous work is particularly acute in family and smallholder farms and agricultural enterprises along the supply chain in the informal economy. However, there are tasks in agriculture that are not hazardous when performed in safe conditions at the legal working age. Appropriate training (including prior to work), supervision, and close monitoring can help children work safely in a safe and healthy environment. This can be beneficial to them.

Of the cotton produced in Pakistan more than 90 per cent is genetically modified (the GM Bt.) Cotton seeds for conventional cotton are not readily available and those available are of limited variety. There are efforts to cultivate organic cotton but the share of organic cotton in Pakistan is less than 0.5 per cent of the total cotton produced in the country. Organic cotton and certification are being promoted by the Government, but their use is not widespread presently.

Challenges to cotton production in Pakistan

There are numerous challenges, the key ones being: unverified and poor-quality seeds and crop; lack of government minimum support price over the years; difficulties in accessing formal financial/loaning facilities; machinery that is unsuitable for smallholder farms and women; underfunded and lagging research and development (with its main source of funding being the difficult-to-recover cess); and the absence of active organizations that focus on the cotton subsector and its workers and that give voice to concerns and advocate for issues important to the development of the subsector. Decent work and respect for fundamental principles and rights at work are seldom given consideration in discussions and policies related to cotton production. The Government announced a support price for cotton in 2021.

Legal, policy and programme framework

International labour standards: The labour rights Conventions of the International Labour Organization (ILO) set minimum standards and their accompanying Recommendations guide social policies. The **Declaration on Fundamental Principles and Rights at Work and its Follow-up** (ILO 2022) provides a framework of ten interdependent and mutually reinforcing fundamental Conventions that enshrine rights and principles in five critical areas, which are⁶:

- a. Freedom of association and the effective recognition of the right to collective bargaining,
- b. Elimination of all forms of forced or compulsory labour,
- c. Effective abolition of child labour, and
- d. Elimination of discrimination in respect of employment and occupation,
- e. A safe and healthy working environment

⁶ In June 2022, the International Labour Conference amended the ILO Declaration on Fundamental Principles and Rights at Work (1998) to include "a safe and healthy working environment" as the fifth fundamental principle and right at work, and to make consequential amendments to the ILO Declaration on Social Justice for a Fair Globalization (2008) and the Global Jobs Pact.

Fundamental principles and rights at work form one of the four strategic objectives of the globally recognized Decent Work Agenda, the other three being: decent employment and income opportunities, effective social protection for all, and tripartism and social dialogue.⁷ Pakistan has ratified eight ILO Fundamental Conventions.

Legal framework: The Constitution of Pakistan guarantees the protection of rights of workers, their families, and their livelihoods by mandating fundamental rights. These present a framework for policy formulation, labour legislation and programmes, including decent work. Pakistan has enacted laws related to the five critical areas of fundamental principles and rights at work and those related to the cotton industry. Gaps remain between the laws and international minimum standards that Pakistan has committed to, enforcement capacity is weak, and agriculture workers, including those involved in cotton production in the informal economy, are largely not protected by law. In its 2023 report, the Committee on the Application of Conventions and Recommendations (CEACR), presented Observations on the application of ILO Conventions and Recommendations in Pakistan, among other countries. It sought the Government's action to amend the federal and provincial Industrial Relations Acts to bring them in compliance with the Conventions so workers, including in the informal economy, can be protected and enjoy their rights under the Conventions (ILO 2023, 198–208, 647–653).

Policy and programme framework: Pakistan's current Decent Work Country Programme (DWCP) IV (2023–27), an outcome of consultations by the Government of Pakistan with representatives of employers' and workers' organizations and other stakeholders, is closely aligned with Pakistan's development framework, **Vision 2025**, provincial development plans, related employment and labour policies and priorities, Agenda 2030 Sustainable Development Cooperation Framework (UNSDCF) 2023–27. These frameworks reinforce the need for partnerships, coordinated and integrated planning, and effective implementation.

The four key priorities identified for implementation through Pakistan's DWCP-IV are: promoting decent work in economic recovery and climate response; expanding social protection coverage; reinforcing adherence to international labour standards and fundamental principles and rights at work; and fostering safe, healthy, gender-responsive workplaces free of violence and harassment.

The third **Textile and Apparel Policy – 2020-25**, approved in February 2022, aims to make Pakistan “one of the major players in global textiles and apparel supply chain”. Measures promoted in the policy include wide-ranging capacity development; reducing cost of doing business, strengthening of textile and apparel value chain; human resource development; and other measures to facilitate resolution of trade disputes and strengthen the role of textiles and apparel associations (Pakistan MOC n.d.). The **Agriculture Transformation Plan** of the Ministry of National Food Security and Research, aims to boost and expand cotton production to improve livelihoods at the farm level and related industries.

Economic and social sector programmes: The Government added numerous programmes to mitigate socio-economic consequences of the Covid-19 pandemic. For instance, cash transfers were increased under the Benazir Income Support Programme (BISP), the **Ehsaas** programme reached millions of beneficiaries through its digital payment system and a package was added for cash transfer to daily wage workers. Programmes of Punjab and Sindh Provinces included interest-free loans to farmers, subsidies for fertilizers and pesticides, a reduction in the mark-up on agricultural loans, and crop insurance (FAO 2020).

Institutions supporting productivity and innovations in the cotton sector include government departments at the federal and provincial level and affiliated bodies, such as the Pakistan Cotton Standard Institute, the Pakistan Central Cotton Committee, the Central Cotton Research Institute, and the Agriculture Extension and Adaptive Research Department. The main research focus is on improved

⁷ Social dialogue and tripartism in the context of labour and work refer to the negotiation, consultation and information exchange between and among representatives of government, employers' and workers' organizations on matters related to the world of work. The aim of social dialogue and tripartism is to promote consensus building and democratic involvement of the main stakeholders in the world of work.

quality of seeds. Major trade associations that advocate vigorously for their members include the Pakistan Cotton Ginners Association, Karachi Cotton Association, and All Pakistan Textile Mills Association.

There is an effort by the Employers' Federation of Pakistan to develop small farmers' associations with support of the ILO and local partners. Similarly, with no specific workers' organizations for workers in cotton production, Pakistan Workers' Federation, with support of the ILO and local partners is promoting local workers' organizations in cotton growing communities.

Formal lending sources include commercial agriculture banks, microfinance and credit institutions and the agricultural development bank.

Key findings and analyses

The findings from the mapping survey provide an overview of the work of farmers and workers in cotton production, and to a smaller degree of their children, together with information on textile and garment unit workers. The findings from the mapping survey are complemented with data and information from additional quantitative and qualitative sources. The analyses look beyond the numbers as they break down the processes and examine the relationships to better assess whether the work is underpinned by respect for fundamental principles and rights at work at the farm, in the market, the ginning factory and within a diverse set of textile and garment units. It looks at the critical, yet at times equivocal, authoritative multidimensional role of the *arthis* (commission agents) at the local market and that of the *beoparis* (intermediaries), at times rescuers, who travel to farms to purchase bundles of seed cotton right at the doorsteps.

Findings on fundamental principles and rights at work

The Declaration on Fundamental Principles and Rights at Work and its Follow-up (ILO 2022) provides the framework of rights at work and the human infrastructure for greater productivity. It consists of freedom of association and the effective recognition of the right to collective bargaining; elimination of all forms of forced or compulsory labour; effective abolition of child labour; elimination of discrimination in respect of employment and occupation; and a safe and healthy working environment.

The findings from the mapping survey that align with these areas of the framework are indicative of the level to which these rights are available to the cotton farmworkers, the farmers and the workers in the textile and garment units surveyed. For each of the four areas, the findings are presented, to the extent possible, in the context of (a) the community, with responses from farmworkers and farmers, and (b) the textile and garment units, with responses from workers.

Freedom of association and collective bargaining: There are few, if any, workers' organizations or trade unions in the informal economy that is home to cotton production. A few associations or non-governmental organizations that exist focus on land rights and other issues, not particularly on rights at work. For workers' organizations to have the authority to enter into collective bargaining they must be registered to do so. Punjab Province does not cover agriculture workers under its collective bargaining law. The mapping survey found a vast majority (72 per cent) of the 300 workers to be in favour of having an organization to take care of the rights of agriculture farm workers. However, they were uncertain about their employers' reaction to their membership, with most believing there would be no reaction, but some believing they would be fired.

Responses from 76 key informants in 14 textile and garment units showed unionization rates and collective bargaining to be low, while threats and fear of repercussions were high, particularly in the cottage industry and power loom units. Some respondents in the export-oriented industry units said they belonged to unions.

Forced labour and compulsory labour: A vast majority (75 per cent) of the 300 workers surveyed in cotton production loaned money from diverse sources, including informal lenders (mainly the village or market traders) or the farm input dealers. When paying off the debt became problematic, the borrowers found ways and means of paying back whether through wage deduction or selling their livestock. Only one in eight workers had a written agreement.

Most workers surveyed in the power loom and cottage industry units, said they had experienced, and had indications of the presence of forced overtime and threats of termination in case of non-compliance with demands made of workers. This was not reported in the case of workers in the export-oriented units. In certain circumstances, debt bondage and work under threat of penalty are indicators of forced labour.

Child labour: Notwithstanding all the attention the issue of child labour has generated across the sectors, the mapping survey reflected a lack of understanding on part the workers. Adult farm workers' perception put legal minimum age for work to be much higher than what it is by law. At the farms surveyed, the average age for girls and boys in cotton production was 15 years. For girls, it ranged from 8 years to 18 years; and for boys from 10 years to 17 years.

Almost half (43 per cent) of the 300 farmers said boys and girls help on the farms. Girls' involvement was mainly in harvesting (as for most adult female workers) and boys in sowing the seeds. Children's work on cotton farms came at a cost to their education, with many dropping out of school during the harvest season. When children's work impacts their education to such a degree that they drop out, it becomes a child labour situation.

Children 15 to 17 years were reported to be working in surveyed power loom and cottage industry units, but not in export-oriented units. The daily earning of girls ranged from 283 to 371 rupees and for boys from 245 to 388 rupees, with girls earning more than boys in harvesting and boys earning more in sowing, hoeing and plant spacing.

Discrimination in employment and occupation: Most workers did not think there was discrimination between girls and boys in accessing primary school. However, one fourth of the farmers said there were access barriers for girls, such as the need for girls to help at home, the tradition that girls do not go to school, and the families' financial constraints.

Twenty-five non-Muslim workers were interviewed as key informants to get a better understanding of discrimination in employment and occupation from their perspective. A majority of 68 per cent (17 workers) said they received equal wages but 32 per cent (8 workers) said they received lesser than Muslim workers. Some workers mentioned what they considered discriminatory social interactions, and discriminatory treatment in school.

Most textile and garment workers in the cottage and power loom industry units had experienced discrimination, including discriminatory treatment against them for access to basic amenities that management were provided. For women, discrimination also involved lack of opportunities to advance from low-paid and low-skilled jobs.

Safe and healthy working environment: Hazardous work perception of the workers did not quite match the dangers present in farm work or their experience with the tasks. While all 300 workers (female and male) surveyed considered tillage, levelling, pesticide and fertilizer application to be 'hazardous' work, most said they had not experienced hazards while performing the tasks, except in pesticide application. None of the workers perceived harvesting (done mainly by females) to be hazardous, but almost half the workers had experienced some form of health issue related to the task.

Farmers and farmworkers in 20 per cent of the surveyed villages noted that health services were not available in their villages, and they had to travel more than 10 km to get them. Access to health care providers for surveyed workers in the textile and garment units depended largely by the type of industry, with the more organized export-oriented units providing access to private medical providers to most of the surveyed workers, while the those in the power loom and cottage industry units paid for public health providers in crowded facilities.

Findings on farm mechanization and education levels

Farm mechanization: Cotton production in Pakistan remains mostly traditional, with limited mechanization, particularly on small farms. Most tasks are done manually by farm workers.

Education of cotton farmers, farmworkers and their children: Some 70 per cent of workers and 40 per cent of farmers on cotton farms had no formal education. The education level of the workers' children on cotton farms surveyed showed an improvement in the "no formal education" category for their parents. Some 45 per cent of workers' children and 43 per cent of the farmers' children had no formal education. The low education level of workers and farmers poses a threat to their personal development and the productivity and advancement of the sector.

Findings on farmworkers in cotton production

Of the 300 workers (female and male) surveyed in Punjab and Sindh Provinces, 72 per cent had casual employment status. Female workers did different types of work on farms throughout the season, working an average of 92 days during the cotton season. The work they did ranged from sowing to harvesting. Farmworkers' wages for casual cotton workers depended on the specific work involved, with female workers earning on average 392 rupees and male workers earned 495 rupees per (working) day. Permanent (salaried) workers were able to get some bonus, but not the casual workers.

Findings on farmers in cotton production

Reflecting on reasons why they opted for cotton farming, many farmers said they did so to carry on family tradition, some considered the income 'decent', while others said they had no other alternative. But, for most (58 per cent), the fact that it was valued in the community was important. Half the farmers engaged their family youth in cotton farming; and the majority said young people liked to farm. Environmental impact on cotton production and producers is a reality that the farmers and farmworkers are familiar with. One third of the farmers were aware of cotton farming's negative impact on soil, air and water.

When the cotton yield failed to meet its target, the economic vulnerability of many smallholder farmers came to the forefront, and they found their income insufficient and their lives unstable. They took out more loans to make ends meet, let the education of their children suffer, or sold livestock to overcome losses. Only 8 per cent of the surveyed farmers considered their cotton farming income sufficient to cover the basic needs of the family. About 88 per cent farmers said that they had to seek other sources of income as well.

Many farmers borrowed from different sources, including commercial banks and informal lenders such as the beoparis, arthis and farm input dealers. They borrowed for different reasons, but a large number (40 per cent) did not indicate whether they borrowed.

Seed cotton price determination at the community level remained elusive as traders deducted subjectively for impurities. Farmers showed no clarity on the value of their cotton. They did not understand how the price of seed cotton was fixed or who was responsible for setting the price.

When it came to selling their newly harvested seed cotton, 73 per cent chose to avoid the trade market and instead made deals with the village beoparis who collected it from their doorsteps. Many of these deals were made in advance of the harvest. Still, a quarter of the farmers sold in the market to the arthis and some 2 per cent sold directly to the ginneries.

Findings on working conditions in textiles and garments units

In addition to the field survey in tier-4, the findings of which are presented above, surveys were conducted with 76 workers in units of textile and garment industries in different tiers of the supply chain to gather data on the prevailing working conditions.

For workers in the textiles and garments units their employment status, working conditions and benefits depend largely on the type of industry they work in. The export-oriented industry units were more organized, offering better opportunities for workers to exercise and avail of their fundamental rights, more job security and better working conditions and benefits than the cottage or the power loom industry units. The export-oriented industry units seemed to maintain higher industry standards and were subject to industry audits, although this alone does not guarantee higher standards as was seen in Pakistan's garment sector factories.⁸ Generally, the standards and compliance declined from the export-oriented to the power loom and even more at the cottage industry units, with more precarious conditions.

Conclusions and recommendations

The results and analyses of the findings of this mapping exercise and review of available studies and information, show decent work deficits and a general lack of knowledge and disregard for fundamental principles and rights at work in the cotton growing communities in the five districts surveyed in Punjab and Sindh Provinces. There is no evidence of the promotion of fundamental principles or rights at work or their application at the community level in the cotton growing communities. This also is true for the textile and garment units surveyed in the cottage industry and the power loom industry units. The export-oriented industry units that were surveyed showed somewhat more respect for and application of fundamental principles and rights at work. While exclusive vertical supply chains may provide swift opportunities for tracking, tracing, verification and transparency, they are not fool proof and sustainable if the overall supply chain, in which they are embedded, continues to be incognizant of the standards and where fundamental principles and rights at work are not respected.

More specific description of the situation of these rights is given in section 5.1 Conclusions.

The support programmes for the agriculture sector are less focused on cotton growing communities and are not adequate in scope and duration for their multiple needs and those of the enterprises along the supply chain in the informal economy.

The vision of **a human-centred decent work approach** presented in the report's recommendations can bring this change about.

The main recommendations are presented to address: legal and policy reform, analytical research and research networks, capacity development, multi-media community-based campaigns, integrated inspection and community-based monitoring of fundamental principles and rights at work, linkages to social protection programmes, and a multi-stakeholder sectoral project.

The recommendations are directed at the ILO, its constituents, and partners (including sectoral and development partners) to take forward in partnership and to support, develop, implement, and promote.

⁸ In 2012, over hundreds of workers lost their lives, and many were injured, in fire at a textile factory in Karachi. Just weeks before that, the factory had been audited and issued a SA 8000 certificate.

Recommendation 1. ILO should support **legal and policy reform** and advocacy for a more equitable and enabling environment for cotton production that fosters respect for fundamental principles and rights at work.

More specifically, the recommendation includes the following sets of action:

- **Advocacy** at different levels to promote review of laws and policies related to fundamental principles and rights at work as they impact the agriculture/cotton growing communities.
- Research, in collaboration with **competent research bodies**, on specific issues in the laws and policies in need of change. (Some research topics listed below would also apply.)
- Support to **employers' and workers' organizations** in their advocacy for legal reform impacting the agriculture/cotton workers and their coverage under the laws, particularly the industrial relations laws.
- Technical support to the **Federal and Provincial Governments** to review and revise laws and policies that restrict freedom of association and collective bargaining, do not adequately address conditions of forced or compulsory labour and child labour, that hinder gender equality, are discriminatory, and do not ensure safe and healthy working environments.

Recommendation 2. ILO should undertake **analytical research** and strengthen **information systems** on fundamental principles and rights at work in the cotton supply chain in Pakistan.

The report illustrates the need for a deeper understanding in the following areas:

- The role of **human infrastructure**, particularly for women and children, in achieving cotton yield targets;
- The role of the intermediaries, particularly the *arthis* and *beoparis*, in the cotton growing community and the trading market;
- The expected and actual impact of existing **cotton taxation (including cess)** on the cotton growing communities and the revenue gaps created by barriers to implementation of the policies;
- Access to formal and informal **credit and loan facilities** by farmworkers and smallholder farmers;
- **Involvement of children** (below 18 years of age) in cotton production and support activities, including domestic and other non-farm/caring responsibilities ensuing from their parents' work in cotton production;
- Safe and permissible tasks that **adolescents** can perform; and
- **Gender-responsive situation analysis.**

Recommendation 3. ILO to convene a **regional meeting of experts** on fundamental principles and rights at work in the cotton supply chain, followed by a **national meeting for policymakers and implementing stakeholders.**

Recommendation 4. ILO should implement a **capacity development programme** on fundamental principles and rights at work in the cotton supply chain at the institutional and individual levels through traditional and innovative approaches.

Gender analysis and training needs analysis should be undertaken prior to developing a programme to meet the needs of the target groups and to integrate gender equality in the programmes.

The target groups for the programme include:

- Concerned government departments at the federal, provincial and district level, including extension service workers, employers' and workers' organizations, trade associations, non-governmental organizations;

- At farm and market levels for workers, farmers, input providers, intermediaries (*beoparis* and *arthis*), members of the market committee, other trade committees, community level non-governmental organizations and other bodies; and
- Agriculture universities and research institutes, particularly those with existing extension agriculture programmes or with a potential to develop programmes that can connect university students and faculty with cotton growers for mutual learning.

Recommendation 5. The ILO to develop localized training programmes, particularly for smallholder farmers, to support them in improving **the business of farming**.

It is recommended that the farmers' training programmes should be needs-based and multidisciplinary to the extent possible but focused on farming from a business perspective. Social partners and local trade and business associations must be trained to deliver the programme and support the growth of the farm business.

Recommendation 6. Support the local communities to launch and implement locally developed multi-media community-based campaigns to promote decent work practices, respect for fundamental principles and rights at work.

Recommendation 7. Support the mandated local government agency and partners in the development and implementation of a simple and do-able **integrated inspection and community-based monitoring** of fundamental principles and rights at work.

Recommendation 8. Based on evidence and experience, continue to support partners in creating and strengthening linkages of farmworkers and farmers to social protection programmes, credit providers and other opportunities that will help improve prospects for securing their fundamental rights at work and supporting them and their families with the social protection needed

Recommendation 9. Support, through a **multi-stakeholder sectoral project**, national partners and domestic and international businesses that are sourcing from Pakistan to improve the cotton supply chain through greater formality with respect for fundamental principles and rights at work in the cotton growing communities and businesses along the supply chain.

► 1. Introduction

1.1. Purpose of the mapping report

This report intends to better understand the socio-economic context and the status of fundamental principles and rights at work in the selected cotton producing communities, and to seek opportunities for their improvement through evidence-based recommendations.

The improvement of fundamental principles and rights at work and of the economic opportunities for local stakeholders in the cotton supply chain are aligned with and contribute to the priorities articulated in Pakistan's Extended Decent Work Country Programme (DWCP 2016-22).⁹ The key aligned priorities are promoting decent work in the rural economy and strengthening compliance with international labour standards through social dialogue.

The International Labour Organization (ILO) commissioned the preparation of this report to the Worldwide Fund for Nature Pakistan (WWF Pakistan) by under two ILO projects, namely, **Promoting fundamental principles and rights at work in the cotton supply chain** (funded by INDITEX),¹⁰ and **Eliminating child labour in the cotton, textile and garment value chains: an integrated approach** (in short, the CLEAR Cotton project) co-funded by the European Union (EU).

1.2. Scope of the report

The cotton supply chain spans many tiers. It starts at the farm (tier-4), where the cotton crop (or cotton bolls as mature cotton pods are referred to) is produced and then moved through to the trade (grain) market and ginning.¹¹ As it leaves the ginning factory, cotton is split into different streams or channels for different end products. The seed is processed into seed oil and the lint part is processed further and finally manufactured into textiles, garments and other products for domestic and international consumption (tier-1). Pakistan also exports raw cotton. (See figure 5 of this report for a conceptual illustration of the tiers and the tasks and people involved.)

The scope of this report encompasses:

- the cotton producing community, key actors involved in a range of tasks, relationships, practices, and issues (including discrimination, forced labour, child labour, and lack of representation and collective bargaining)
- the market where seed cotton is traded;
- the ginnery, to the extent that it links with the farm and the market through the sale and purchase of seed cotton; and
- a few textile units, mainly in tiers-3, a few in tier-2 and one in tier-1, for an overview of their working conditions from the perspective of fundamental principles and rights at work.

⁹ For an explanation of fundamental principles and rights at work, see section 3.1.1. of this report.

¹⁰ Industria de Diseño Textil SA (INDITEX), one of the largest fashion retailers, collaborates with the ILO through a global public-private partnership initiative aimed at promoting respect for fundamental principles and rights at work in the cotton sector.

¹¹ The community trade market is also referred to as grain market.

1.3. Methodology

The research methodology involved undertaking desk review and conducting key informant interviews during field surveys. Primary and secondary data were collected and analysed. The mapping exercise involved desk review, field work, data analysis and preparation of the report.

Desk Review: Information was collected and examined from publicly available sources on national statistics, international statistical databases, reports, relevant policies, laws, and media reports.

Fieldwork: The study area consisted of:

- three districts in Punjab – Vehari, Bahawalpur, Rahim Yar Khan – for interviews with farmworkers and farmers; and two cities – Faisalabad and Lahore – for interviews with textile units, and Multan for stakeholder interviews.
- two districts in Sindh – Mirpur Khas and Sanghar; and the city of Karachi for interviews with textile and garment unit workers.

Field enumerators had experience in research in rural Punjab and Sindh and were well versed in local languages including Punjabi, Urdu, Saraiki, and Sindhi. The team included female surveyors to facilitate access to female workers, and specialized staff for data compilation and analysis. The team attended a briefing and planning meeting and participated in a comprehensive week-long training programme. ILO data collection tools, including protocols for semi-structured interviews, themes for focus group discussion and survey questionnaires for workers and farmers were adapted, pre-tested and refined. Each of the stakeholders interviewed was assigned a geographic location accessible through geographic information system.

Systematic random sampling technique was used to identify communities of farmers and farmworkers. Ten villages were randomly selected from each of the five districts. For purposes of the study (and keeping with actual practice), the farmer was considered the decision-maker on cotton crop issues for single- and multiple-owner farms.

A total of 728 interviews were conducted, including key informant interviews and focus group discussions at the community level (also referred to as tier-4 in this report), stakeholder interviews in various locations of the two provinces, and key informant interviews with industrial units in tiers-1, 2, and 3.

The interviews and focus group discussions were as follow:

► **Table 1: Interview respondents and focus group discussions**

Respondents	Gender/group	Total	Districts
Farmers	300 male farmers average age 39 years; range 30-70; most 35-40 years	300 individual interviews	Bahawalpur, Mirpur Khas, Rahim Yar Khan, Sanghar, Vehari
Workers	150 female workers 150 male workers average age 37; range 20-61 years; most 30-45 years	300 individual interviews	As above
Farmers, workers	5 male farmers, 5 female workers, 5 male workers, 1 semi-structured interview	16 focus groups	As above
Stakeholders (Various – see, below and annex 3)	4 females 32 males	36 individual interviews	Bahawalpur, Mirpur Khas, Multan, Rahim Yar Khan, Vehari

Respondents	Gender/group	Total	Districts
Textile units - 4 export-oriented - 5 power loom - 5 cottage industry	Gender – not available 26 persons in 4 export-oriented; 25 persons in 5 power loom/domestic factories; 25 persons in 5 units cottage industries	76 individual interviews	Lahore, Faisalabad, Karachi

Source: Mapping data.

Interviews and focus group discussions were conducted in 50 villages across a total of five districts in Punjab and Sindh Provinces. Stakeholders interviewed were provincial and local government officials (including labour, agriculture and children's agencies), cotton inspectors, and extension service workers, cotton research institutes, ginners, farmers and agriculture workers' organizations (NGOs), trade associations, grain market commission agents, agriculture credit banks, input companies (seeds, pesticides and fertilizers), laboratories, development non-governmental organizations, and programme organizations (WWF, BCI, Cotton Connect).

Other than Mirpur Khas, the selected districts are among the highest cotton producing areas in the two provinces.

Industrial units in Lahore, Faisalabad and Karachi were randomly selected and categorized as:

- export-oriented industries,
- power loom industries, serving mainly the domestic market, and
- cottage industries, serving the domestic market and supplying components to the export-oriented market.

Data Analysis: The software statistical package for social sciences (SPSS v25) was used for data entry and analysis. The range of data reliability at $0.7 \leq < 0.9$ was good. The average reliability statistics data on the farmers was Cronbach's Alpha 0.766, and for workers it was 0.721. The statistical measures used to test the reliability of the collected data (according to the equation of reliability developed by Cronbach in 2004) were as below:

$$\square = \frac{N \cdot \bar{c}}{(\bar{v} + (N-1) \cdot \bar{c})}$$

Where: N = the number of items; $\bar{c}$ = average covariance between item-pairs; $\bar{v}$ = average variance.

Range of Reliability: ≥ 0.9 Excellent; between ≥ 0.7 and < 0.9 Good reliability; between ≥ 0.6 and < 0.7 Acceptable; between ≥ 0.5 and < 0.6 Poor; < 0.5 Unacceptable.

Focus group discussions and stakeholders' interviews were transcribed, descriptively analysed, and summarized into manageable form for reporting. Summaries were prepared according to the themes. Considering data reduction, a key for analysis, concept maps were created to facilitate analysis.

1.3.1. Data and information sources

Findings of the survey, undertaken by WWF-Pakistan for the **Mapping of the Cotton Supply Chain at the Community Level** and information from other related sources including reports, studies, databases and reports in the media, form the basis of the analyses, discussions, conclusions and recommendations presented in this report.

ILO knowledge, operational experience and its engagement with the business community on rights-based standards and approaches on the subject in Pakistan and globally informed the Report and provided a credible reference point.

1.3.2. Target audience

The primary target audience of this report are development partners including government, non-government and private sector policy makers, researchers and implementers, employers' and workers' organizations, trade associations, trade unions, officials of the ILO, and individuals with an interest in fundamental principles and rights at work in cotton supply chain, enforcement in rural areas, and improving socio-economic conditions of cotton producing communities, particularly those practicing traditional agriculture.

1.3.3. Limitations of the report

The Report is challenged by limitations of the mapping exercise. Among these are: (a) the reduced scope and investigation and a lack of depth in coverage of fundamental principles and rights at work; (b) data that mostly is not sex-disaggregated, nor age-disaggregated in the case of children; (c) the difficulties in accessing and collecting rights-based information in a traditional setting; and (d) the impact of COVID-19 pandemic on cotton production and the communities, along with impeded, and at times halted, mobility that slowed the pace of data collection and report preparation.

The findings provide an overview of fundamental principles and rights at work in the districts and units where the survey was conducted cannot be generalized to the entire cotton supply chain. Nonetheless, the findings and context are valuable. They point to opportunities and gaps, resulting in sound conclusions and recommendations for impactful action by multiple stakeholders.

1.4. Structure of the report

Section 1 introduces the mapping exercise, providing its purpose and scope. It explains the methodology used for collecting data and information and identifies geographical areas where surveys and focus group discussions were conducted with the cotton farmers, workers and other stakeholders in tier- 4. It further informs about the survey in selected textile and garment units in the tiers-2, 3 and 1. The section identifies the primary target audience for the report. In conclusion, the Introduction points to the limitations of the study.

Section 2 sets the socio-economic context for cotton production in Pakistan, on the one hand its significance to the national economy and on the other the vulnerability of smallholder cotton growing families, farmers and workers, in particular women and children.

It discusses farm size, the issues related with cotton seed quality, organic cotton and certification. It explains why human infrastructure is an important factor that contributes to the yield but is seldom highlighted in discussions and reports. The section ends with a discussion of the numerous and varied challenges facing cotton growers in Pakistan.

Section 3 introduces the framework of international labour standards, the Declaration on Fundamental Principles and Rights at Work and its Follow-up (ILO 2022) and briefly explains its four pillars, which are the focus of this report. The section highlights the national legal framework, touching on the fundamental rights and principles guaranteed by the Constitution of Pakistan. It then notes related legislation, including the industrial relations and employment of children's laws of Punjab and Sindh, looking particularly at coverage of agricultural worker (or absence of coverage), the minimum age for employment and prohibited hazardous work for children.

The section discusses the strategic programme frameworks that underpin action on fundamental principles and rights at work in cotton production. The global Decent Work Agenda and Pakistan's Extended Decent Work Country Programme III are highlighted, followed by a look at key social sector initiatives by the Federal Government and the Provincial Governments of Punjab and Sindh.

Institutions supporting productivity and innovations in cotton production are mentioned, together with trade associations, employers' and workers' organizations, and non-governmental organizations. The section concludes with a look at institutions that provide credit to cotton producers and their access to credit.

Section 4 presents the findings of the mapping undertaken in selected areas of five districts in Punjab and Sindh Provinces of Pakistan. The findings throw light on the tier-4 farmworkers and farmers, and on fundamental principles and rights at work and working conditions in a limited number of industrial units.

The section presents multi-sourced analyses based on the findings of the mapping survey and other related sources including reports, studies, databases and reports in the media. It discusses key issues as well as the work involved at different stages of cotton production from preparing the soil, sowing the seeds to harvesting, weighing and trading. The approach provides a focus on the people involved – the workers, farmers, traders, ginnery owners, textile unit workers – and their business relationships. The role of key players is discussed at some length.

Section 5 proposes a set of recommendations based on the multi-sourced analyses. They are aimed at the ILO, its tripartite constituents (government, employers' and workers' representatives) and partners in Pakistan, who are seeking to improve fundamental principles and rights at work and the socio-economic conditions of the cotton growing communities.

A **Glossary** at the end of the Report provides brief explanations of key concepts used and local language terms that have been retained.

Additional and explanatory information is provided in **Annexes** as follows:

- Annex 1 lists districts and villages in Punjab and Sindh Provinces where the mapping surveys were undertaken;
- Annex 2 lists districts and villages where focus group discussions were conducted;
- Annex 3 lists stakeholders interviewed;
- Annex 4 gives cotton producing districts in the two provinces; and
- Annex 5 gives additional charts pertaining to the mapping exercise.

► 2. Socio-economic context for cotton production

2.1. The socio-economic context

The economy and COVID-19 pandemic: The Islamic Republic of Pakistan has over the years faced challenges arising from its economy but none like the unprecedented difficulties posed by the 2020 COVID-19 pandemic, together with impacts of the country's stabilization programme. The Government's COVID-19 response helped to mitigate severe impact of the pandemic to some extent. It provided economic stimulus¹² and construction packages, an expanded social safety net for the vulnerable, monetary policy support and targeted financial initiatives that helped with an increase in domestic economic activity.

The early 2020 complete lockdown kept workers away from the fields, disrupted transportation and access to farm inputs and sale. It exacerbated the already challenging situation of smallholder farmers and workers. Agriculture workers were allowed by the Government to return to the fields in mid-April. However, lack of transport made it difficult for workers and producers to move, leaving farmers in many cases without the skilled workers they needed and reduced chances for crops and produce to be taken to the market. Moreover, many rural markets where produce and crops are traded, were unable to meet the required standard operating procedures for heightened hygiene and remained closed (FAO 2020).

The economy started to revive and regained its pre-COVID-19 trajectory with 3.94 per cent growth in 2021 fiscal year, recovering from 2020 fiscal year growth of negative – 0.47 per cent and surpassed the 2.08 per cent of 2019 fiscal year.¹³ The Government reported broad-based recovery across sectors, underpinned by sector-specific measures. Agriculture sector growth of close to 2.77 per cent contributed to the overall Gross Domestic Product (GDP) growth. (Pakistan FD 2021). Continued adverse climatic conditions, including floods, drought and excessive heat, together with macroeconomic imbalances and global and downturn in global economy resulted in another decline in the economic growth rate – to 0.29 per cent – in 2022/23 and the agriculture sector's growth to 1.55 per cent. Cotton was particularly affected presenting a decline of 41 per cent in cotton production. The sector's overall growth was enabled by livestock and other crops.

The compounded impact of multiple shocks - including the 2022 and previous catastrophic floods, high food and energy prices and the overall macroeconomic environment – was set to increase poverty in Pakistan to over 39 per cent (US\$3.65/day) in 2023 – a 5 per cent increase from 2022. Even with increased levels of targeted cash transfers to those impacted, the country saw an additional 12.5 million people below the poverty line in 2022 (WB 2023).

In addition to adverse climatic and macroeconomic conditions, Pakistan development challenges include low and stagnant literacy rate – particularly the female literacy rate; inadequate primary and secondary education outcomes; low public expenditures on education, low-skilled agriculture labour, lack of opportunities for advancement and gender inequality. Many of these challenges are systemically embedded and impact cotton production, cotton growers and businesses as well. To a large extent, the human development challenges arise from and propagate in the large informal economy, that is home to

¹² Amounting to Rs 1,240 billion.

¹³ Pakistan's financial year cycle is 1 July through 30 June.

cotton production, and is largely unregulated, uninspected, devoid of workers' representation and voice, and a catalyst for vulnerabilities to child labour, forced labour, and discrimination.

Cotton and its contribution to the national economy: Pakistan's agriculture sector is critical to the country's economic and human resource growth, with its contribution of almost 23 per cent to Pakistan's GDP¹⁴ and its over 37 per cent of total employment generation – the largest category of employment in 2022/23 (Pakistan FD 2023). However, the sector's ability to provide employment and livelihood opportunities has come under strain. Agriculture employment has declined in recent years, with male agriculture employment at 28 per cent of total male agriculture employment; and female employment at just over 67 per cent of the total female employment.¹⁵

Unlike its competitive major crops (wheat, rice, sugarcane and maize), cotton did not meet its production target, nor did it improve on its previous year's growth. There was an almost 23 per cent reduction in the number of cotton bales compared to the previous year, from over 9 million bales in 2019/20 fiscal year to just over 8 million bales in 2020/21 fiscal year and under 5 million bales in 2022/23. Factors contributing to this were a decrease in the area cultivated, unpredictable monsoon rains and pests. Wheat production, which has high growth, benefited from the Government's sectoral comprehensive targeted financial support which provides subsidies and has increased the minimum support price of wheat (Pakistan FD 2021; Pakistan FD 2023). The cotton stakeholders advocated for greater subsidies and cotton support price over the years and in 2021 the Government announced a minimum support price for cotton as well.

Pakistan was the fourth largest producer globally in 2018/19 (Pakistan API 2019). For the 2020/21 crop season, sources show it at fourth position (Pakistan MOFS&R API 2020), while others show it sliding to fifth (USDA 2024b). In 2023, it dropped further to seventh position, but regained its fifth position according to current data for 2023/24, (Pakistan MOFS&R API 2022, USDA FAS 2024a, and USDA FAS 2024b). Pakistan's global position for yield has been sliding, with 33rd position noted for the 2022-2023 season. (Pakistan MOFS&R API 2022).

The health and vitality of the cotton sector has wide impact on the national and local economies and livelihoods. The largest industrial sector in Pakistan is the textile sector, contributing 8 per cent to the GDP and 60 per cent to foreign exchange earnings. It accounts for some 40 per cent of the industrial workforce, employing 10 million people. The integrated cotton and textile sector has 1,050 ginneries, 430 textile mills and 350 cottonseed crushers and oil refiners (USDA and GAINS 2021). Pakistan is both an importer and exporter of cotton. It imports better grades of cotton for production of high quality textile products for export. The decline in cotton crop led to about 23 per cent decline in cotton ginning in 2022/23 (Pakistan FD 2023). In 2021, the Chair of the Pakistan Cotton Ginners' Association reported that 60 per cent of ginners had shut their factories in the previous three years due to the lower domestic production of seed cotton. There also were reports of hard-pressed cotton farmers switching to other crops, such as wheat and sugarcane that are considered to be more reliable harvest output and more support from Government (Dilawar and Mangi 2021).

The impact of lower-than-expected seed cotton production in the country, has negative impact on workers not just on the farms but along the supply chain in the various textile and garment industry units. Pakistan is both an importer and exporter of cotton. It imports better grades of cotton for production of high quality textile products for export. The decline in cotton crop led to about 23 per cent decline in cotton ginning in 2022/23 (Pakistan FD 2023). In 2021, the Chair of the Pakistan Cotton Ginners' Association reported that 60 per cent of ginners had shut their factories in the previous three years due to the lower domestic production of seed cotton. There also were reports of hard-pressed cotton farmers

14 Pakistan's GDP for 2022 was 394 billion US dollars according to The World Bank Data. <https://data.worldbank.org/indicator/NY.GDP.MKTP.CD?locations=PK> (consulted in February 2024).

15 Data from the World Bank Data webpage, "Employment in agriculture – Pakistan" – modeled ILO estimates at. (The agriculture sector in this data consists of activities in agriculture, hunting, forestry and fishing.) https://data.worldbank.org/indicator/SL.AGR.EMPL.ZS?locations=PK&most_recent_value_desc=true (accessed March 2024).

switching to other crops, such as wheat and sugarcane that are considered to be more reliable harvest output and more support from Government (Dilawar and Mangi 2021).

Studies note lack of skills and technology as being major internal factors that affect textile production. Many of these production units are in the informal economy with the workers having low skills, thereby posing a challenge to the introduction of advanced technology (Hussain et al. 2020). Karachi, which is the largest industrial hub of Pakistan, absorbs a large number of migrant workers from Afghanistan, Bangladesh and Myanmar, along with interprovincial migrants, particularly from Khyber Pakhtunkhwa Province. They live mostly in the over 100 migrant concentrated areas in the city and many work in the textile sector (PILER 2017). These migrants too are impacted by the success or failure of the cotton yield.

Cotton by-products: Not only cotton but its by-products have significant economic value for Pakistan. Cotton seeds are used to produce edible oil for which it meets 74 per cent consumer requirement. After extraction of oil from the cotton seeds, the raw material is used for 3 to 4 million tons of poultry and livestock feed (Pakistan MOFS&R 2019). Waste stems and sticks are used to produce an estimated 4 to 5 million tons of firewood mainly for use in households. Cotton linters (the small fuzz left on the cottonseed after the cotton is removed) are used to produce paper and other products.

Workforce: Pakistan's 2022 population was estimated at over 235 million and its growth rate at nearly 2 per cent – with an almost equal per cent of male and female populations.¹⁶ The labour force participation of those 15 years and over was 29 per cent for the female¹⁷ and 80 per cent for the male population in 2019.¹⁸ The ratio of female to male labour force participation rate was 30 per cent (2022) according to The World Bank modelled ILO estimate.¹⁹

Pakistan had a pre-COVID-19 working population of nearly 56 million, or 35 per cent of 10 years or older (estimated January to March 2020). From April to July 2020 it declined to just over 35 million or 22 per cent (estimated), as people either lost their jobs or were unable to work). However, the working population numbers were largely recovered in the following months of August to October 2020 to 52 million, or 33 per cent, as people returned and the Government's COVID-19 response programme started to have impact. Of the workers who lost their jobs or could not work, 74 per cent were in the informal economy. Some 3 per cent of agricultural workers were affected (Pakistan MOPD&SI PBS 2020).

Although in decline, employment in agriculture at 36 per cent of the total employment continued to be the largest category of employment in 2022. The decline is reflected in both the male and female numbers, with male employment in agriculture at 28 per cent of total male employment; and female population at just over 67 per cent of the total female employment.²⁰ It shows that of women who are working (and are counted in national statistics), most are in agriculture.

Pakistan has 59 per cent of its population in the age group of 15-59 years, with 27 per cent of this in the 15 to 29 years of age range. Over 31 per cent of youth (aged 15 to 24 years) are neither in school nor captured in employment statistics (UNDP 2021).

The United Nations Development Programme's Human Development Index (HDI) ranked Pakistan 154 out of 189 countries for 2019, with an overall HDI value at 0.557 — an increase of 38.6 per cent between

16 Data from The World Bank Data webpage, "[Population, total - Pakistan](#)" (consulted March 2024).

17 Data from the World Bank Data webpage, "[Ratio of female to male labour force participation rate \(%\) \(national estimate\) - Pakistan](#)" (consulted March 2024).

18 Data from the World Bank Data webpage, "[Labor force participation rate, male \(% of male population ages 15+\) \(modeled on ILO estimate\) - Pakistan](#)" (consulted March 2024).

19 Data from the World Bank Data webpage, "[Ratio of female to male labor force participation rate \(%\) \(modeled on ILO estimate\) - Pakistan](#)", (consulted March 2024).

20 Data from the World Bank Data webpage, "[Employment in agriculture - Pakistan](#)" (modeled on ILO estimate). The agriculture sector in this data consists of activities in agriculture, hunting, forestry and fishing, (consulted March 2024).

1990 and 2019. It is among countries with low gender equality in HDI achievements between women and men – its female HDI at 0.456 is lower than for males at 0.612 (UNDP 2020).

Literacy, education and skills: Providing quality education and requisite skills to the large proportion of the young population for the realization of economic gains is critical but remains a challenge. Skilled labour force is about 28 per cent of the workforce (UNDP 2020b). Federal and Provincial Governments' public expenditure on education was estimated at 1.5 per cent of the GDP in 2019-20, — a decline of 0.8 from 2018-19. The closure of educational institutions and the lockdowns due to COVID-19 pandemic may have contributed to this decline, but inadequate allocation towards primary and secondary education remains a longstanding obstacle in many South Asian countries. The literacy rate of the population (10 years and above) in Pakistan in 2019-20 remained stagnant at about 60 per cent (Pakistan Bureau of Statistics 2021). The World Bank estimates for 2014 pin female literacy at 44 per cent and male literacy at 69 per cent of their respective populations of 15 years and above.²¹ Province-wise analysis show Punjab with the highest literacy rate of 64 per cent followed by Sindh with 58 per cent, Khyber Pakhtunkhwa with 55 per cent (excluding Merged Areas) and 53 per cent (including Merged Areas), and Balochistan with 46 per cent.²²

Climate change: Compounding challenges to Pakistan's socio-economic development are the increasingly disastrous impacts of climate change. The Climate Risk Index (CRI)²³ of Germanwatch, which advocates for global equity and the preservation of livelihoods, places Pakistan among the top ten countries most affected by climate change during 2000–2019. It shows the country's losses per unit GDP as 0.52 per cent (Germanwatch 2021). Since 2010 Pakistan has experienced the impact of climate change through severe and intense episodes of floods, high temperatures and windstorms. It is also confronted with increased cyclonic activity, landslides and melting of glaciers in the country's northern areas.

These climatic changes can result in devastating impacts on cotton and agricultural productivity and quality as well as on the overall socioeconomic condition of farmers, workers and other people along the cotton supply chain, particularly the vulnerable communities. Small landholders, who form more than 80 per cent of the total farming community are particularly vulnerable (Saif 2017).

2.2. Cotton production in Pakistan

Cotton is an immensely critical cash crop, providing livelihood for millions of families and raw material for Pakistan's largest industrial sector, that is, the textile sector. Covering 14 per cent of arable land, cotton is a **kharif** crop that depends on the monsoon rains (April to June). Its yield and quality are affected by the timing and intensity of the rains, among other factors. Grown mostly in Punjab and Sindh, Punjab produces 68.5 per cent of the country's cotton (Samee et al./FAO 2015). It accounts for 65 per cent and Sindh for 35 per cent of the cotton planting area in Pakistan (USDA and Gain 2021).

21 Data available on the World Bank page "[Literacy rate, adult male \(% of males ages 15 and above\) – Pakistan](#)", (consulted October 2021).

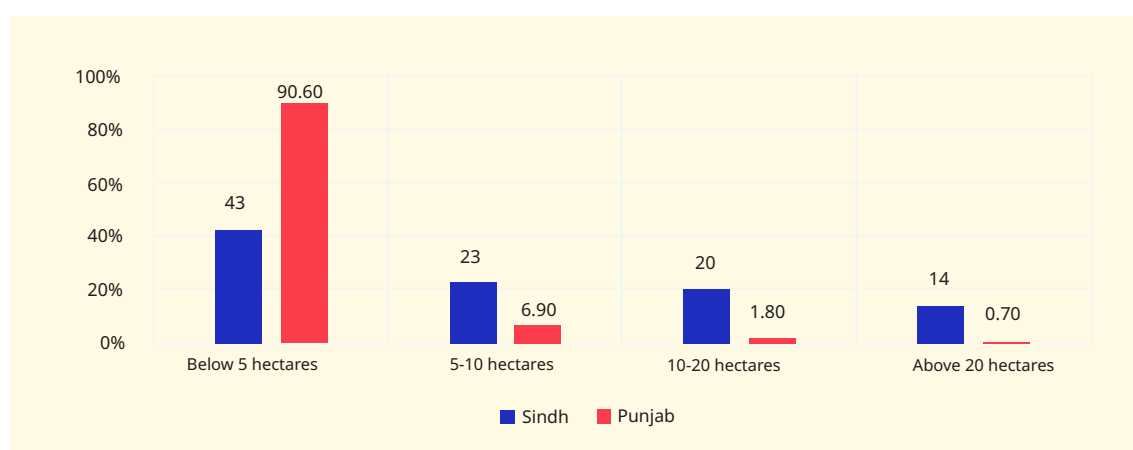
22 Merged Areas refers to the Pakistan's seven Federally Administered Tribal Areas (FATA) that were merged in 2018 with the neighbouring province of Khyber Pakhtunkhwa.

23 Global Climate Risk Index, according to Germanwatch, analyses to what extent countries have been affected by impacts of weather-related loss events (storms, floods, heat waves, among others).

Cotton farm size

More than 90 per cent of cotton is produced by small farmers cultivating less than 5 hectares of land (USDA and GAIN 2021). Over 90 per cent of cotton farmers in Punjab and 43 per cent in Sindh have less than 5 hectares of land. In Punjab, about 64 per cent of these farmers have land less than 2 hectares, and 24 per cent have under 0.5 hectares. Comparatively, 57 per cent of farms in Sindh are 5 hectares or more while in Punjab just over 9 per cent of farms fall within this range (Pakistan MOFS&R 2019) (see figure 1).

► **Figure 1: Cotton farm size, Sindh and Punjab, hectares, 2020 (percentage)**



Source: Pakistan MOFS&R (chart combines provincial data from slide 43).

Types of cotton produced

Pakistan produces conventional, genetically modified (popularly known as Bt. cotton), organic, and sustainable cotton.

More than 90 per cent of the cotton produced in Pakistan is genetically modified (GM) or Bt. Cotton seeds for conventional cotton are not readily available and those available are of limited variety. As a result, farmers tend not to produce conventional cotton. The share of organic cotton in Pakistan is less than 0.5 per cent of the country's total cotton produced. This is consistent with the low share of certified organic cotton globally, which accounts for less than 1 per cent of global cotton cultivation (Baloch et al. 2021).

The Government and international organizations, such as Centre for Agriculture and Bioscience International (CABI) and Better Cotton, are collaborating with partners in Pakistan to assist with the more sustainable cotton farming, including organic inputs.

Certification systems

The mapping survey reported at least four organic cotton certification systems being implemented in Pakistan to various degrees:

Milliner Cotton Standard is a local standard developed by Artistic Milliner and WWF-Pakistan. Initially, 400 farmers were engaged in Rahim Yar Khan district.

Reel Cotton Standard was implemented in Sindh Province in two districts (Ghotki and Dadu) where 22,000 farmers were trained on use of the standard. There was a premium for cotton ginners. Farmers saw it as an opportunity for learning and economic benefit.

Organic Cotton Standard was implemented in two districts of Balochistan. During the 2019/20 cotton season, farmers received 200 to 300 rupees per 40 kg seed cotton premium for certified organic cotton.

Better Cotton Standard System was implemented in almost all the cotton-producing districts of Punjab and Sindh Provinces, with over half a million cotton growers during 2020. Public and private sector implementing partners trained farmers in use of the standard. Farmers did not receive premium in this programme, but most of the 300 farmers surveyed for the mapping exercise were satisfied with their access to training. In 2018-19, close to 369,300 BCI farmers produced 906,000 tonnes of Better Cotton – up from 701,000 the previous season, making Pakistan the second largest producer of Better Cotton, after Brazil. Better Cotton was grown on over a million hectares (BC 2020).

WWF-Pakistan, in collaboration with the Agriculture Extension Department Balochistan, worked with 4,000 cotton growers in Lasbela and Barkhan districts to produce certified organic cotton. In 2019-20 cotton season, 10,000 certified organic cotton bales were produced, an increase over 4,000 produced the previous year.

The Federal Government plans to further expand cotton production in Balochistan and Khyber Pakhtunkhwa Provinces, which present low pest pressure and reportedly high yields (Pakistan Finance Division 2021).

2.2.1. Cotton crop yield and impact on livelihoods and industry

Although Pakistan is a major cotton producing country, its per acre yield is well below its potential.²⁴ Cotton policy analysis for the 2022/23 cotton crop noted that Pakistan ranked 33rd globally in terms of yield, while it ranked higher for production and area (Pakistan MOFS&R API 2022). Boll weight of 2 to 3 gm per boll, is less than half that in the United States, Egypt and Australia and half of that in Turkey (ITC 2011). Low yield and low cotton boll quality have negative impact on earnings and livelihoods for farmers and farmworkers in the cotton producing communities as well as workers and businesses in other associated industries along the supply chain. Cotton crop production generates 85-90 billion rupees in rural employment. (See table 2.)

► **Table 2: Estimated rural workforce earnings generated from cotton production.**

Respondents	Gender/group
Planting of Cotton	8-10 billion
Picking	60-65 billion
Spraying	15 billion
Weeding and Hoeing	4-5 billion
Total	85-90 billion

Source: Pakistan MOFSR 2019

Factors contributing to yield impacts are often categorized as **technological** (agricultural practices), **biological** (weeds, pests, insects, diseases) and **environmental** (climatic conditions, soil fertility, water quality and reliability, topography) (Liliane and Charles 2020). **Human infrastructure** – seldom highlighted in discussions and policies on cotton yield – impacts all aspects of cotton production. Through its strengths and weaknesses, it is critical to the quality and quantity of the yield. In this report, the concept of 'human infrastructure' encompasses the farmers, farmworkers and their families, together

²⁴ Yield – the harvested cotton crop generally is measured in kilograms per hectare (kg/ha).

with those who provide farm inputs, credit, training and other services. The adequacy or inadequacy of health, education, literacy level, knowledge, information, know-how and working conditions of the people who form this infrastructure, and their fundamental rights at work, can have positive or negative impacts on the yield. For instance, studies have suggested the harmful impact of pesticides on workers' health, with one study arguing that agriculture productivity can be impacted in significantly negative ways that cause "damage of the labour". The study notes a negative outcome to be lower farm production due to reduced availability of sick labour, with some workers attending to the sick and others taking on additional work load on the farm. Another outcome noted is reduced availability of time for household tasks. Contributing in important ways to a sound and robust human infrastructure are governmental, non-governmental and private sector support systems and programmes (Rahman et al. 2019).

Policy and administrative factors contributing to low yield include the following when they are insufficient or inadequate: research and development (including seed technology and farm machinery for small farms), extension services, supportive measures, regulations and enforcement.

2.2.2. Vulnerability of cotton growing families

Failure of cotton crop to meet its target yield is a huge burden for the farming community, particularly for families (workers and farmers) relying entirely or mainly on income from the crop and also having to pay off loans. Crop insurance for cotton is not common, making the families, and more generally the community, vulnerable to multiple and cumulative impacts that result from a range of factors including reduced earnings and uncertainty of livelihoods, to nutrition, health and education of their children.

Pakistan's vulnerable employment in 2022 was at 56 per cent of its total employment. The same year, 74 per cent of female employment and 51 per cent of male employment was estimated as vulnerable.²⁵

Women in cotton production

Women's contribution to cotton production in Punjab and Sindh is massive (as it is to agriculture in general in the country), but largely "unrecognized, unpaid, underrated, and overlooked" and it does not rise to the level of decision-making. In Sindh, cotton production is considered a women-led activity as women participate in all phases – weeding, thinning, hoeing, applying manure, cotton cleaning, waste stick removal and cotton storage. According to one estimate 76 per cent women surveyed in Sindh allocate their time "mostly" and another 23 per cent "occasionally" to cotton picking, and only 33 per cent to cotton marketing/selling (Samee et al./FAO 2015, 115–128).

Women face multidimensional constraints from the wider social, cultural and traditional spheres that impact their work and productivity in cotton production. They are entrusted with the triple-burden of daily family chores, childcare and farm work. Tradition holds them back from having a voice in decision-making regarding family and work. Lack of contact with the agricultural extension workers (who have limited capacity and travel long distances) prevents female farmworkers from learning about issues affecting them and their work. It also limits information and technology transfer to the female workers. Other constraints include lack of training and awareness opportunities on the farm; lack of education/literacy; and no encouragement to try machinery or new tools and techniques. Access to finance is difficult as women generally are not holders of land titles, even if the family owns the land. Ownership of land is mainly with the male members of the family.

Notwithstanding the constraints and challenges, there is increasing recognition of the role of women in agriculture, and the benefits accrued to their families and the national economy as a result of their work. Increasingly, training programmes and loaning facilities are being directed towards them. There

²⁵ Data from The World Bank Data webpage, "Vulnerable employment – Pakistan" – modeled ILO estimates at https://data.worldbank.org/indicator/SL.EMP.VULN.ZS?end=2022&locations=PK&most_recent_value_desc=true&start=1991&type=shaded&view=chart (accessed March 2024).

are efforts to empower female cotton workers to take on roles with greater responsibilities and even to become independent farmers. The Better Cotton Initiative (BCI) in Pakistan reported working with 140 female farmers in Punjab and Sindh Provinces, helping them overcome cultural, financial and other challenges and to grow cotton more sustainably (BC Pakistan 2018). For the large part, traditional and gender barriers remain for females, whether they be farmers or farmworkers.

Children in cotton production

Children 5 to 17 years of age in cotton production are among the 112 million children in child labour in the agriculture sector worldwide – that is 70 per cent of the children in child labour globally. Over 64 million of those in child labour in agriculture are in hazardous work (ILO and UNICEF 2021). Agriculture is one of the three most dangerous sectors for work-related fatalities, non-fatal accidents and occupational diseases.

Child labour persists in farming in Pakistan, as in many countries globally, due to inadequate laws and poor enforcement, lack of education, family poverty at the farm level, traditional and labour-intensive methods of farming, attitudes and perceptions about the role of children. With little information and many misconceptions, it is difficult for families and those who work with children to grasp or acknowledge the hazardous nature of children's work that is detrimental to their health and often illegal. The problem of child labour and hazardous work is particularly acute in family and smallholder farms and agriculture enterprises along the supply chain in the informal economy. However, there are tasks in agriculture that are not hazardous when performed when performed in safe conditions at the legal working age. Appropriate training (including prior to work), supervision, and close monitoring can help children work safely in a safe and healthy environment. This can be beneficial to them.

2.2.3. Challenges to production of cotton crop in Pakistan

Some challenges – such as lack of focus on the inadequate and fragile human infrastructure that is a critical contributing factor towards yield, and the vulnerability of women and children – facing the cotton growing sector have been discussed above, other key ones are noted below. They are interconnected and overlapping.

Sole buyer: One of the key challenges is the lack of competition at the industry level. The All Pakistan Textile Mills Association (APTMA) is the sole buyer of cotton in the country. According to sources, including the Ministry of Food Security and Research, this monopoly results in tremendous losses to the farmers (Pakistan MOFS&R 2019, slide 12).

Unverified and poor-quality seeds and crop: Latest seed development technology is lacking in the public and private seed developing institutions. Reliable and verifiable varieties of seeds are not commonly available to farmers. The Seed Certification Department has insufficient regulatory staff, resulting in a flourishing market of uncertified and unreliable seeds. The current system of marketing of seed cotton is based on weight and there is no premium for cotton of good quality. As such, farmers are not incentivized to produce high quality cotton. As a fallback measure, in case of failure due to poor seed quality, some farmers use different varieties in the same season. The private sector fully dominates the seed market, with little inspection, monitoring or verification. Plant breeder rights are not regulated, and multinational companies have little confidence in investing in the area.

Lack of government support price and access to formal financial/loaning facilities: Cost of production is rising while its profitability for the farmers is declining. Most production takes place on small marginal farms where farmers often are not able to access loans from formal financial institutions. They borrow from local traders but that has its shortcomings, including the farmers' reluctance to spend more on farm inputs, such as fertilizers.

Government support policies for competitive crops have negative impact on cotton production. In the case of sugarcane and wheat, the Government announces support prices, which benefits their farmers. This led some cotton farmers in core cotton areas to switch to sugarcane. Pakistan's cotton is

not competitive in the world market where others have subsidies. After prolonged advocacy by cotton stakeholders in 2021, the Government announced minimum support price for cotton as well.

Machinery that is unsuitable for smallholder farms: The Agriculture Engineering Department focuses on machinery, with local production meeting demand to considerable extent. However, the machinery available requires a tractor for it to operate. Many smallholder farmers do not possess tractors. There is a lack of machinery and mechanized tools for use on small farms. As noted previously (in section 2.2.2.) female workers say they are left behind when it comes to training on and use of machinery. Once female workers start to use the machinery there will be a demand for farm machinery and tools to be adapted to meet their needs.

Research and development that is underfunded and lagging: Public sector research institutions depend on cess, set at 50 rupees per bale of 170 kg each. The budget of the Pakistan Central Cotton Committee, the apex cotton research and development body, depends on cess. However, collection is not efficient, with millions of rupees pending to be recovered from textile units. Some 166 mills defaulted on over 620 million rupees from 2012 to 2017 (Pakistan MOFS&R 2019, slide 31). The Pakistan Central Cotton Committee qualified for funds under the Covid19 pandemic programme, which was an immediate rather than sustainable solution.

In support of its industry members, The All Pakistan Textile Mills Association advocates vigorously for withdrawal of cess on imported cotton on the grounds that the industry is significant to Pakistan's economy and livelihoods of millions of people (Daily Times 2021).

No active organizations for cotton farmworkers and subsistence farmers to voice their concerns and advocate for issues important to them. The Pakistan **Kissan Rabita Committee** (PKRC), representing small farmers and peasants is not specific to cotton farmworkers or to their fundamental rights at work. The *Hari Committee* in Sindh, with a long history of a struggle-based movement supporting peasants against injustices, is a political organization. Furthermore, there are questions of whether it can respond effectively to current realities of the *haris* (tenants) since the dynamics of farm ownership, tenants and casual work have changed enormously in recent years (Sangi 2017). The Hari Committee too is not specific to the cotton farming community.

▶ 3. Legal, policy and programme framework

3.1. Legal framework

3.1.1. International labour standards

Fundamental Principles and Rights at Work Framework

International labour standards, the legal treaties shaped by Government, employers' and workers' representatives from ILO member States, are legally binding once ratified by Member states.²⁶ The ILO's labour rights Conventions and their accompanying Recommendations set minimum standards and guide social policies. Ten of the Conventions, designated as Fundamental Conventions, enshrine rights and principles in five critical areas. Conventions addressing these areas constitute the framework of the **Declaration on Fundamental Principles and Rights at Work and its Follow-up** (ILO 2022). Member states are obliged to respect the provisions of these Conventions.

They are interdependent and mutually reinforcing. Addressing one critical area leads to positive impacts in the other critical areas; conversely, deficits in one area, lead to negative impacts in other areas. The framework of decent work and fundamental principles and rights at work validates and builds on the value of this interdependence.

The Fundamental Conventions are:

- ▶ [Freedom of Association and Protection of the Right to Organize Convention, 1948 \(No. 87\)](#)
- ▶ [Right to Organize and Collective Bargaining Convention, 1949 \(No. 98\)](#)
- ▶ [Forced Labour Convention, 1930 \(No. 29\)](#)
- ▶ [Abolition of Forced Labour Convention, 1957 \(No. 105\)](#)
- ▶ [Minimum Age Convention, 1973 \(No. 138\)](#)
- ▶ [Worst Forms of Child Labour Convention, 1999 \(No. 182\)](#)
- ▶ [Equal Remuneration Convention, 1951 \(No. 100\)](#)
- ▶ [Discrimination \(Employment and Occupation\) Convention, 1958 \(No. 111\)](#)
- ▶ [Occupational Safety and Health Convention, 1981 \(No. 155\)](#)
- ▶ [Promotional Framework for Occupational Health and Safety, 2006 \(No. 187\)](#)

²⁶ The Conventions lay out the basic principles that countries are expected to use in updating their national legal frameworks and implementation mechanisms. Guidance on implementation of Conventions is provided in accompanying Recommendations, which are not legally binding.

Freedom of association and the effective recognition of the right to collective bargaining: the framework for legal standards and action for this area are provided by the following Conventions:²⁷

Freedom of Association and Protection of the Right to Organise Convention, 1948 (No. 87) sets forth the right for workers and employers to establish, join or organize organizations of their own choosing without previous authorization. These include federations and confederations, which may in turn affiliate with international organizations of workers and employers.

Right to Organise and Collective Bargaining Convention, 1949 (No. 98) stipulates that workers enjoy adequate protection against acts of anti-union discrimination, which can include: requirements that a worker should not join a union or relinquish trade union membership for employment, or dismissal of a worker because of union membership or participation in union activities. The Convention also provides for the right to collective bargaining, through which employers and trade unions can establish fair wages, working time, training, occupational health and safety and equal treatment and working conditions. The Convention provides the basis for sound labour relations and when its implementation is inclusive it can lead to reduced inequality and enable labour protection.

The elimination of all forms of forced or compulsory labour: Forced labour differs from exploitative or unsatisfactory working conditions. The framework for legal standards and action for this area are provided by the following conventions:²⁸

Forced Labour Convention, 1930 (No. 29), Abolition of Forced Labour Convention, 1957 (No. 105), and the 2014 Protocol to the Forced Labour Convention. The three elements of forced labour are:

- **Forced labour can occur in work or service** of all types, in any activity, industry or sector (including in the informal economy);
- **Menace of any penalty** refers to a wide range of penalties used to compel someone to work; and
- **Involuntariness** refers to the absence of free and informed consent on part of the workers to take on the job and a lack of their freedom to leave the job at any time.

Work is considered forced labour when workers are coerced to work using violence or intimidation, or by more subtle means as noted in [ILO standard indicators of forced labour](#). These are: restricted mobility/movement of workers (e.g., guards, high fences), isolation, exploitation of workers' vulnerabilities, deception, physical and sexual violence, intimidation and threats, retention of identity documents, withholding wages, debt bondage, abusive working and living conditions, and excessive or mandatory overtime.

Bonded labour, a subset of forced labour, is a significant problem in many of the South Asian countries, including Pakistan, where it is prevalent in agriculture and brick kilns.

The effective abolition of child labour: The framework for legal standards and action for this area are provided by the following Conventions:²⁹

ILO Convention on Minimum Age to Employment, 1973 (No. 138), which provides the most comprehensive and authoritative international definition of minimum age for admission to work or employment, calls on Member States to set:

27 See the ILO webpage on ILO instruments by subject and status at: https://normlex.ilo.org/dyn/nrmlx_en/f?p=NORMLEXPUB:12030:0::NO::

28 See the ILO portal on Forced Labour at www.ilo.org/topics-and-sectors/forced-labour-modern-slavery-and-trafficking-persons#ils

29 See the ILO webpage on ILO instruments by subject and status at: https://normlex.ilo.org/dyn/nrmlx_en/f?p=NORMLEXPUB:12030:0::NO::

- a general minimum age for admission to work or employment of at least 15 years of age (Article 2.3) (14 years of age in less developed countries), and
- a higher minimum age of not less than 18 years for employment or work which, by its nature or the circumstances in which it is carried out, is likely to jeopardize the health, safety or morals of young persons, i.e., hazardous work (Article 3.1).

Light work: Convention No. 138 allows less developed countries to permit light work for children younger than the minimum age. For those with minimum age at 15 years, light work could be 13 to 14 years, and for those minimum age at 14 years, light work may be 12 to 13 years. Light work is defined as work that does not interfere with children's schooling, or their ability to benefit from it, and that is not harmful. If countries choose to allow light work, they must determine what activities are considered light work, and the hours and the circumstances under which these activities may be carried out.³⁰

ILO Convention on the Worst Forms of Child Labour, 1999 (No. 182), supplements the Minimum Age Convention No. 138 by emphasizing the subset of worst forms of child labour, requiring immediate action. For purposes of the Convention, worst forms of child labour are as follow (Article 3):

- a. All forms of slavery or practices similar to slavery, such as the sale and trafficking of children, debt bondage and serfdom, as well as forced or compulsory labour, including forced or compulsory recruitment of children for use in armed conflict
- b. The use, procuring or offering of a child for prostitution, for the production of pornography or for pornographic performances
- c. The use, procurement or offering of a child for illicit activities, in particular for the production and trafficking of drugs as defined in relevant international treaties
- d. Work, which, by its nature or the circumstances in which it is carried out, is likely to harm the health, safety or morals of children.

As defined by international standards, child labour refers to the subset of children's work that is injurious, negative or undesirable to children and that should be targeted for elimination.

Furthermore, **The Convention on the Rights of the Child (CRC)** recognizes the child's right to be protected from economic exploitation and from performing any work that is likely to be hazardous or to interfere with the child's education, or to be harmful to the child's health or physical, mental, spiritual, moral or social development (Article 32.1). To achieve this goal, the CRC calls on State Parties to set minimum age for admission to employment, having regard to other international instruments (Article 32.2).

The elimination of discrimination in respect of employment and occupation: The framework for legal standards and action for this area are provided by the following conventions:³¹

Equal Remuneration Convention, 1951 (No. 100) requires the principle of equal remuneration for men and women for work of equal value to be applied to all workers, basic wage or salary and any additional emoluments payable directly or indirectly, whether in cash or in kind.

Discrimination (Employment and Occupation) Convention, 1958 (No. 111) defines discrimination as "any distinction, exclusion or preference made on the basis of race, colour, sex, religion, political opinion, national extraction or social origin, which has the effect of nullifying or impairing equality of opportunity or treatment in employment or occupation". Other grounds covered in the Convention are **discrimination**

30 From the ILO webpage on "International Labour Standards on Child Labour" available at: www.ilo.org/topics-and-sectors/child-labour#ils

31 See the ILO webpage on ILO instruments by subject and status at: https://normlex.ilo.org/dyn/nrmlx_en/f?p=NORMLEX_PUB:12030:0::NO::

in access to education, vocational training, employment and occupations, and **discrimination in terms and conditions** of employment.

The list of prohibited grounds of discrimination can be extended after consultation with representatives of employers' and workers' organizations, and relevant bodies. Some of the **additional prohibited grounds** recently included in national legislation in different countries include disability, age, real or perceived HIV status, sexual orientation and gender identity.

Ratifying States are required to declare and pursue a national policy designed to promote equality of opportunity and treatment in respect of employment and occupation, with a view to eliminating any discrimination in these fields.

A safe and healthy working environment: among key instruments are the following conventions.

Occupational Safety and Health Convention, 1981 (No. 155) and its Protocol of 2002, requires a national occupational safety and health policy, action by governments and within enterprises to promote occupational safety and health and to improve working conditions. The Protocol calls for a periodic review of the requirements and procedures for recording and notification of occupational accidents and diseases, and for the publication of related annual statistics.

Promotional Framework for Occupational Safety and Health Convention, 2006 (No. 187), promotes national preventive safety and health culture, the systematic treatment of occupational safety and health and coherent national policies through dialogue between government, workers' and employers' organizations.

Priority and other Conventions

Four Conventions, significant for enabling a **governance framework** for implementation of the international labour standards, are designated as priority instruments. These are: Labour Inspection Convention, 1947 (No. 81), Employment Policy Convention, 1964 (No. 122), Labour Inspection (Agriculture) Convention, 1969 (No. 129), Tripartite Consultation (International Labour Standards) Convention, 1976 (No. 144).

Furthermore, Recommendation No. 202 on National Social Protection Floor, Recommendation No. 203 on Forced Labour and Protocol of 2014 are important policy guidance documents on the respective subjects and in regard to the agriculture sector. ILO Recommendation No. 204 on Transition from the Informal to the Formal Economy, 2015, provides guidance including the transition of workers and economic units from the informal to the formal economy and is particularly relevant to the cotton supply chain.

Pakistan has ratified eight ILO Fundamental Conventions; two of the four ILO Governance (Priority) Conventions (Labour Inspection Convention (C081) and Tripartite Consultation (International Labour Standards) Convention (C144)); and 26 of the 177 ILO Technical Conventions.³²

Pakistan also has ratified several UN Conventions on human rights, including UN Convention on Rights of the Child (CRC), Convention on the Elimination of All Forms of Racial Discrimination (CEDAW), International Covenant on Civil and Political Rights (ICCPR), International Convention on the Elimination of all Forms of Racial Discrimination (CERD), International Covenant on Economic, Social and Cultural Rights (ICESCR).

32 From ILO webpage on "Ratifications for Pakistan", available at https://normlex.ilo.org/dyn/normlex/en/f?p=NORMLEXPUB:11200:0::NO::P11200_COUNTRY_ID:103166

3.1.2. National legal framework

Fundamental Rights and Principles guaranteed by the Constitution of Pakistan

The Constitution of Pakistan guarantees the protection of rights of workers, their families and their livelihoods by mandating fundamental rights, which present a framework for policy formulation, labour legislation and programmes, including decent work.

Among the rights are the prohibition and elimination of all forms of exploitation, slavery, all forms of forced labour, trafficking, and child labour below 14 years of age in any factory or mine or any hazardous employment. It guarantees the right to assemble peacefully and without arms; to form associations or unions; the right to equality; and prohibition of gender discrimination. It allows for “any special provision for the protection of women and children” and ensures skills development and training.

Furthermore, it calls for “equitable adjustment of rights between employers and employees, and landlords and tenants”, for social security by compulsory social insurance or other means for all employed persons. For all citizens unable to earn a livelihood due to infirmity, sickness or unemployment, it mandates the provision of basic necessities of life (food, clothing, housing, education and medical relief) irrespective of sex, caste, creed or race.

Laws in Pakistan related to fundamental principles and rights at work

Pakistan has enacted laws related to the critical areas of fundamental principles and rights at work and to the cotton industry. Gaps remain and provisions of the national laws do not always meet the international minimum standards. Enforcement capacity is weak. Agriculture workers, including those involved in cotton production in the informal economy, are not protected by law to a large extent.³³ In its 2023 report, the Committee on the Application of Conventions and Recommendations (CEACR), presented Observations on the application of ILO Conventions and Recommendations in Pakistan, among other countries.³⁴ It sought the Government’s action to amend the federal and provincial Industrial Relations Acts to bring them in compliance with the Conventions so workers, including in the informal economy, can be protected and enjoy their rights under the Conventions (ILO 2023, 198–208, 647–653).

Coverage of agriculture workers is limited in Pakistan’s laws and capacity for implementation in general is weak, further limiting coverage.

Sindh Industrial Relations Act, 2013 (SIRA-2013) regulates the formation of trade unions and their activities, relations between employers and workers, and the avoidance and settlement of differences and disputes between them. A unique feature of the Act is recognition of agriculture and fishery workers as industrial workers and thereby granting them the freedom and the right to join a trade union of their choice, form trade unions and collectively bargain with their employers. It is applicable to all agriculture workers in Sindh, including those working on cotton farms, whether the farms be small, medium or large. It also calls on trade unions, to include women in the executive and office bearers’ positions to the same proportion in which they are employed in the establishment.

33 ILO, “Database of national labour, social security and related human rights legislation: Pakistan”, Natlex (October 2021)

34 CEACR, composed of eminent jurist, provides an impartial and technical evaluation of the application of international labour standards in ILO member States. For more information, see ILO webpage on Committee of Experts on Application of Conventions and Recommendations”, <https://www.ilo.org/global/standards/applying-and-promoting-international-labour-standards/committee-of-experts-on-the-application-of-conventions-and-recommendations/lang-en/index.htm>

The law entitles female agriculture workers to equality in remuneration, non-discrimination (based on gender, land ownership, caste, religion, ethnicity and residential status), and forming or being associated with unions and associations. The law further sets standards for working conditions and provides for time off and different types of leave to female agriculture workers. It entitles them to access government programmes, credit facilities, social security and asset transfers, and protects them from harassment and abuse.

Not all industrial relations laws in Pakistan cover agriculture sector workers. Being able to join and form associations under industrial relations laws does not automatically entitle workers to collective bargaining – except when it is specifically granted as in Sindh under SIRA 2013 and in Balochistan under the 2015 amendment to its industrial relations law.³⁵

Sindh Women Agriculture Act 2019. This law, legislative milestone, gives rural women workers' rights like collective bargaining, social welfare, child health, community development, access to economic resources and public goods and services.

The Punjab Restriction on Employment of Children Act, 2016, under Section-2 (C) and **The Sindh Prohibition of Employment of Children Act, 2017**, applies to persons who have not attained the age of 15 years (children) and persons who have completed the 14 but not the 18 years of age (adolescents). It applies in agriculture establishments and a 'workplace where any economic activity is being carried on'.

The hazardous work activities included in the respective Schedules to the two Acts offers some protection to children working in agriculture, as noted below, although not all are relevant to cotton production:

- Power driven cutting machinery, like saws, shears, guillotines, agricultural machines, thrashers, fodder cutting machines;
- Mixing and manufacture of pesticides and insecticides and fumigation;
- Exposure to all toxic, explosive and carcinogenic chemicals (specific examples are provided in the Schedules);
- Lifting and carrying of heavy weight (15 kg and above) specially in transport industry; and
- Tobacco processing and manufacturing including, *niswar* and *biri* making.³⁶

In addition to the hazardous work listed in the Schedules, the laws' coverage is extended to other worst forms of child labour as stated in ILO Convention No. 182 on the Worst Forms of Child Labour. These are (a) slavery or practices similar to slavery, forced or compulsory labour and debt bondage, (b) prostitution, pornography and pornographic performances, and (illicit activities, such as production and trafficking of drugs. Both Acts prohibit the engagement of children below 15 years of age in all forms of child labour.

The laws set minimum age for employment and hazardous work, regulate the conditions of work for adolescents, define hazardous work, constitute a provincial committee on child labour, address the question of age verification; and extend the applicability of laws offering social security and benefits to adult and adolescent workers. The laws describe penalties for non-compliance including fines and prison time, require appointment of inspectors, and assigns responsibility of the 'occupier' towards the adolescents for completion of education. They seek protection against discrimination.

Other related laws are on Prevention of Trafficking in Persons Act, 2018, the bonded labour system (The Sindh Bonded Labour System (Abolition) Act, 2015 and The Punjab Bonded Labour System (Abolition) Act, 1992 (Amendment Act, 2012), which abolish bonded labour, and free the labourers from obligation that requires them to work for the creditor in a Bonded Labour System.

³⁵ Paycheck.pk "Labour Laws – Trade Union",

³⁶ *Niswar/naswar* is a mix of ground tobacco and other substances commonly chewed in some areas of Pakistan; *Biri/Bidi* are locally made cigarettes.

The Sindh Occupational Safety and Health Act, 2017, Sindh Occupational Safety and Health Rules, 2019 and The Punjab Occupational Safety and Health Act (2019), provide coverage for occupational safety and health at workplaces against the risk of workplace injury. The law promotes a safe, healthy and decent working environment for the physical, physiological and psychological needs of all persons at work. It is comprehensive, but does not cover agriculture,³⁷ farms, or the workers in the field, but the 2019 Rules for Sindh do specify ginning factories.

Laws related to cotton production and ginning

The Market Committee Act of 1939 established the grain market committees when canal colonies were established and vast areas in the River Indus Plains were under cultivation. Grain markets were established to facilitate trade of local agriculture produce.

The Punjab Agricultural Produce Markets (General) Rules, 1979 gave broader authority to the committees established under the Market Committee Act of 1939.

Cotton Cess Act, 1923 (No. XIV of 1923) provides for the creation of a fund for the improvement and development of the growing, marketing and manufacture of cotton. The Pakistan Central Cotton Committee (PCCC) is constituted under this the Act.

The cess assessed by the Federal Government and levied and collected on all cotton either exported from the Provinces of Pakistan to any place outside Pakistan or consumed in any mill in the country. The Act establishes that the proceeds of the cess and any other funds received by PCCC are for the purpose of meeting its expenses (as approved by the Federal Government) and for promoting agricultural and technological research in the interests of the cotton industry in Pakistan.

Cotton Ginning and Pressing Factories Act, 1925 (Act XII of 1925), Pakistan. It aims to regulate cotton ginning and cotton pressing factories. cotton" means raw cotton, whether baled or loose, which has been ginned.

Cotton Standardization Ordinance, 1994, re-promulgated in 2002, establishes Institute in Karachi with sub-offices in other parts of Sindh and Punjab to focus on principles of standardization, testing and quality control.

The Cotton Standardization Ordinance, 2002 promotes quality control of cotton in the country. It aims to transform cotton marketing and pricing system to internationally recognized standards and practices.

The Punjab Cotton Control Ordinance, 1966, regulates the processing and sale of cotton, and other related matters.

3.2. Policy and programme framework

3.2.1. Agenda 2030 and Decent Work

2030 Sustainable Development Agenda

Decent Work and Economic Growth (Goal 8) are central to the 2030 Sustainable Development Agenda, that is implemented through 17 Goals for sustained and inclusive economic growth and development. Goal 8 aims to drive progress, create jobs and improve living standards. Promoting jobs and enterprise,

³⁷ Under a broad interpretation of the law, a place "where work is carried out, by one or more persons and includes premises used for the storage of tools, machinery, equipment or substances" could be applied to storage facilities for input providers or farm machinery spaces, but it does not apply to the fields or farms. In the case of Sindh's law, there is no reference to the Sindh Industrial Relations Act, 2013, which could have provided some links to agriculture and fisheries as the 2013 Act recognizes them.

guaranteeing rights at work, extending social protection and promoting social dialogue are the four pillars of the ILO Decent Work Agenda, with gender as a cross-cutting theme.

In Pakistan, Goal 8 has become crucially urgent following the devastation of Covid-19 and the climatic conditions, particularly the floods and rain and in the context of the country's macroeconomic challenges impacting millions of lives and the regional and national economies.

Global Decent Work Agenda: strategic objectives

Decent Work is productive employment that fetches a fair income; it is performed at a safe and secure place, where workers have a voice, where they can organize themselves to have representation, and where they can participate in the decisions that affect their work and lives. Underlying Decent Work is the principle of equality of opportunity and treatment for all.

The four strategic objectives at the core of ILO's globally recognized Decent Work Agenda are:

- Setting and promoting standards and fundamental principles and rights at work
- Creating greater opportunities for decent employment and income.
- Enhancing the coverage and effectiveness of social protection for all.
- Strengthening tripartism and social dialogue.³⁸

A decent work approach to cotton production, and more broadly to the agriculture sector, involves an emphasis on extending social protection and rights at work, giving a voice to and improving the organization, the working conditions and environment, the productivity and the earnings of workers – including those on smallholder farms, larger plantations, in small businesses and cooperatives (ILO 2019b).

The Covid-19 pandemic put millions of workers in the informal economy worldwide at risk of losing their livelihoods, in addition to the millions that had already lost their jobs. In June 2021, delegates from 181 countries representing their governments, workers and employers at the International Labour Conference, unanimously adopted the Resolution, **A global call to action for a human-centred recovery from Covid-19 crisis that is inclusive, sustainable and resilient**. The Call to Action asks countries to implement broad-based and socially inclusive policies, and concrete action. It calls for specifically addressing inequalities highlighted by the crisis, particularly for groups that were disproportionately impacted and the small and microenterprises that provide most of the world's employment (ILO 2021).

- Among wide-ranging actions, it underscores the need for inclusive economic growth and employment, including through more resilient supply chains that contribute to:
- decent work,
- sustainability of enterprises along the supply chain, including micro-, small- and medium-sized enterprises,
- environmental sustainability, and
- protection of and respect for human rights in line with the UN Guiding Principles on Business and Human Rights, the ILO Tripartite Declaration of Principles concerning Multinational Enterprises, and social policy supported by sustainable international trade and investment.

³⁸ Social dialogue and tripartism, in this context, refers to the negotiation, consultation and information exchange on matters related to the world of work between and among representatives of government, employers' and workers' organizations. The aim of social dialogue and tripartism is to promote consensus building and democratic involvement of the main stakeholders in the world of work.

Pakistan Decent Work Country Programme

The Ministry of Overseas Pakistanis and Human Resource Development (MOPHRD), through consultations with the Employers' Federation of Pakistan (representing the employers) and the Pakistan Workers (representing the workers) and other stakeholders, and with technical support of the ILO, develops and implements Pakistan's Decent Work Country Programme (DWCP). The current DWCP IV (2023–27) is closely aligned with Pakistan's development framework, **Vision 2025**, provincial development plans, related employment and labour policies and priorities, Agenda 2030 Sustainable Development Cooperation Framework (UNSDCF) 2023–27. These frameworks reinforce the need for partnerships, coordinated and integrated planning, and effective implementation.

The four key priorities identified for implementation through Pakistan's DWCP-IV are as follow:

- Promoting decent work in economic recovery and climate response;
- Expanding social protection coverage:
- Reinforcing adherence to international labour standards and fundamental principles and rights at work: and
- Fostering safe, healthy, gender-responsive workplaces free of violence and harassment.

3.2.2. Economic and social sector initiatives

Federal Government

At the onset of the Covid-19 pandemic, the Government expanded existing social protection programmes and launched new ones particularly for workers in the informal economy as they were not covered by formal social protection programmes or insurance. This included the cotton workers and smallholder farmers. To mitigate the socio-economic consequences of the pandemic, numerous programmes were expanded and new ones were launched during the pandemic.

Among measures taken were the following (Pakistan FD 2021; 2023):

- **Benazir Income Support Programme (BISP)**, a targeted cash transfer programme, focuses on poor women. It aims to eradicate extreme and chronic poverty and to empower women. Its database, the National Socio-Economic Registry (NSER), has socio-economic indicators of some 35 million households throughout the country. In 2022/23 BISP released an almost 266 million rupees in conditional and unconditional cash transfers to some 9 million beneficiaries. The amount disbursed in 2019/20 (over 232 million rupees) was more that doubled from the previous year (close to 109 million rupees). The estimated amount transferred since its inception in 2008/09 is 1,638 billion rupees.
- **Pakistan Poverty Alleviation Fund (PPAF)**, started in April 2000, supports communities through various initiatives. During 2022/23 1.6 billion rupees were disbursed to its partner organizations in 149 districts across the country. Programmes included institutional development/social mobilization; livelihood enhancement and protection; water and infrastructure; education, health and nutrition; interest free loans; and flood emergency response.
- **Pakistan Baitul Mal (PBM)**, provides financial assistance at the district level to widows, children who have been orphaned, persons with disabilities, and those in need. To serve its target populations, PBM also implements programmes, such as operating schools for rehabilitation of children withdrawn from hazardous work, homes for children who have been orphaned, homes for senior citizens (60 years and above), shelter homes, and food delivery to the poorest.

- **Ehsaas**, launched as a multisectoral social protection and poverty eradication initiative in 2019 and was expanded in the post-Covid19 pandemic context. It was used, in particular, for the emergency cash disbursements. The poverty eradication strategy of Ehsaas is focused on making the governance system work for a more equal society; implementing safety net programmes for the disadvantaged; supporting jobs and livelihoods; and enhancing human capital development.

The third **Textile and Apparel Policy – 2020-25**, approved in February 2022, aims to make Pakistan “one of the major players in global textiles and apparel supply chain” through the measures (Pakistan MOC n.d.) including the following:

- Wide-ranging capacity development encompassing technology up-gradation, infrastructure, small and medium enterprises, marketing, integration in the global value chain, domestic commerce, technical and fourth generation textiles, research and development, and regulatory regime and international compliance.
- Reducing cost of doing business, including through subsidies and lower rates for electricity, the continuation of customs duty-free machinery, and rationalizing the tariff regime.
- Strengthening of textile and apparel value chain through sector specific initiatives. In the case of the cotton sector, the policy seeks to address issues pertaining to profitability in cotton farming specifically by introducing latest seed technology and increasing yield.
- Human resource development to facilitate increased participation of women by providing them with training, day care centres and other facilities and reviewing labour laws. Training programmes are also foreseen for other workers along the textiles and apparel value chain.
- Other measures include facilitating resolution of trade disputes and strengthening the role of textiles and apparel associations.

The Agriculture Transformation Plan of the Ministry of National Food Security and Research, aims to boost and expand cotton production to improve livelihoods at the farm level and related industries. Measures to do so include supporting development of quality seeds, enhancing biological control of cotton pests, ensuring timely issuance of subsidy for pesticides, seeds and fertilizers through **kissan cards**. It is also attempting to increase production in Balochistan and explore production in Khyber Pakhtunkhwa Provinces, where cotton yields are reportedly high and pest pressure low (Pakistan FD 2021).

Provincial – Punjab and Sindh

Punjab Labour Policy 2018 is wide-ranging and touches on all critical labour issues. It establishes a one-system software for centralized transparency in inspection with online supervisory monitoring inspection system. It also establishes a Policy Implementing Unit to monitor implementation of policies in all related sectors for the labour market information and analysis system.

First Sindh Labour Policy 2018: A Framework of Industrial Relations, Social and Economic Well-being of the Peoples of Sindh, an ambitious and wide-ranging policy with commitments to comply with international labour and other conventions. The Policy, developed through tripartite consultations with the employers’ and workers’ representatives, covers many critical labour areas, ranging from home-based work to child labour, bonded labour, women, social security to old-age pension, for which it says measures will be taken. It sets guidance for governance and coordination.

Empowerment of Kissan through financial and digital inclusion, Punjab: Commonly referred to as the **Kissan** (farmers) E-Credit Scheme, this five-year (2016-21) initiative provides interest-free loans to small farmers through two commercial banks/participating financial institutions (Zarai Taraqati Bank, ZTBL – Agricultural Development Bank and National Bank of Pakistan) and three micro-finance institutions (MFIs) (Telenor Micro Finance Bank, National Rural Support Programme and Akhuwat Islamic

Microfinance. Farmers having land of up to about 12.5 acres are eligible and those with less than 5 acres are also eligible for interest/markup subsidy. Farmer registration is through the computerized Punjab Land Records Authority (PLRA). Interest free loans of 40,000 rupees per acre for kharif crop (including cotton) are provided. Productivity and farmers' profitability are the main objectives of the scheme, but it also aims to increase farmers' literacy and their overall financial inclusion. Furthermore, it brings the under- and non-banked farmers into the banking network – 70 per cent of the loan was to be disbursed to farmers with no previous credit history.

E-voucher based fertilizer and cotton seed subsidies, Punjab: The cotton seed subsidies cover Multan, D. G. Khan and Bahawalpur and are provided on selected certified cotton seed varieties. The objective of the subsidies is to promote certified cotton seed varieties and the use of phosphate and potash fertilizers to improve yield.

Initiatives of the Punjab Government also include pilot testing of innovative technologies to improve water use efficiency, design, development and testing of farm machinery and implements, enforcement drives against adulteration of agricultural inputs, and crop insurance. As many of these are not specific to cotton, advocacy on behalf of the cotton producers and facilitation in helping them avail and make full use of the subsidies is needed.

3.3. Institutions supporting productivity and innovations

3.3.1. Public sector

Ministry of National Food Security and Research

The Ministry of National Food Security and Research (MNFS&R) is responsible for implementing, enforcing, developing, and executing policy on agriculture, rice, livestock, fishing, and farming. Cotton crop is included in the portfolio.

► The Pakistan Central Cotton Committee

The Pakistan Central Cotton Committee (PCCC), under the Ministry of National Food Security and Research, is the apex research and development body for cotton crop. Established under the Cotton Cess Act, 1923 its mission is to improve and develop the production, marketing and manufacturing of cotton and cotton by-products through an extensive programme of research and development in all aspects. It is responsible for the review and monitoring of the research and development activities of cotton research institute and cotton research stations under it. Furthermore, it coordinates with national and international stakeholders.

► The Central Cotton Research Institutes

The PCCC is responsible for three leading research and development institutes, namely, the **Central Cottage Research Institutes (CCRI)** in Multan and Sakrand; and the **Pakistan Institute of Cotton Research and Technology (PICR&T)**, Karachi

The CCRI maintains a gene pool of varieties from around the world, developed 23 cotton varieties, production technologies and pest management strategies, and advised cotton farmers in improving their production. Private seed companies are also involved in seed development, together with CCRI, to meet the demand for new and better seeds. There are more than 170 approved varieties.

The PCCC is also responsible for seven **Cotton Research Stations (CRS)** across the cotton growing areas; and a separate directorate in Lahore to cover the spinning mills in the area and throughout Pakistan.

At the national level, the PCCC coordinates with various institutions involved with cotton at different stages of the supply chain or in research, including All Pakistan Textile Mills Association (APTMA), Karachi Cotton Association (KCA), Pakistan Cotton Ginners Association (PCGA), Trade Development Authority of Pakistan (TDAP), Pakistan Bureau of Statistics (PBS), Provincial Agricultural Research Departments, Agricultural Policy Institute (API) and State and agricultural banks.

To meet its financial needs, the PCCC collects cess on export through the Federal Board of Revenue and cess on consumption of cotton through the Directorate of Cotton Cess Management. Cess collection has been a challenge with the removal of the mandatory status for collection of the cess from textile manufacturers and is a topic of ongoing debate and advocacy. To make up partly for the loss of resources from cess, the PCCC qualified for the Covid19 pandemic support.

► **Agriculture Policy Institute (API)**

Agriculture Policy Institute (API), an organization under the Ministry of National Food Security and Research, focuses on broader agricultural policies, conducts studies on emerging policy issues and annual input-output field surveys in main cotton and other crop producing areas. It is also responsible for responses to questions from the Senate and National Assembly on the agricultural sector.

► **Federal Seed Certification and Registration Department**

The Department, under the Ministry of National Food Security and Research, is responsible for seed certification, registration of seeds, inspection and quality control of the seeds for cotton and other crops. The introduction of uncertified and poor quality seeds is one of the major issues impacting the farmers and their cotton yield.

Ministry of Commerce and Textiles

The Ministry of Commerce and Textiles impacts not just the exports and imports of cotton but the livelihoods and employment of the millions of people in cotton production, textiles and related sectors and industries. It promotes economic growth and commerce development by facilitating trade liberalization, improving exports and reducing the cost of business. The Ministry is responsible for making policies for the industrial and export sectors and plays a vital role in tariff policy and its implementation.

► **Textile Industry Division** formulates strategies and programmes, enabling the textile sector to become globally competitive, to upgrade its supply chain and maximize value-addition.

► **Pakistan Cotton Standards Institute (PCSI)** is a body corporate under the Ministry's Textile Division. The institute is responsible for standardization of cotton and bringing it at par with the internationally accepted standards for improved competitiveness and ensuring better returns to cotton growers, ginnerers, spinners, exporters and the national economy. It establishes cotton standards, devises quality control measures, conducts grading of seed cotton and classification of lint cotton, prequalifies private inspection companies for certifying quality, handles arbitration, and makes recommendations to provincial governments.

Ministry of Overseas Pakistanis and Human Resource Development

The Ministry of Overseas Pakistanis and Human Resource Development (MOPHRD) holds the mandate for labour related policies and programmes at the federal level and coordinates with provincial labour departments. It also convenes and participates in tripartite meetings and activities, bringing together representatives of employers' and workers' organizations in Pakistan together with government and technical agencies. With support of the ILO Country Office for Pakistan, it oversees the development, implementation and reporting of Pakistan's Decent Work Country Programme and participates in the ILO's labour standard setting and implementation work. The Ministry is also responsible for policies and programmes for overseas Pakistanis.

Agriculture Extension and Adaptive Research Department

Government Agriculture Extension and Adaptive Research Departments in Punjab and Sindh have a vast networks of extension workers to train farmers. They develop and disseminate production technologies and plans for crops, including cotton; organize farmer capacity building programmes at the village level; regulate inputs and provision for crops; and issue licenses for cotton ginning factories.

University of Agriculture, Faisalabad

The University of Agriculture, Faisalabad, through its research and development, and innovation and commercialization programmes undertakes research on various aspects of cotton production, including seeds, and cultivation and irrigation.

As a member of the International Centre for Development and Decent Work (ICDD), the university contributed to the fight against poverty and global hunger through collaborative research, education (scholarships, exchange visits) and scientific meetings and workshops for development of decent work in the partner universities. The ICDD, is a global multidisciplinary network led by the University of Kassel in Germany in collaboration with the ILO and partner universities to contribute to the global action against hunger and poverty through research and education in support of Sustainable Development Goal 8.³⁹

Ayub Agricultural Research Institute, Faisalabad

Ayub Agricultural Research Institute (AARI) develops new varieties of crops, technologies for food safety, value addition, conservation of natural resources and introduction of new plants. It has also been involved in developing a Cotton Research Sub-station in south Punjab in Rajanpur/DG Khan, where the livelihood of the people is dependent on cotton crop. The district also has a vast ginning industry.⁴⁰

3.3.2. Trade associations of cotton

Pakistan Cotton Ginners Association

Pakistan Cotton Ginners Association (PCGA), the apex body representing more than 1,300 ginners across the country, is the mandated voice of ginners in Pakistan. It provides information, including district-wise monthly cotton statistics on bales; monitors legislative and other measures affecting the cotton trade; makes representation to local, provincial and central authorities; arbitrates in the settlement of disputes arising out of transactions; and regulates conditions of employment in the industry. PCGA is not a member of the Employers' Federation of Pakistan (EFP), however, it collaborates with EFP in organizing some events.

Karachi Cotton Association

Karachi Cotton Association (KCA) is a registered association recognized across Pakistan. It maintains a code of trading practices in the form of rules, regulations, by-laws and standards for classification of cotton. It provides arbitration for settlement of disputes. It serves as a forum where buyers and sellers can obtain information on ruling cotton prices and current market conditions globally.

³⁹ University of Kassel, Germany, "International Center for Development and Decent Work (ICDD)".

⁴⁰ For more information, see the Ayub Agricultural Research Institute webpage at: https://aari.punjab.gov.pk/adp_rajanpur_new (March 2024).

All Pakistan Textile Mills Association

All Pakistan Textile Mills Association (APTMA) is a registered national trade association representing the textile, including spinning, weaving and composite mills representing the organized sector through 223 textile companies in Sindh, Khyber Pakhtunkhwa, Punjab and the Federal Capital Territory). APTMA is affiliated with the Federation of Pakistan Chambers of Commerce and Industry (FPCCI). It promotes and protects the trade, commerce and manufacturers of Pakistan in general, the cotton trade in particular and takes measures for promoting, supporting or opposing legislative and other measures affecting the trade or commerce of its members.⁴¹

3.3.3. Employers' and workers' organizations

Employers Federation of Pakistan

Employers' Federation of Pakistan (EFP), the apex body for employers of Pakistan, represents employers' interests on policies, legislation and issues related to industrial relations, economic growth, employment generation, environmental issues, human resource development and decent work in Pakistan, among others. As an ILO constituent, the EFP participates in setting labour standards, and developing policies and programmes to promote decent work in the country and globally. The EFP partners with the ILO in Pakistan, together with the Ministry of Overseas Pakistan and Human Resource Development and Pakistan Workers Federation (also ILO constituent members from Pakistan) to develop, promote and support the implementation of the Pakistan's Decent Work Country Programme.

EFP membership of close to 900 includes the All Pakistan Textile Mills Association along with textile industries. Its members are mainly from the field of manufacturing, pharmaceutical, chemical, energy, services (including airlines, educational institutions, telecommunications, and banks), food and beverages, chambers of commerce and industry, and rubber, plastic and leather.⁴²

Pakistan Workers' Federation

Pakistan Workers' Federation (PWF), an umbrella organization of the trade unions representing over a million workers from different sectors, promotes workers' rights and legislative reforms in Pakistan. PWF is an ILO constituent representing workers from different sectors. Together with the Ministry of Overseas Pakistanis and Human Resource Development and the Employers' Federation of Pakistan, the PWF promotes decent work in Pakistan and globally. PWF represents workers as member of the governing bodies of EOBI, Social Security and Workers' Welfare Fund.

Pakistan National Textile Leather Garments and General Workers Federation

Pakistan National Textile, Leather, Garments and General Workers Federation (PNTLGWF), is the biggest union in the textile sector in Pakistan with a membership of over 12,500. It operates out of Multan, Punjab (CNV International 2016). PNTLGWF is affiliated with IndustriALL Global Union, representing workers in 140 countries in mining, energy and manufacturing sectors. PNTLGWF has collaborated with the ILO Pakistan office in farm level awareness raising for cotton picking women and girls in southern Punjab (ILO 2019a).

⁴¹ For more information, see APTMA webpage at: <https://aptma.org.pk/> (March 2024)

⁴² For more information, see EFP webpage at: <https://efp.org.pk/about-us/> (March 2024)

The Sindh Agriculture and Fishing Workers Union

Pakistan's first trade union for Agriculture and Fishing Workers (SAFU) was registered in Karachi in 2015 under the provisions of Sindh Industrial Relations Act, 2013. By extending coverage of the Sindh Industrial Relations Act, 2013 to the fisheries and agriculture sectors, Sindh became the first province in Pakistan to give women and men in these sectors not only the freedom to organize and participate in trade unions but also to participate in collective bargaining. At the time of its registration, the union had 400 members, of which about half were women from across Sindh.

3.3.4. Non-governmental agencies

The Pakistan Kissan Rabita Committee

The Pakistan Kissan Rabita Committee (PKRC), formed in 2003 with over 5,000 small farmers and peasants in Chishtian in southern Punjab, focuses on land rights for introduction of food sovereignty, displacement of small farmers (to make room for development or mega-projects), and democratic rights. It is based in Lahore and a member of La Via Campesina, an international peasants' movement with over 180 organizations in over 80 countries.

Farmer Associates of Pakistan

Farmer Associates of Pakistan (FAP) sponsored by large and progressive farmers, is a membership-based non-governmental organization of large farmers. Its stated objectives are (a) to support the socially, economically and technologically backward areas of the agrarian sector to be modernized, and (b) to organize itself into a "coherent economic entity". It focuses on technical support, public relations and liaison, the legal environment, to ensure laws are in line with its vision of twenty-first century farming. It has an executive membership of close to 30 and a wider membership that includes other categories. It does not categorize itself as a trade organization.⁴³

3.3.5. Credit Institutions

Farmers have access to credit for cotton crop through formal and informal credit resources. About 61 per cent farmers said that they have access to while 39 per cent said that they did not borrow money for cotton crop.

Zarai Taraqiati Bank (Agricultural Development Bank)

Zarai Taraqiati Bank (ZTBL), a rural financial institution, focuses on the small farmers with a view to helping them to increase their productivity and income levels through reliable credit and advisory services. It aims to strengthen agriculture value added commodity chains. According to its 2022 Annual Report, ZTBL has a country-wide network of 501 branches, serves some half a million clients annually and has over a million accumulated account holders. In 2022, ZBTL disbursed 86 per cent of its agricultural credit to subsistence growers (ZBTL n.d., 248).

43 For more information on FAP see: <https://www.brecorder.com/news/577541> (March 2024), and <https://www.pakissan.com/2001/english/spotlite/fap/FAP.shtml>.

State Bank of Pakistan

State Bank of Pakistan, under the Agricultural Credit Scheme, provides indicative targets to 21 banks on annual basis for the purpose of credit to farmers. Its Initiatives for the Promotion of Agriculture Financing, in collaboration with the federal and provincial governments, includes insurance programmes for crops and livestock, e-credit schemes, adoption of automation of land record for agriculture financing, implementation of credit guarantee scheme for small and marginalized farmers, and workshops, trainings, capacity building, and awareness raising (Pakistan FD 2018).

Microfinance Institutions/finance institutions

For the smallholder farmers, the main source of credit has been the *arathi* and *beopari*, but microfinance institutions (MFIs) in rural areas of Punjab are gaining ground. On the face of it, loans from the *arathi* or *beopari* seem a good option. There is no interest, the process is simple, and the loan can easily be rescheduled. The downside for the farmers is that the rescheduled loan starts a new cycle and is linked to the future crop. It confines sale by the farmer to the same *arathi*, without the freedom for the farmer to explore other options. It also means higher commission for the *arathi*. The microfinance comes with an interest but also the freedom to sell the crop to anyone (Rana 2018).

National Rural Support Programme, Punjab

National Rural Support Programme (NRSP), a non-profit organization under the Companies Act, 2017, has massive outreach in the rural areas through a network of close to 240,000 community organizations that work with over 3.70 million households in poverty. NRSP is one of the Participating Financial Institutions (PFIs) implementing Government of Punjab's Interest-free Agri. E-Card Scheme for 150,000 smallholder farmers with up to 2.5 acres (1.01 hectare) of land and tenants/ sharecroppers of up to 5 acres (2.02 hectare) in selected districts of Punjab.

National Rural Support Programme Microfinance Bank Ltd.

NRSP Microfinance Bank Ltd. operates in accordance with the Microfinance Ordinance of 2001 with the social mission to provide financial services to poor and marginalized clients including small and landless farmers, and government and private institutions. Through its 155 branches in 53 districts, it offers micro-credit and micro-insurance to the 'uninitiated', including the smallholder and landless farmers.

3.3.6. Access to training programmes

Training and awareness raising programmes are arranged by the Agriculture Extension Department, non-governmental organizations, and development agencies, including the ILO. Training is also organized by the private sector on specific inputs, particularly pesticides and fertilizer companies. Training by pesticide and fertilizer companies also promotes their products. Training programmes by non-governmental organizations generally are funded by companies buying certified products. ILO programmes tend to be on occupational safety and health and fundamental principles and rights at work.

Access to these programmes is limited for many of the workers and farmers. It is particularly limited to casual workers and female workers.

► 4. Key findings and analyses

4.1. Community level cotton supply chain, Punjab and Sindh Provinces

A typical cotton supply chain in Pakistan's Sindh or Punjab province is an enormous, complex and somewhat messy system. It begins with the preparation of the field for sowing the seed cotton (tier 4), followed by local or international trading, some is processed and heads to factories (tier 1) where it is manufactured into textiles, garments or other products for in-country consumption or export, and some is transformed to by-products.

► Figure 2: Community level cotton supply chain (tier-4)

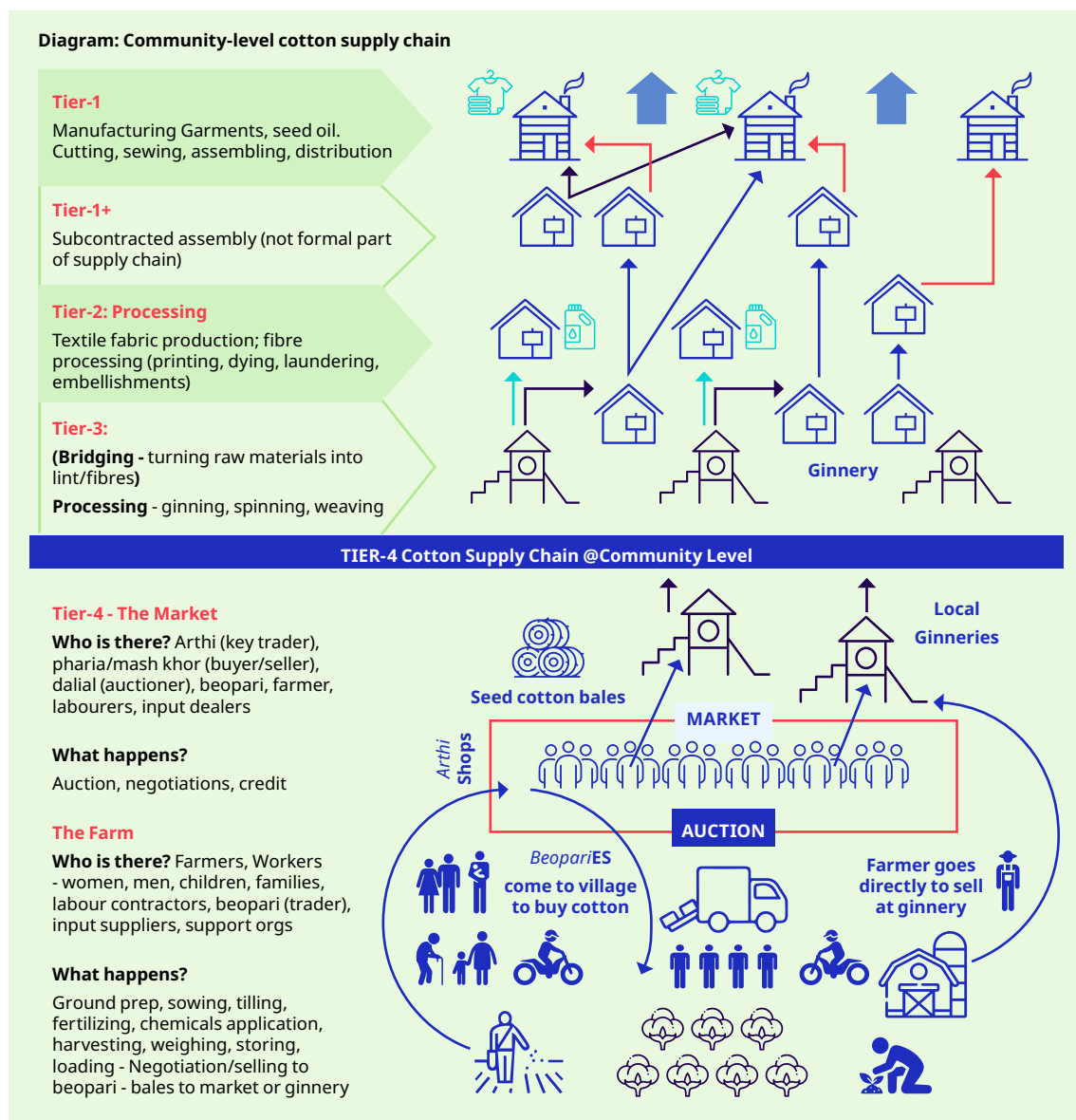


Figure 2 sketches the core actions in the various tiers, that often flow from one to the other and are not always distinct.

Below is a descriptive breakdown of the activities and some processes involved in cotton production at the community level from getting ready to sow the seed cotton through harvesting and trading at the local market and ginneries. Also presented are insights into the relationship prism and dynamics that impact the nature of cotton production in the selected communities in Punjab and Sindh Provinces.

The information and analysis in this sub-section (4.1) draws on Rana 2018, in addition to the mapping exercise information.

4.1.1. At the farm – producing the cotton

Preparation of the soil

The cotton supply chain starts with the farmworkers preparing the soil for seed cotton after the crop of the previous season (such as wheat) is harvested and the field is cleared. Soil preparation involves processes that are repeated multiple times, such as irrigating the soil (*rauni*), ploughing, levelling for more irrigation efficiency, planking to compact the topsoil to preserve moisture.

Cultivation

Cultivation is the next step when the seed sowing (planting) takes place. Growing cotton is a delicate process involving intense and, at times, arduous and hazardous manual labour, particularly on small farms with little or no mechanization. It involves weeding, hoeing, fertilizing the soil and spraying pesticides.

This stage also requires the farmers or land managers to be knowledgeable about the various needed inputs (fertilizers, pesticides, water, equipment and labour, among others) and their potential risks. They must be vigilant in checking and monitoring the situation on the farm, the weather conditions and the flow of irrigation water. Importantly, they must also be effective managers to engage workers with the right skills and provide an enabling human infrastructure with respect for fundamental principles and rights at work to maximize productivity.

Cotton picking

In Punjab, there is an **informal contracting system** for cotton pickers. An elderly woman heads a group of cotton pickers comprised of 15 to 20 women. She bargains with the farmers for wages and transport. The group members do not compensate the head or pay her commission. However, the farmer pays her 50 to 75 rupees per day in addition to her wage, as a compensation for the service.

In Sindh and in Punjab's Rahim Yar Khan District, the farmers' contractors hire cotton pickers through a more **formal contracting system**.

The duration of the cotton-picking season varies by the region: in lower and central parts of Sindh it is July through October; and in Punjab and upper Sindh, it is mid-August through November.

As July–September is a period of intense heat, the cotton pickers start work early and end by 11 a.m. In October and November, when the heat has subsided, cotton picking continues to 4 p.m.

The daily quantity of seed cotton picked depends on the duration of the daily picking time: At the start of the picking season, when the picking time is shorter because of the heat, a picker (usually woman) generally picks 25 to 30 kg a day. When the heat subsides and the time increases, a picker can pick around 40 kg per day.

The pickers collect the cotton in big pouches (*johli/kanbalbharree*), formed by tying a large piece of cloth around the waist. When the *jholi* is full (or work is done) it is detached from the body, knotted into a bundle, carried on the head, and then deposited in a pile at a designated place on the farm. A cotton picker usually fills 5 to 6 loads of *jholis* a day, each weighing 5 to 6 kg. At the end of the workday, all the cotton that has been picked and collected is tied into a large bundle in a bigger cloth or empty fertilizer bags made of synthetic fibre.

Weighing, loading and payments

Punjab: To determine the weight of the seed cotton, the farmer weighs the pickers with the seed cotton load on their head; then weighs them without the load. The weight is documented in the register by the farmer, who provides each picker with a paper slip that has the weight recorded on it. It is the responsibility of the farmers to transport picked cotton from the field to the storage place. Payment is made directly to the picker after the farmer sells the cotton. The farmers (usually of small farms) or the farm managers (in case of large farms), go to the doorsteps of the pickers to make the payments. No commission is involved.

Sindh: In the case of *haris* (tenants), the cotton of each *hari* is generally stored separately at the landlord's place and sold separately to the *beoparis*. It is weighed, the weight is recorded in registers by the *kamdars/munshis*⁴⁴ who issue weight slips to the pickers. After the *haris* have sold the cotton, they pay the contractors, who, after deducting their commission, pay the pickers. Payments include loading and transportation charges. Labour charges for loading are about 15 rupees per 40 kg of seed cotton, while transportation charges are up to 3.50 rupees per 40 kg, depending on the distance involved.

During the 2020 cotton season, unit price for cotton picking was 300 rupees to 400 rupees for 40 kg of cotton picked. The payment is made in cash in Sindh and most of Punjab, but in Punjab's Mianwali and Sargodha districts, it is in-kind payment.

4.1.2. At the farm – selling the cotton to the *beoparis*

The *beoparis* (local intermediaries), are important players at the community-level in the cotton supply chain. They have a business relationship with many of the farmers to whom they provide advance credit/loans at the start of the cotton production season. These loans are mainly for farm crop inputs (such as seeds, fertilizers). Recovery of the loans is tied to the post-harvest sale of seed cotton. Typically, one *beopari* has 35 to 50 farmer clients in a village.

The *beoparis* deal mostly with the small farmers, often at the farmer's doorstep, as many of the farmers lack resources to arrange for transport to the off-farm sale points, which are the local market or individual ginners. At harvest time, the farmers inform the *beoparis* about availability of their harvest. On a typical trading day, a *beopari* buys around 8,000 kg (200 maunds)⁴⁵ of seed cotton from different farmers. The rate offered to the farmers by the *beoparis* is the ginner rate. For *beoparis* who work on commission basis, they get 1 to 1.5 per cent commission for the ginner, otherwise they get 20 rupees per 40 kg (a maund).

4.1.3. At the trade market (*mandi*) – selling and buying the cotton

The commodity markets of Punjab are spread across the province with an average distance of some 30 km between any two markets. A market can have 100-250 *arthis*. Punjab has 244 agricultural wholesale markets for grains, fruits, and vegetables and 135 market committees overseeing the markets within their notified areas. Key players in the trade market are the *arthis* (key market traders), farmers, *beoparis*

44 **kamdars/munshis**- persons responsible for accounts and administrative matters

45 1 maund (Pakistan) = 40 kg

(local village traders), *pharias/masha khorsins* (those who both sell and buy), *the dallal* (auctioneers), purchase agents of the processing factories, including ginning agents), brokers, transporters, *palledars* (labourers), and farm input dealers, among others.

Market committees

The responsibilities of the market committees include operating the market and its facilities, levying fees on produce or crops sold in the market, managing the budget and expenditure, arbitration of disputes between the sellers and buyers, and fixing the daily market rates for the produce and crops. The levies are collected through the *arthis*. The committee charges buyers 2 rupees per 100 kg seed cotton traded from the buyers. The *arthis* are represented on the market committees and have an influential role. Successive governments have tried to revamp the colonial era bodies established under Market Committee Act of 1939 but have had limited success (Asif 2021).

Key market players: the pharias/masha khors

The *pharias* (*masha khors* in Sindh), are the next important players in the market after the *arthis* and operate as both buyers and sellers. They have close relationship and collaboration with the *arthis*, renting their space and even taking short-term advance credit. The *pharias/masha khorsins* buy large consignments, which they either divide into smaller portions for re-sale at the market and outside, or they sell in large portions. Unlike the *arthis*, they are not licensed, and they find themselves somewhat marginalized in the market system where the influence of the *arthis* is overwhelming.

Key market players: the arthis

The *arthis* are considered to be the most powerful players in the market, embodying generational authority, social capital, and forward and backward linkages. They possess business and marketing acumen and operate as an institution in its own right in the arcane, undocumented and informal economy. They are represented on the market committees and organized in market-based associations at the district and provincial levels, which also serve as lobby groups. Services to members include publishing directories of *arthis*. They have shops (owned or rented) in the market where they conduct business.

The 1979 Rules (66(a)) specify commission rates for specific commodities, but the *arthis* can charge differently because commission is calculated based on subjectively assessed value rather than volume. They are responsible for collecting the market committee levy from the buyers who purchase from them in the market.

At the *arthis* shops, the cotton is heaped client wise. During peak season, the cotton remains in the shop for 12-14 hours. The next day there is bidding by the *pharias/masha khor* or the purchasing agents (buyers) and cotton is sold to the highest bidder. The *arthis* pay for the *dallals'* (auctioneers') charges 0.08 per cent to 0.12 per cent and labour charges of 0.50 per cent of the total sale.

The buyers bear the cost of loading, charged at 8 to 10 rupees per 40 kg (maund), the cost of transportation to the *arthis'* shop, and the market committee fee of 2 rupees per about 49 Kg (quintal). The buyer and seller (*beopari/farmer*) pay a commission of 1 per cent each to the *Arthi*.

During the cotton season, the **daily procurement at the arthi's shop** amounts to 8,000 to 56,000 kg (200 to 1,400 maunds). For the entire cotton season, this can be 800,000 to 4,000,000 kg (20,000 to 100,000 maunds) of seed cotton.

Cotton procurement by *arthis*

Generally, the *arthis* procures seed cotton from a variety of sources: (a) *beoparis* brings the cotton to the *arthis*' shop in the market. Transportation and 0.5 per cent unloading costs are borne by the *beoparis*; (b) farmers who bring the cotton to the *arthis*' shop in the market. Transportation and 0.5 per cent unloading costs are borne by the farmers; (c) *arthis* also buys cotton from the farmers' doorstep, as the *beoparis* do; or through ginners, when the cotton is deposited directly in the ginnery by the *beoparis* and farmers and payment to them is made by the *arthis* in the market shop.

Price setting for seed cotton

Price setting in the village by the *beoparis* is based on appearance of the seed cotton. Trash, moisture, sheen and colour are considerations that go pricing decisions. Some deduction rates are mandatory, and vary by region.⁴⁶ When not mandatory, the deduction rate is mutually agreed between the farmer and the *beopari*, who then determines the price in accordance with the rate expected at the ginning factory, which is a lower rate than the market rate.⁴⁷

Price setting in the trade market by the *arthis*. The market committee is responsible for fixing the daily price of seed cotton in the market. Nonetheless, the price offered to the farmer can vary as a result of a number of points through which subjective decisions creep in. For instance, the buyer at the market auction and the farm does a visual and manual inspection, checking for colour, trash, moisture and sheen and then applies deductions to the price.

National factors affecting community and market pricing of cotton: The government regulates the export of cotton and textile products through duties and subsidies, which affect the price of cotton in the local market. For instance, in 2019, the Cotton Crop Assessment Committee's projected shortfall of 30 million bales led to the government allowing duty free import of cotton. This in turn resulted in a significant decline in the price of cotton in the local market, from 4,000–4,500 rupees to 2,800–3,500 rupees for 40 kg of cotton. The lower price impact is passed onto the farmers and ultimately to the cotton pickers.

Taxes and commission

The *beopari* and *arthis* pay 12 per cent income tax on the total amount of their commission and 0.003 per cent withholding tax. These are deducted at the point of sale by the buyer (ginner) and expected to be paid to the government. The *arthis*, charges commission on sale to the ginner. The *beoparis* charges 1 to 1.5 per cent commission from the ginner when the seed cotton is delivered directly to the ginnery. (The ginning factories taxes and fee are noted below.)

Issues observed at the market

These included the lack of storage facilities for seed cotton. The existing storage space is generally uneven, sometimes even unpaved. The seed cotton heaps are not covered, increasing the likelihood of contamination with garbage, human hair and other waste material. Barber shops in the market are a potential source of hair contamination. It is not clear whether or not there are fire extinguishing arrangements. Buyers do not demand that cotton be arranged or sold grade-wise, leaving the *beoparis* and *arthis* with little incentive interest in quality parameters (Rana 2018).

⁴⁶ For example, in Punjab's Vehari district it is one kg per 40 kg and in Bahawalpur it is one half kg per 40kg respectively. In Sindh it is 1kg per 40kg.

⁴⁷ If no commission is involved, the *beopari* gets 20 rupees per 40 kg (maund) from the ginner.

4.1.4. At the ginnery – from seed cotton to lint

Generally, ginning factories operate from October to January in Punjab, from July to November in lower Sindh and September to January in upper Sindh.

Ginning capacity of a single set mill is 30, 000 bales during the season but when seed cotton yield does not meet its target, availability to the ginnery declines, slowing the operation of the mills, often resulting in closures and job losses.

Ginnery key players

Among key players at the ginnery are the owner/manager, selector, seasonal labour through contractors, permanent staff, farmers, *beoparis* (local community traders), and transporters. Visiting the ginnery from textile groups/units are the general manager, spinning master and assistant spinning master.

A standard ginning factory has up to five permanent staff members while bigger factories have up to ten permanent staff. During the ginning season the number of seasonal workers, who come through contractors, increase from 30 to 50.

Occupational injuries and accidents

The machinery used in ginneries can result in injuries and accidents and is particularly dangerous for children even if they are older (15 to 17 years old). Protective clothing and masks are needed for adults too because of the high level of lint that causes respiratory issues.

Buying seed cotton, quality check and pricing

A ginning factory procures cotton from the farmers, *beoparis* and the *arthis* at the market. A **cotton selector** represents the ginning factory when buying in the market. When seed cotton is brought to the factory directly by the farmers or the *beoparis*, the cotton selector checks for trash to determine the price deduction to be applied. It is then weighed. Its price is determined at the rate of the day, which depends on factors including price of lint, price of cotton seed and expected ginning out turn (GOT).⁴⁸ Daily arrivals are low at the beginning of season but during peak season they can amount to over 200,000 kg (5,000 maunds) per day. One lot of lint consists of 100 bales (160 kg). It is stored lot-wise in the open, and like cotton, without being graded for quality.

Heap making

Heap making is a technical process used to produce lint of consistent quality. To avoid variation in lint, workers layer the different qualities of cotton and then scramble them to mix the lint thoroughly to maintain consistency after ginning. Preparation of the heap takes a day and conditioning another 36 hours.

Selling lint

Buyers of lint are mainly **textile groups**. They purchase through the **brokers** who employ (salaried) cotton selectors to negotiate and purchase according to the required specifications from the textile groups. The reference point for price of lint is the international market rate, but price at the ginnery takes into consideration the quality of lint and the fibre characteristics at the selected ginning station. Inspection of lint is mainly visual and physical, while some mills use meters for moisture level assessment.

⁴⁸ GOT - Ginning out turn is the percentage of ginned lint obtained from a mass of seed cotton through the ginning process which that separates from the cotton seeds from the lint (white fluff).

The characteristics that are checked include colour, fibre length, fibre strength, moisture, uniformity of fibre, fineness, and the amount of long and short fibres.⁴⁹

Taxes and commission

Taxes paid by the ginning factories include general sales tax, withholding tax, income tax, market committee fee, excise duty, Employees' Old Age Benefit Institution contributions, Workers Welfare Fund contributions, and annual license fee.

Brokers commission is 0.25 to 0.50 rupees per 100 rupees, paid either by the ginner or shared by the lint buyer and the ginner. The selectors' fee, paid by the ginner, is around 2,000 rupees per lot in Sindh and 1,000 to 1,500 rupees per lot in Punjab. Most of the cotton is bought from October through November as good quality cotton is available during these months.

4.2. Overview of findings from the mapping exercise

The survey findings for workers, presented below, are based on key respondent responses from 150 female workers and 150 male workers. The average age of the workers was 37 years in the age range of 18 to 61 years; most were in the range of 30 to 45 years.

The survey findings for farmers are based on key respondent responses from 300 male farmers. The average age of the farmers was 39 years in the age range of 30 to 70 years. As with the workers, most farmers were in the range of 35 to 45 years.

Of the 300 farmers interviewed, only 5 per cent said income from cotton farming was their entire income. For 57 per cent of the farmers income from cotton farming accounted for over 50 per cent of their entire income; and for 8 per cent farmers, it was 20 per cent of their entire income. This shows that most of the farmers who were interviewed had other sources of income as well. For most, the additional income came from other crops (not specified but possibly wheat or sugar cane) or livestock. For a smaller number it came from another business or job.

The workers and farmers surveys were conducted in 50 villages across five districts in Punjab and Sindh Provinces of Pakistan. The districts in Punjab are Vehari, Bahawalpur, Rahim Yar Khan and in Sindh they are Mirpur Khas and Sanghar.

The findings for textile and garment units are based on key respondent interviews with 76 workers in Faisalabad and Lahore (Punjab) and Karachi (Sindh) (see table 1 for more details).

Throughout section 4, the findings from the mapping survey are boxed and numbered. The findings from the mapping exercise presented in sections 4.3 address aspects of the five areas of fundamental principles and rights at work as they emerge from responses at the farm level and within the textiles and garment units. The finding in sub-sections 4.4 through 4.6 pertain to issues at the farm level; and those in section 4.7 pertain to workers in textiles and garment units.

The findings in sub-sections 4.3 through 4.7 are drawn from the mapping surveys. Where limited additional relevant information is provided in subsections 4.3 – 4.7, it is marked as "Related information" and referenced.

The findings below are categorized as indicated below.

⁴⁹ Types of lint available are Better Cotton (BCI cotton), Real or Prime Mark Cotton, Organic Cotton and Conventional cotton.

4.3. Findings on fundamental principles and rights at work

The Declaration on Fundamental Principles and Rights at Work and its Follow-up (ILO 2022) provides the framework of rights at work and the human infrastructure for greater productivity. It consists of:

The mapping survey findings 1 to 13 align with the five areas of the rights at work framework and give an indication of the extent to which these rights are available to the cotton farmworkers, farmers and workers in the textile and garment units surveyed.

For each of the five areas noted below, the findings are presented as pertaining to:

- the community level, with responses from farmworkers and farmers respectively, and
- the textile and garment units, with responses from workers.

4.3.1. Freedom of association and collective bargaining

At the community level

Finding 1

A vast majority (72 per cent) of the 300 workers wanted an organization to take care of their rights. However, they were uncertain about their employers' reaction to their membership, with most believing there would be no reaction and some believing they would be fired.

None of the farmworkers interviewed were members of any workers' organization. The 300 female and male workers were asked if they were to become members of a workers' organization what would be the reaction of their employers (contractor or farmer). The responses varied: 29 per cent said there would be no reaction, while 11 per cent said becoming members would not be allowed. Others said there would be anger, or their wages would be stopped, or that they would be fired. Some even said the reaction would be positive.

Almost half (47 per cent) the farmers believed local government could facilitate the community's access to social rights; almost an equal percentage considered village leaders could do so.

Another 43 per cent of farmers considered the head of the community (**numberdar**) councillor, social leader or tribal chief) as the most suitable person for facilitating access to social rights for the community.⁵⁰ Others considered local politicians as most suitable or non-governmental organizations. Some believed there was no role for community leaders in the matter.

Related information

Forming and registering trade unions remain a challenge in the formal sector, where they are generally opposed by employers and workers are fearful of forming or joining them. To become a collective bargaining agent, a trade union must be registered. This condition presents additional obstacles (ETI 2017). In the informal economy, that is home to cotton production, there are no workers' organizations

⁵⁰ *numberdar* – traditional title, often hereditary, of powerful landlord families of a village or town, having state-privileged power and status

or trade unions for cotton workers. The few associations or non-governmental organizations that exist for agriculture focus on land rights and other issues, not particularly on labour rights.

In the textile and garment units

Finding 2

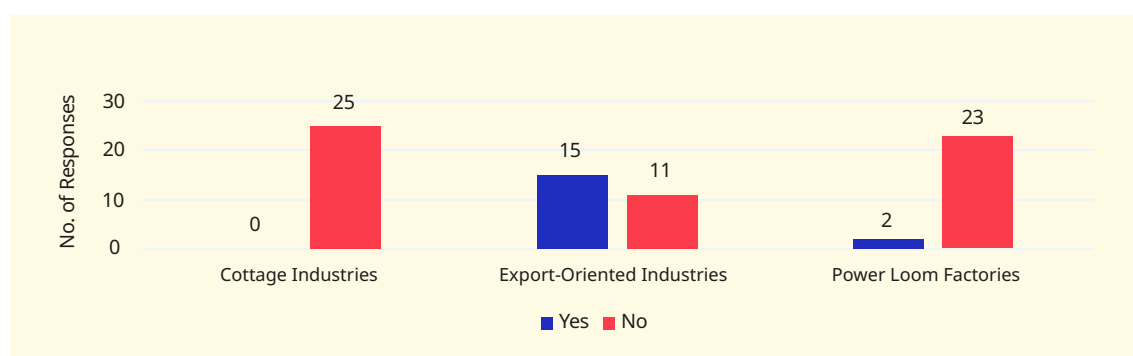
In the textile and garment units surveyed, unionization rates and collective bargaining were low while threats and fear of repercussions were high, particularly in the cottage industries and power loom categories.

Among trade unions and employees' unions operating in the industrial units surveyed were: Pakistan Power Looms Employees Union, Faisalabad Workers and Staff Trade Union, Lahore Union Zahid Garments Industry, Pak Labour Union, and Azhar Garments Pak Employees Union.

- **Export-oriented industry units:** More than half of the respondents (15 of 26) reported having the liberty to choose or belong to a union.
- **Power loom units:** Two (of the 25) respondents reported belonging to a union.
- **Cottage industry units:** None of the 25 respondents belonged to a union.

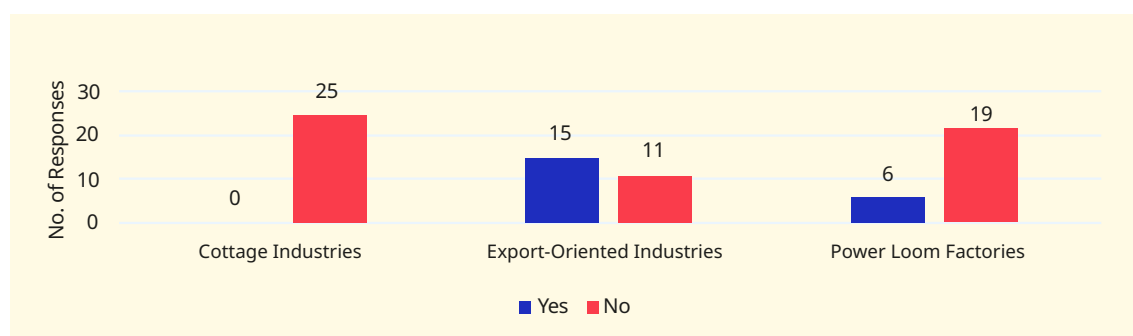
As noted above, the responses are based on 76 key respondents.

► **Figure 3: Union membership in surveyed units, Sindh and Punjab, 2020 (number of workers)**



Source: Mapping data.

► **Figure 4: Collective agreement in surveyed units, Sindh and Punjab, 2020**



Source: Mapping data.

4.3.2. Forced labour or compulsory labour

At the community level

Finding 3

A vast majority (75 per cent) of the 300 workers surveyed in cotton production incurred debt by taking credit advances from a variety of sources, including informal lending lenders and of farm input dealers, binding them to their lenders in the case of informal lenders. Some borrowers experienced problems paying back the advances. Few had written agreements.

Some 75 per cent of the 300 (female and male) workers surveyed said they received advance payment. For some workers (and farmers), paying back the advance is problematic. Some 13 per cent said they had to sell livestock to pay back. Others (60 per cent) paid back through wage deduction against loans and 27 per cent were re-paying in instalments. All of them said that they could not quit their jobs without first clearing the advances. Only 13 per cent workers had written agreements for their loans.

Almost all workers (98 per cent) said they could take leave if needed; 16 per cent workers said that they could quit their job if they wanted to; and some 5 per cent workers said that they were not working at a place of their choice.

Related information

Advance credit from the *arthis* (trade market intermediaries): The *arthis* provide collateral-free loans to farmers with whom they have longstanding relationship. They also provide loans to the *beoparis* (the village intermediaries). The process is swift with no requirement for documentation. The loans, based on the farmers' area under cultivation, help the farmers with their production needs and general consumption purposes (household needs, expenses related to children's weddings and other family events). Although these advances from the *arthis* carry no interest or markups, they are conditional. The farmers (or *beoparis*) must then only sell to the *arthis* who have advanced the credit. If the product sells below the expected value or the farmers/*beoparis* need credit for other purposes, the balance is rolled over to the next crop. With the rollover, the *arthis* charge commission above the stipulated rate. Amidst this scenario of the *arthis* coming to the rescue of the farmers/*beoparis*, their relationship is further forged, and the dependency of the borrower increases (Rana 2018).

Article 2 of the ILO Forced Labour Convention, 1930 (No. 29) defines forced labour as "all work or service which is exacted from any person under the menace of any penalty and for which the said person has not offered himself voluntarily." Hence, debt bondage is a form of forced labour. It is clear that cotton production involves the giving of advances to farmers and workers. More research is needed for an in-depth investigation into the nature of the loans and consequences of delayed repayments or inability to pay. The presence of a single indicator in a given situation may in some cases imply the existence of forced labour.

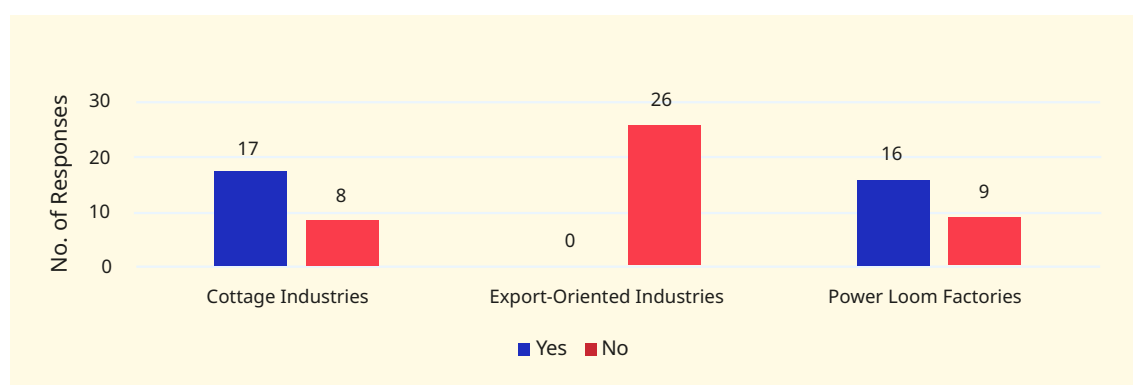
In the textile and garment units

Finding 4

Most workers surveyed in the power loom and cottage industry units, said they had experienced, and had indications of the presence of, forced overtime and threats of termination in case of non-compliance. This was not so in the case of workers in the export-oriented unit.

- **Export-oriented industry units:** None of the 26 workers reported having experienced or seen indications of the presence of force or compulsory labour in their units in the form of forced overtime or work extracted under the threat of penalty.
- **Cottage industry and power loom units:** A large number of the 50 respondents from power loom factories and cottage industries reported having experienced forced and compulsory labour themselves and also indicated the presence of it in their units. They said they were forced to work over-time and faced threats of termination in case of noncompliance.

► **Figure 5: Has worker experienced forced labour or compulsory labour, 2020**



Source: Mapping data.

4.3.3. Child labour

At the community level

Finding 5

Adult farm workers' perception of legal minimum age for work highlights their misinterpretation of the law, with many of them saying minimum age for work was much higher than it is by law.

Most of the survey respondents indicated 18 years as the minimum age for work, others mentioned up to 20 years.

► **Table 3: Surveyed farm workers' perception of minimum working age in Sindh and Punjab, 2020**

Age (Years)	10	12	14	15	16	17	18	19	20	21	22	23	24	25
% Response	0.67	1	2	8	5	15	32	15	14	2	2	0.33	2	1

Source: Mapping data

Related information

There is no indication that the farm workers were aware of minimum age for employment or work (15 years of age) or minimum age for hazardous work (18 years of age) set by the child labour laws in Punjab and Sindh. It also is common for people to indicate an age above the legally permissible age to give an impression of compliance with the laws.

Finding 6

Children's work on the cotton farm

Less than half (43 per cent) of the 300 farmers said boys and girls helped on the farms. Girls' involvement was reported mainly in harvesting (as for most adult female workers) and boys in sowing the seeds.

Children's work on cotton farms came at a cost to their education, with many dropping out of school during the harvest season.

The average age for girls and boys in cotton production was 15 years. For girls, it ranged from 8 years to 18 years; and for boys from 10 years to 17 years.

Tasks: The survey noted that children were involved in sowing, plant spacing, hoeing and harvesting. Boys' involvement in sowing was high (93 per cent of the boys). As with adult women on cotton farms, almost all the girls (96 per cent) were involved in harvesting, while 20 per cent of the boys were. (Also see the 2017 findings of the National Rural Support Programme below.)

Wages: The girls in harvesting earned 283 rupees per day, which was higher than the boys' earnings of 243 rupees in harvesting. Other than in harvesting, the boys' daily earnings were more than the girls' daily earnings. The most difference in daily wages was in harvesting, where girls earned 38 rupees more per day than the boys. In sowing, the boys earned 31 rupees more than the girls.

School drop-out: During the harvest season, the mapping survey reported a 10 per cent drop-out of children from schools.

► **Table 4: Children's work and daily earnings, by gender, according to surveyed farmworkers, Sindh and Punjab, 2020**

Tasks	Girls %	Daily earnings PKR	Boys %	Daily earnings PKR
Sowing	41	288	93	319
Hoeing	56	371	38	388
Plant Spacing	54	288	60	309
Harvesting	90	283	20	245

Source: Mapping data.

Related information

According to NRSP research (n.d.) in four districts of south Punjab (Bahawalpur, Bahawalnagar, Rahim Yar Khan and Rajanpur), all children were involved in cotton farming. Children of tenants/**haris** were involved from morning to evening and those of smallholder farmers were involved in farm work after school hours. NRSP notes that after fifth grade (around age 10) most children dropout, either because of the quality of schooling or non-availability of elementary schools and they do full-time work on or off-farm. NRSP research findings state that sexual and vocal abuse exist but there is reluctance in talking about it. Among health hazards that children and farm workers are exposed to, it lists snake bites, wounds on feet and fingers, eye infections, suffocation leading to fainting, general weakness, malaria, pneumonia, fever, typhoid and scabies.

The NRSP research shows a child can pick on average 15 to 20 kg per day for work of 9 to 10 hours. Based on interviews with teachers, the NRSP reports that children's attendance during sowing and picking (harvest) drops by almost 50 per cent. This is considerably higher than the 10 per cent drop during the harvest reported by the mapping survey. Reasons for the variation could include difference in distance to school, with longer distance making it difficult to combine the long hours of work with travel to and back from school. Other reasons could be family and community pressure to work (or attend school), availability of other family members to help on the farm during surge periods. Nonetheless, it confirms a considerably big dropout from school when children are heavily involved in cotton sowing or harvesting.

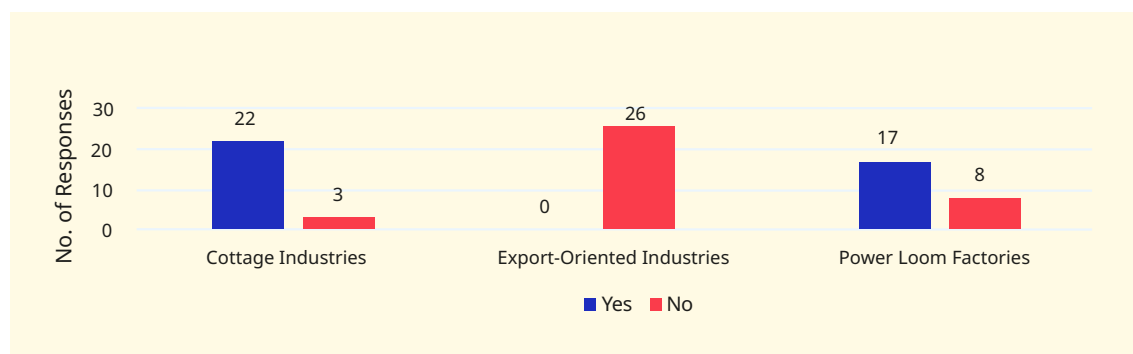
In the textile and garment units

Finding 7

Children of 15 to 17 years were reported to be working in power loom and cottage industry units, but not in export-oriented units.

- **Export-oriented industry units:** Presence of children was not reported in the export-oriented industries.
- **Cottage industry and power loom units:** According to the adult respondents, children (15 to 17 years of age) were present in large numbers in the cottage industry and power looms units surveyed. Whether or not their work involved hazardous child labour would depend on the nature of the work and working conditions. Most of these children did not have written contracts because they did not possess the computerized national identity cards. Some were reported to have started working after accompanying their parents to the factory.

► **Figure 6: Children (15–17 years) working in surveyed textile and garment units, Sindh and Punjab, 2020(numbers)**



Source: Mapping data.

4.3.4. Discrimination

At the community level

Finding 8

Gender-differentiated access to education for children of surveyed farm workers

Most workers (75 per cent) did not think there was differentiated access to primary school for girls and boys. However, one fourth of the surveyed farmers said there were access barriers for girls.

The 25 per cent who said that access was different for boys and girls explained the difference. Of these, most (32 per cent) said that they lacked resources, 30 per cent said that education of girls was not common in the community and 27 per cent said that girls must help at home. Around 11 per cent said a lack of separate schools for girls in the community hindered access.

Related information

“Lack of resources” for girls’ education would have two interrelated angles – one that the family cannot afford the education costs, and the other that the girls are needed to support the family with work on the farm (or at home) so they can earn more money, particularly where payment is piece-rate based.

Finding 9

Gender and religion-based differentiation

Most non-Muslims interviewed said they did not experience discrimination based on religion, but some said they did. Discrimination was reported in regard to wages, social interaction and treatment at school

Wages for most work in cotton farming are paid based on unit of work completed (for instance, weight of bolls of cotton picked, cans of chemical sprays applied). The survey did not find much difference between the wages of male and female workers. There were slight differences in what was reported. For instance, in the case of sowing, male workers earned 478 rupees per day while female workers earned 470 rupees – a difference of 8 rupees. For plant spacing, male workers earned an additional rupee per day. Similarly, in the case of hoeing, male workers earned an additional eight rupees and in harvesting, men earned an additional three rupees per day. Only 2 per cent females said that they received less wage than male workers.

Complaint mechanism: Workers said they have recourse in case they get less than the agreed wages or if they have complaints regarding their work. Some 31 per cent said they talk to the employer, and 43 per cent approached the head of the community, but 26 per cent said they remained silent. The complaints were reported to have been addressed in 63 per cent of the cases.

Responses of non-Muslims surveyed:

Wages: A majority (17 workers) of the 25 non-Muslim workers surveyed said they received wages equal to those of the Muslim workers, but eight workers said that they received lesser wages than the Muslim workers.

Social interaction: Asked if they could drink water from the same utensil as the Muslims, 13 said they could, while 12 said they could not. Asked if they could sit on the same bed as the owner, 11 non-Muslim workers said they could and three said they could not.

School experience: Children of 11 non-Muslim workers did not go to school while those of 14 went to school. The reason cited for not sending their children to school was poverty.

Of the non-Muslim workers, eight said that their children were not discriminated against at school; five non-Muslim workers said their children were discriminated against at school.

The discrimination reflected in some responses could be on the ground of religion or class.

In the textile and garment units

Finding 10

Discrimination in employment and occupation in surveyed garment units

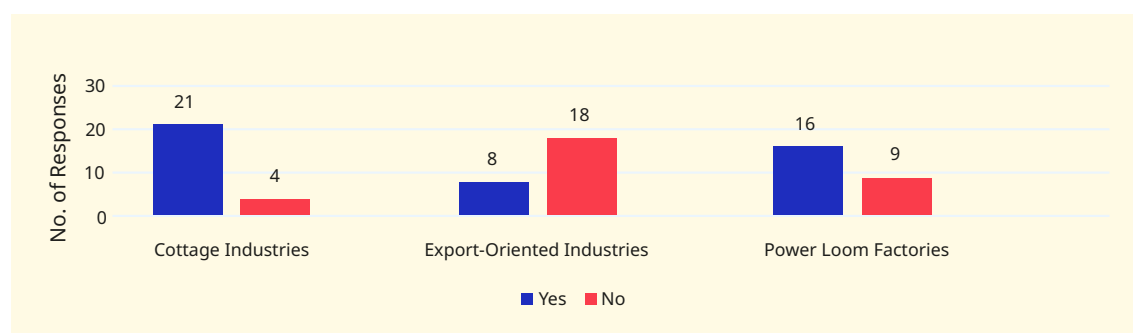
Most workers in the cottage industry and power loom units said they had experienced discrimination, including discriminatory treatment for access to basic amenities. Women mentioned being stuck in low-paid and low-skilled jobs.

The concept of maternity leave is not well-developed and in some small-scale industry units, pregnant women were reported terminated on ground of inefficiency.

- **Export-oriented industry units:** Most of the workers in the export-oriented industry units responded negatively to being faced with discrimination based on gender, religious or ethnic identity, pertaining to hiring, promotion and dismissal, and on the grounds of connection with any union.
- **Cottage industry and power loom units:** Most of the respondents (21 out of 25 in cottage industry units and 16 out of 25 in power loom units) said they had experienced discrimination at some point while working there. Women faced discrimination in the form of low-paid and low-skilled jobs. Workers noted the marked difference in provision of basic amenities between the managerial staff and the workers, reporting they did not have access to clean drinking water and sanitation facilities.

(See figure 9.)

► **Figure 7: Surveyed garment unit workers' responses on discrimination at work, Sindh and Punjab, 2020(number of workers)**



Source: Mapping data.

4.3.5. Safe and healthy working environment

At the community level

Finding 11

Farmworkers surveyed “perceived” tillage, levelling, pesticide and fertilizer application to be ‘hazardous’ tasks generally. However, their “experience” did not match their perception, as most indicated they had not experienced these workplace hazards, except in pesticide application.

In addition to tillage, levelling, and pesticide and fertilizer application, irrigation, hoeing, plant spacing, and sowing were also considered hazardous by some workers. The one tasks that none of them considered to be hazardous was harvesting – done entirely or mostly by female workers.

Yet, when asked about hazards in cotton production experienced by the workers themselves, one in almost six respondents surveyed expressed having experienced allergy on their hands and one in five respondents had experienced cough because of harvesting. Other illnesses mentioned in connection with various tasks included: tiredness (mostly when sowing or hoeing), allergy on hands (mostly when sowing or harvesting), cough (mostly when harvesting), injury – unspecified (mostly when hoeing or tilling). Other issues included back pain, fever, blisters on hand, headaches, eye infections and respiratory problems.

Training programmes on health and safety issues, more along the line of awareness raising were organized by the Agriculture Department and NGOs. The fertilizer and pesticide providers provided some training, but most seemed to be promotional.

This shows a disconnect between workers’ perception of what is hazardous in general, their perception of what is hazardous in their experience and their actual hazardous experience.

► **Table 5: Surveyed farmworkers' perception of tasks being hazardous or not, Sindh and Punjab, 2020 (percentage of workers)**

Type of Work	Hazardous (%)	Not hazardous (%)
Tillage	100	0
Levelling	100	0
Pesticides application	100	0
Fertilizers application	97	3
Irrigation	81	19
Hoeing	73	27
Sowing	19	81
Plant Spacing	19	81
Harvesting	0	100

Source: Mapping data.

► **Table 6: Workplace hazards experienced by farmworkers as reported by them, Sindh and Punjab, 2020 (percentage of workers)**

Tasks	No hazard	Tiredness	Headache	Injury	Fever	Allergy on hands	Cough	Blisters on hands	Foot Injury	Back Pain	Eye Infection	Respiratory Problem	NA
Fertilizers Application	19	8	1	0	0	12	1	0	0	8	0	0	50
Harvesting	13	4	5	1	7	17	19	0	0	5	0	3	26
Hoeing	34	15	0	18	1	0	10	3	18	0	0	0	0
Irrigation	32	9	0	2		0	0	0	0	0	0	0	58
Leveling	45	0	0	2	3	0	0	0	0	0	0	0	50
Pesticides Application	5	4	13	0	4	1	8	0	0	0	7	0	58
Plant Spacing	36	5	1	0	9	9	0	20	0	19	0	0	0
Sowing	49	17	0	0	4	20	9	0	0	0	0	0	0
Tillage	63	12	3	16	6	0	0	0	0	0	0	0	0

Source: Mapping data

Related information

Studies note the lack of awareness of risks, particularly to women, pregnant women and children, that are posed by pesticides, even when they are not the ones applying them. There is a lack of knowledge of personal protection measures, particularly among women. Studies also show cotton picking experience and some education contributed to women applying protective measures, while those without literacy tended not to apply protective measures but did traditional treatments (Memon 2019). Pesticide application by children under 18 years of age is prohibited by law in Punjab and Sindh Provinces.

Formal training on health and safety, if any, does not seem to have been adequate. There have been efforts to raise awareness of cotton farm owners and pickers, including by the Federal Government in collaboration with non-governmental organizations and Punjab and Sindh Provinces, about the need for protective clothing, but it does not seem to have had adequate impact (Shaikh and Tunio 2017).

Finding 12

Farmworkers and farmers in 20 per cent of the surveyed villages noted that health services were not available in their villages and they had to travel more than 10 km to get them. Public health facilities providing free treatment were the first choice for a majority (78 per cent) of the surveyed workers. A majority of the workers reported that they or their families covered the cost of treatment for illness.

Asked how the farmer/manager/employer would respond in case of an accident, 43 per cent said they would be taken to the hospital, 33 per cent said they would be granted leave, 13 per cent said they would get money for treatment, while 4 per cent said they would get no sympathy and 6 per cent noted that no accidents happen.

In case of treatment for illness, the majority (58 per cent) said they shouldered the cost, while 20 per cent said the family picked up the cost. Farmers/owners picking up the cost was reported by 14 per cent of the workers, and 5 per cent said the contractor paid for the treatment. Parents were noted as paying the treatment cost by 3 per cent of the surveyed workers.

In the textile and garment units

Finding 13

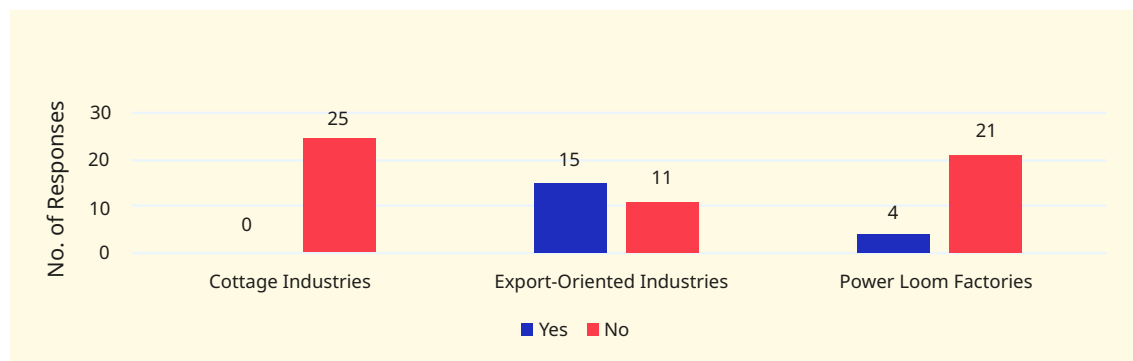
Access to health care providers for surveyed workers in the textile and garment units depended largely by the type of industry, with the more organized export-oriented units providing access to private medical providers to most of the surveyed workers, while the those in the power loom and cottage industry units paid for public health providers in crowded facilities.

- **Export-oriented industry units:** all respondents in export-oriented industries had basic access to health care services. Respondents reported that a percent of medical costs is covered jointly by the employer and worker. Basic medical care was available to almost half of the workers.

Majority of the export-oriented industry units respondents used designated private medical providers in the vicinity, while others used the publicly funded health providers' system.

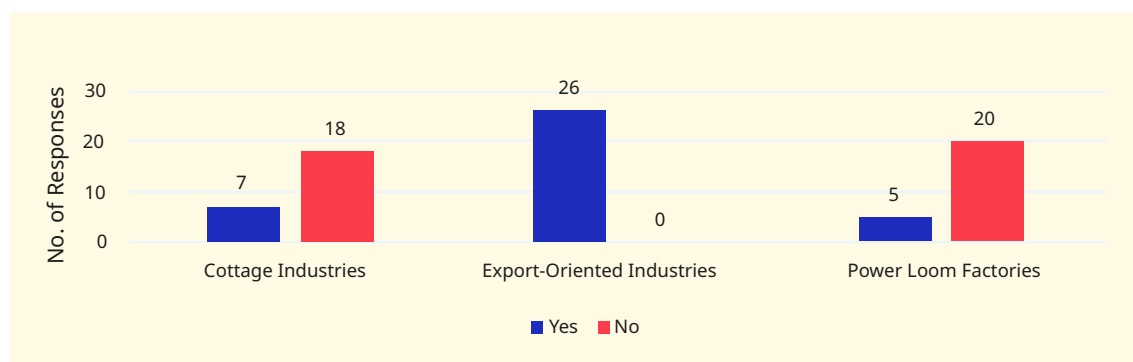
- **Power loom and cottage industry units:** Only four out of 50 respondents in these units said they had access to nearby medical facilities. The medical coverage for workers of the power loom and cottage industry sector was funded through the public health system. Although accessible and cheap, it was reported as overcrowded, under-funded and overworked. Most of the workers were required to pay the fees when using the facilities.

► **Figure 8: Surveyed textile garment unit workers health insurance, by type of industries, Sindh and Punjab, 2020**



Source: Mapping data.

► **Figure 9: Basic health services availability in surveyed garment unit workers' communities, by type of industries, Sindh and Punjab, 2020**



Source: Mapping data.

4.4. Findings on cotton production and trading

Findings 14 to 16 in this sub-section are related to: arm mechanization; education level of cotton farmworkers and farmers, and of their children.

4.4.1. Farm mechanization

Finding 14

Cotton production in Pakistan remains mostly traditional, with limited mechanization, particularly on small farms. Most tasks are done manually by farm workers.

The table below gives an indication of the limited machinery used on the smallholder farms, like those surveyed under this mapping exercise.

► **Table 7: Machinery and implements used on smallholder farms, according to surveyed workers, Sindh and Punjab, 2020**

Tasks	Machinery Used/Manually	Implements used
Ploughing	tractor	plough, disk harrow
Levelling	tractor	levellers, laser levellers
Furrows making	tractor	planter
Sowing	tractor	drill, if sowing is not done on furrows
Plant Spacing/Thinning	manually	no implement required
Hoeing	manually	hoe
Irrigation	manually	spade
Fertilizer Application	manually	no implement required
Pesticides Application	tractor	boom sprayer
Pesticides Application	manually	sprayer tank loaded on back

Source: Mapping data.

Related information

Farm mechanization, important for growth acceleration and increased productivity in the agriculture sector, is encouraged by the Federal Government. Its relief package allows import of farm machinery and equipment at reduced tariff.⁵¹ The domestic tractor industry's contribution to meeting the needs of farm mechanization is increasing. Some 612,000 tractors are in use around the country, but they do not fulfil the need.⁵² Mechanization needs of small and marginalized farms differ from those of larger farms.

4.4.2. Cotton farmworkers and farmers: education level

Finding 15

The surveyed workers and farmers on cotton farms showed limited or no education. Seven out of ten workers had no formal education.

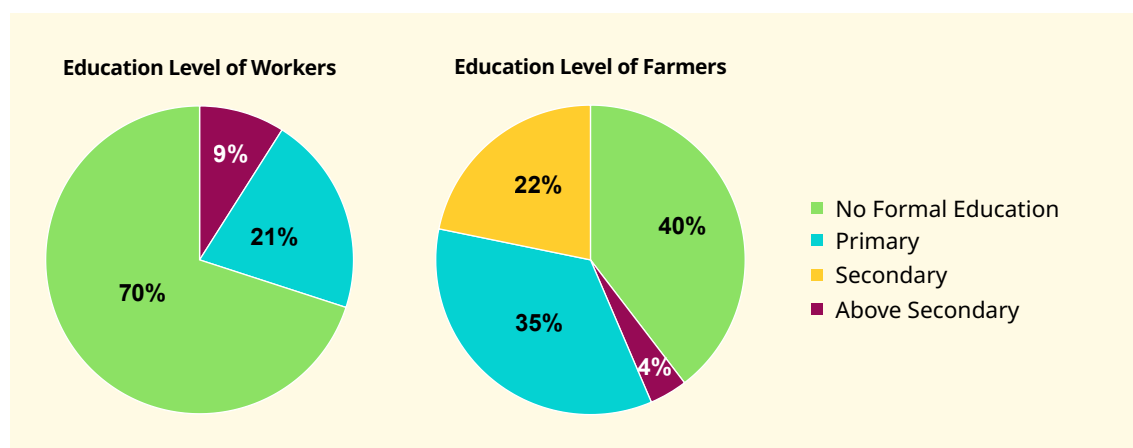
The low education level of workers, but also of farmers, contributed to their lack of opportunities for advancement.

- Some 70 per cent of workers had no education and 21 per cent had only primary education.
- Of the farmers, 35 per cent had no education and 40 per cent had only primary only education.
- The education situation declined further at the post-primary level, with 22 per cent of farmers and 9 per cent of workers having education up to the secondary level.
- Of the farmers, 4 per cent had education above secondary level. None of the workers interviewed had education beyond secondary level.

⁵¹ With custom duty of up to 2 per cent and Goods and Services Tax of 7 per cent.

⁵² The 612,000 tractors provide about 0.09 horsepower (HP) per acre – below the required power of 1.4 HP per acre.

► **Figure 10: Education level of surveyed workers and farmers, Sindh and Punjab, 2020 (percentage)**



Source: Mapping data.

4.4.3. Cotton farmworkers and farmers: their children's education

Finding 16

Among the surveyed farmers and workers, the education level of workers' children on cotton farms improved in comparison to their parents' level, but the level of the farmers' children did not improve compared to their parents.

Some 45 per cent of workers' children had **no formal education** – the reduction of the rate by 25 percentage points showing an improvement over their parents' rate (of 75 per cent). Of the farmers' children, 43 per cent had no formal education, which was 5 per cent more than the rate of their parents for the category of "no formal" education.

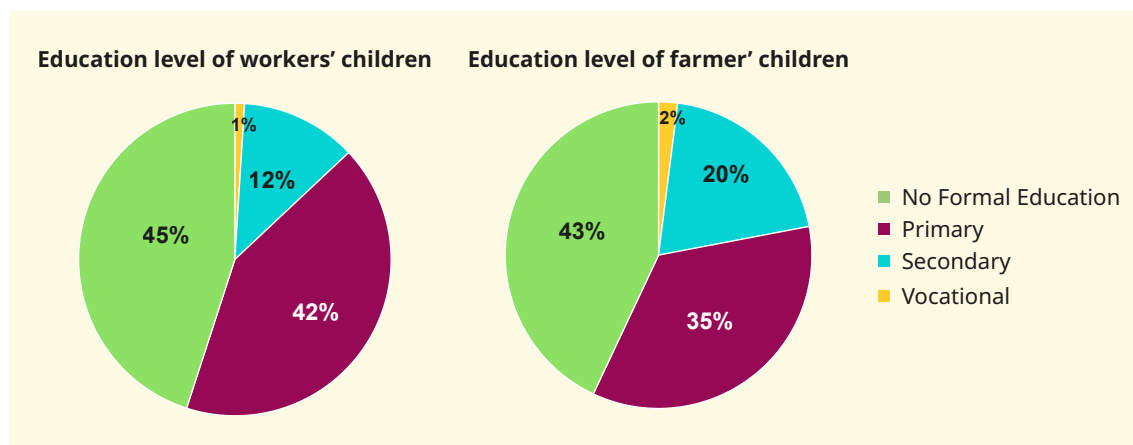
The workers' children **primary completion** rate of 42 per cent was higher than that of the farmers level as well as 2 per cent higher than the rate of the farmers' children.*

In the **secondary education** category too, there was an improvement, though small, for the children of workers from the older generation. Secondary education level of farmers' children was slightly lower than the level of the farmers. In case of the workers, the secondary education level for their children was higher but lower than the level for farmers' children. About 2 per cent of children of farmers and 1 per cent of children of workers had vocational training.

Factors contributing to the downward trend for the farmers' children's education were not clear from the mapping research, whether it could be the children's contribution to their family farm work or something else.

*Note: The national primary completion rate is over 73 per cent (2019). The World Bank, "Primary completion rate, total (% of relevant age group) - Pakistan", 2021.

► **Figure 11: Education level of children of surveyed workers and farmers, Sindh and Punjab, 2020 (percentages)**



Source: Mapping data.

4.5. Findings on farmworkers in cotton production and trading

Findings 17 and 18 in this sub-section are related to the following aspects: employment status and types of work of cotton farmworkers.

4.5.1. Cotton farmworkers: employment status

Finding 17

Casual work was dominant in cotton farming in Punjab and Sindh Provinces, with 72 per cent of 300 workers surveyed being casual.

Casual workers were hired on needs-basis during the cotton season. Some 36 per cent of permanent workers included mostly sharecroppers who worked on part-time basis in Punjab. Unpaid family members worked on small land holdings.

Male workers were mostly sharecroppers, and they worked on the farm where they were employed. All female workers interviewed were casual on daily wage basis. They moved from farm to farm, following earning opportunities.

On average, a cotton farm engaged at least one unpaid family member and employed one daily waged worker. During sowing and harvesting, at least 20 seasonal workers were engaged for sowing and harvesting/picking. In the case of a sharecropper, who received 50 per cent of the crop, the entire family worked.

4.5.2. Cotton farmworkers: types of work

Finding 18

From sowing to harvesting, female workers do a variety of tasks on cotton farms throughout the season.

Some tasks in cotton farming are specific to female workers, others to male workers and some are performed by both. Women were not reported for tasks like tillage, fertilizer application, pesticide application and irrigation.

In Punjab, women work mostly as cotton pickers. Less than 5 per cent male workers in Punjab and 20 per cent in Sindh are engaged in cotton picking.

Tasks performed by both male and female workers included cotton sowing, weeding, hoeing, and thinning (plant spacing),

Of those surveyed, on the average men work 47 days on cotton crop while women work 92 days. Women had more working days for two reasons. Firstly, they moved from farm to farm in search of work, secondly, women were engaged in cotton picking which continues for almost three months. On average, workers worked 6 hours a day. This is because most of the tasks in cotton farming are performed in hot months from April to August. They work until 11 a.m., then rest and start work again after 3 p.m.

► **Table 8: Types of work performed by farmworkers, by gender, according to surveyed workers, Sindh and Punjab, 2020 (number of days)**

Type of Work	Female average days	Male average days
1. Soil preparation		
Tilling	0	5
Levelling	0	1
3. Harvesting		
Cotton picking	57	4
2. Cultivation		
Sowing	10	4
Hoeing	16	9
Plant spacing	9	4
Fertilizers application	0	7
Pesticide application	0	9
Irrigation	0	4
4. Post-harvest		
Weighing	-	-
Loading	-	-
Total average working days per season:	92 days	47 days

Source: Mapping data.

Related information

The work is performed methodically, following the traditional script, at least in the smallholder farms, as it moves from one phase to the next.

4.5.3. Cotton farmworkers: wages and bonus

Finding 19

For casual cotton workers, wages are based on task-specific wage rate, with female workers earning on average 392 rupees and male workers earned 495 rupees per (working) day.

Sharecroppers are paid share of the cotton crop in-kind – one eighth or one tenth in cotton crop on part time basis in Punjab and 50 per cent in Sindh. Average earnings of sharecroppers during the 2019 cotton season were 45,000 rupees. Earnings were lower than expected because yields were low.

Wages to other workers are paid in cash. For most workers, their payment is on wage rate basis. Rates depend on the type of task. The rate for sowing, thinning, hoeing, and weeding is determined on per acre basis. For pesticide, the rate is determined by per tank sprayed. For cotton picking, the rate is determined by quantity of cotton picked.

On average, workers (female and male) for the cotton season in 2019 worked 71 days and earned 427 rupees per working day, with a seasonal earning of 30,317 rupees.

A male worker worked 47 days and earned an average of 495 rupees per (working) day, with a seasonal earning of 23,263 rupees.

A female worker worked 97 days and earned an average of 392 rupees per (working day), with seasonal earning of to 36,064 rupees. The gender pay gap is reflected with the female worker getting an average of 103 rupees less per day than a male worker and 35 rupees below the combined average for male and female workers.

	Average daily wage (PKR)	Seasonal earnings (PKR)	Season days worked
Female & male	427	30 317	71
Male	495	23 263	47
Female	392	36 064	97

In south Punjab, **decisions on spending of income earned from picking** rested with the women who did the picking and earned the money. They were paid directly, and they spent the money, as they wanted to. Nevertheless, it mostly was spent for household needs. In case of **contracted labour**, the payments were made to the head of the family – often a male.

Finding 20

Some bonus was given to permanent cotton farmworkers, but not the casual workers.

There is no system of bonuses in cotton farming for workers. They get only the agreed upon wages. However, for permanent workers employed on monthly salary, additional benefits include 40 kg wheat per month and in some cases fodder for one head of livestock.

4.6. Findings on farmers in cotton production and trading

Findings 21 to 27 in this sub-section are related to the following aspects: reasons for choosing to do cotton; participation of farmers' youth; farmers' perceptions of the environmental impacts; economic vulnerability of smallholder farmers; price determination according to farmers, and selling and buying of seed cotton.

As noted previously, the survey findings for farmers presented below are based on key respondent responses from 300 male farmers in 50 villages across five districts in Punjab and Sindh Provinces of Pakistan. The districts in Punjab are Vehari, Bahawalpur, Rahim Yar Khan, and in Sindh they are Mirpur Khas and Sanghar. The average age of the farmers was 39 years in the age range of 30 to 70 years. As with the workers, most farmers were in the range of 35 to 45 years.

The key informant farmers included a diverse set from smallholder farmers to large farmers, owning land ranging from less than 2 hectares for cotton crop to 10 to 30 hectares; and from cotton crop income ranging from 10,000 rupees (as a small percentage of total income, with additional earning more from other crops) to over 6 million rupees with major investment in cotton over land of over 10 hectares.

Of the 300 farmers, 280 were also involved in farming additional crops, close to half raised livestock and others were involved in business or held a job as a source of additional income. Many combined two additional sources of income, mostly cultivation of other crops and raising livestock. Only 11 farmers depended on cotton farming income alone, most cultivating 3 to 4 acres, with three farmers cultivating 5 to 10 acres. Most of the farmer (176) cultivated cotton on land below 5 acres, 84 farmers did so on 5 to 9 acres, 35 farmers on 10 to 39 acres, and five farmers on 40 to 60 acres of land. The highest total income reported in the range of 5 million to 6.5 million rupees was by three farmers cultivating land of 60 acres (two farmers) and 22 acres (one farmer). All three had additional crop and/or livestock income amounting to 35 to 46 per cent of their total income. Land size and additional income sources were correlated.

Farmers with income between 10,000 to below 100,000 rupees cultivated land below 5 acres and most counted on additional income sources. The findings also show farmers with 1 to 2 acres of land made over 100,000 rupees annually when they had robust income from both cotton farming and additional sources of other crops and/or livestock.

4.6.1. Reasons for choosing to do cotton farming

Finding 21

Many farmers chose cotton farming to carry on family tradition, some considered the income 'decent', but for most, the fact that it was valued in the community was important.

Some 35 per cent of farmers said they started cotton farming, to continue with the family tradition and because they saw it as their family's bread and butter. Almost the same per cent (37 per cent) said they were in cotton production because there was no alternative. Yet, other farmers (28 per cent) considered cotton production as a source of 'decent' revenue.

Most of the farmers (58 per cent) agreed that cotton farming was an activity that was valued in their communities.

4.6.2. Participation of farmers' youth

Finding 22

Half the surveyed farmers engaged their family youth in cotton farming; and the majority said young people liked to farm.

Some 50 per cent of the 300 key respondent cotton farmers engaged the family's young people in cotton production with them. To a question on the youth's liking to be involved in farming, a majority of the farmers (65 per cent) said that the young members of their families liked to engage in cotton farming.

4.6.3. Environmental impacts – farmers' perceptions

Finding 23

One third of the farmers were aware of cotton farming's negative impact on soil, air and water.

One third of the surveyed farmers were aware of the adverse environmental impacts of cotton farming on water, soil and air. Sources of information for farmers included training programmes of Agriculture Department and non-governmental organizations working on capacity building of farmers.

About 28 per cent of the surveyed farmers were involved in sustainable cotton production and working according to Better Cotton Standard System.

4.6.4. Economic vulnerability of smallholder farmers

Finding 24

When the cotton yield failed to meet its target, the economic vulnerability of many smallholder farmers came to the forefront, and they found their income insufficient and their lives unstable.

Of the surveyed farmers, some 32 per cent needed to take out loans to make ends meet, and 27 per cent said the education of their children suffered. Some said they sold livestock to overcome losses from cotton crop failure as they did not have finances to cover expenses for the next crop. Only 8 per cent of the surveyed farmers considered their cotton farming income sufficient to cover the basic needs of the family.

As many of the farmers are smallholders, they needed to supplement their incomes. About 88 per cent farmers said that they had to seek other sources of income as well.

When farmers were asked if their income from cotton farming was stable during the past few years only 23 per cent farmers responded in the affirmative.

4.6.5. Farmers' credit sources

Finding 25

Farmers borrowed from different sources, but a large number (40 per cent) did not indicate whether they borrowed at all.

Loans: Of the 300 farmers interviewed, 120 did not respond to the question, while 80 per cent of the 180 cotton crop farmers who said they did borrow, did so from Zarai Taraqiati Bank, the agriculture development bank. They also borrowed from NRSP Microfinance Bank, other commercial banks, microfinance institutions and some 6 per cent (of those borrowing), did so from the informal loan providers (the intermediaries in the village and trade market – *beoparis* and *arthis*). Another source of credit for the farmers was the input dealers, particularly the pesticide and fertilizer dealers.

It is likely that many of the 40 per cent who did not respond to the question borrowed from the informal lending sources. It is also likely that some farmers borrowed from multiple sources.

Advance credit for farm inputs: The most expensive and major inputs required for cotton crop include fertilizers and pesticides.

Pesticides: of the farmers surveyed, some 77 per cent had **access to pesticides**. Of the farmers buying pesticides, 80 per cent did so on credit from pesticides dealers, 14 per cent buy from the intermediaries (*beoparis*), and the rest from traders in the grain/trade market (*arthis*) and or directly from the pesticide companies.

Fertilizers: some 67 per cent farmers had access to **fertilizers**. Of the farmers buying fertilizers, 67 per cent bought them on credit from (fertilizer) dealers, 17 per cent buy from the *arthis* (traders/ commission agents) and the rest from the intermediaries (*beoparis*) or other sources.

Canal water: about 78 per cent of the surveyed farmers had access to **canal water** provided by the Government Irrigation Department

All credit sources used for pesticides and fertilizer were informal lending sources.

4.6.6. Price determination according to farmers

Finding 26

Trading price at the community level remained elusive as traders deducted subjectively for impurities, with most farmers not aware of how the price was set or who set it.

During the survey, when farmers were asked as to **how the price of seed cotton was fixed**, Some 28 per cent said it was market determination, 26 per cent thought it was mutually agreed between the farmers and the traders (*beopari* and *arthis*), while others said it depended on the quality of the cotton. Some said it was based on international pricing and others said they did not know.

Asked about the **government's role** in cotton seed price setting – 95 per cent said no.

As to whether there was **guarantee of minimum price** of seed cotton – 98 per cent said no.⁵³

53 At the time of the mapping exercise in 2020, the government did not announce support price of cotton; as it for sugarcane and wheat. In 2021, a support price for cotton was announced.

4.6.7. Selling and buying seed cotton

Finding 27

For the majority of farmers (73 per cent), their cotton deal and sale are done at their doorstep through informal community channels – the beoparis.

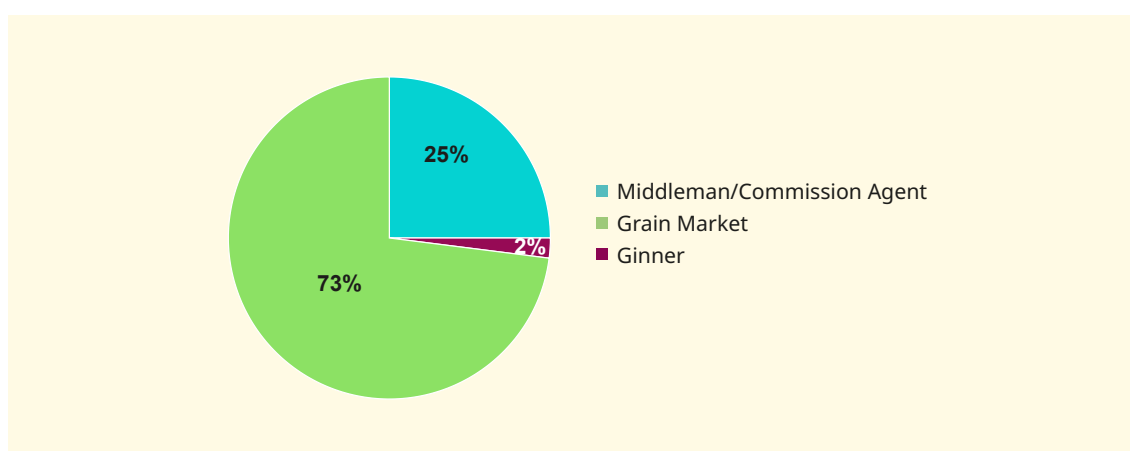
Of the 300 farmers surveyed, some 73 per cent sold seed cotton to the *beoparis* in the village (middleman/commission agent), about 25 per cent to the *arthis* in the market (*mandi*), and 2 per cent directly to the *ginners*.

Of the farmers who sell to the *beoparis*, about 15 per cent said they sold in advance of the harvest (based on the cultivated crop and expected value), others said they sold the crop produce at their doorstep, mainly to the village trader, the *beopari*, whom they know well.

Selling and buying in the trade market and ginnery:

(For related discussion of buying and selling, see sub-section 4.1.3 of this section.)

► **Figure 12: Buyers of farmers' seed cotton according to surveyed farmers, Sindh and Punjab, 2020 (percentage of buyers)**



Source: Mapping data.

4.7. Findings on working conditions in textiles and garments units

In addition to the field survey in tier-4, the findings of which are presented above, surveys were conducted in units of textile and garment industries in different tiers of the supply chain to gather data on the prevailing working conditions. A sample size of 76 workers was composed of units at different levels as follows:

- 26 interviews in 4 units in export-oriented industries units (1 unit in tier-1, 3 units in tier-3),
- 25 interviews in 5 units in power loom/domestic factories units, and

- 25 interviews in 5 units in cottage industries units in tier-3, supplying to export-oriented industries, such as cut to pack, buttons and zips.

All these industries were randomly selected and were in Faisalabad, Lahore and Karachi.⁵⁴

The findings from the mapping exercise show that the employment status of the workers, their working conditions and benefits depend on the type of industry they work in. The export-oriented industry units were more organized than the cottage industry and power loom units. The export-oriented industry units are often considered to be better set with higher industry standards as they are subject to industry audits. However, this alone does not guarantee higher standards as has been seen in Pakistan's garment sector factories. Generally, the standards and compliance are seen to decline from the export-oriented to the power loom and even more in the cottage industry units. This also reflects the move away from the formal economy to the informal economy enterprises.

Findings 28 to 31 in this sub-section are related to the following aspects: employment status; salary and working hours; overtime and benefits, and provident fund and housing.

4.7.1. Employment status, wages, overtime and benefits

The findings from the mapping exercise show that the employment status of the workers, their working conditions and benefits to be more secure and better in the surveyed export-oriented industry units than in the power loom and cottage industry units where, with no terms and conditions being established and documented, the workers were not aware about their rightful employment conditions and benefits. Responses place more workers in the export-oriented industry units with better contractual arrangements on full-time basis, minimum or over minimum salary, and benefits (such as housing fund) than workers in the power loom and cottage industry units. More of the export-oriented unit workers also indicated overtime compensation and bonuses than workers in the other units. The overall trend of the responses corresponds to the general view of the working conditions in these units.

With more than half the workers not responding, the findings show a reluctance on part of the surveyed workers to engage on the issue of wages and other employment related matters.

Employment status

Finding 28

- **Export-oriented industry units:** all 26 workers in the survey were hired on full-time basis.
- **Power loom units:** the majority were on daily wages, some were on contract basis, and a few were permanent. there were no written work contracts.
- **Cottage industry units:** most were on daily wages, while a few were on contract through labour contractors. Family members of some workers also worked without pay. There were no written work contracts.

⁵⁴ Some survey participants marked more than one option, which may be reflected in the number of responses.

Salary and working hours

Finding 29

The responses show

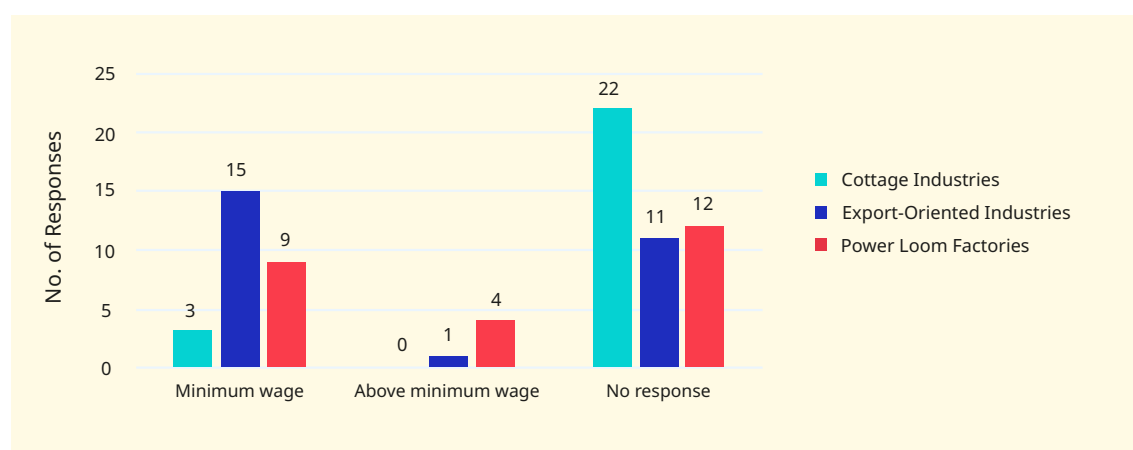
- **Export-oriented industry units:** almost half the workers were paid the minimum wage. The rest did not respond. Analysis of the individual salaries showed the average pay as 15,778 rupees.*
- **Power loom units:** some were paid minimum wage with only a handful being paid higher than the minimum wage. The rest did not respond.
- **Cottage industry units:** only a few were paid the minimum wage. The rest provided no response. Their average earnings ranged from 20 to 250 rupees per piece, depending on the type of work involved. Whether or not this made up to the minimum wage was not clear.

Of the cottage industry unit workers surveyed, 88 per cent did not respond to the question on wages. The no-response rate was high in other two categories as well: 48 per cent for the power loom and 42 per cent for export-oriented industry units.

Workers in cottage industries and power loom factories refrained from complaining about wages to avoid reprisals and the possibility of losing their jobs. This likely led to many opting for 'no response'. Some did complain about short lunch and bathroom breaks.

Note: *Minimum wage at the time (2020) was 15,500 rupees in Punjab Province, and 17,000 in Sindh Province.

► **Figure 13: Basic salary of surveyed of textiles and garment unit workers, by unit type, Sindh and Punjab, 2020 (number of workers)**



Source: Mapping data.

Note: one worker in the export industry registered more than one response, so adding to 27 instead of 26.

Overtime and benefits

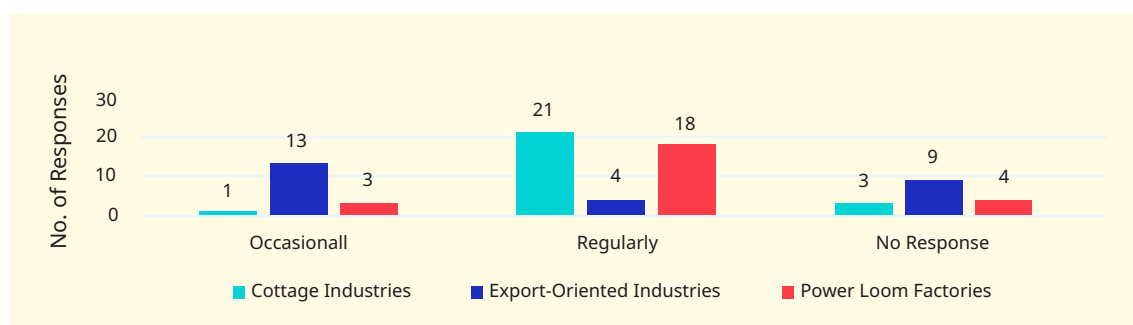
Finding 30

- **Export-oriented industry units:** overtime and productivity bonus were common, with 21 out of the 26 respondents responding in the affirmative.
- **Power loom factories:** only some reported receiving overtime. Most respondents did not respond.
- **Cottage industry:** similarly, the majority respondents did not respond to the question on other forms of remuneration. Only a few reported receiving productivity bonuses.

The workers in power loom factories and cottage industries indicated they were under pressure to work overtime and if they did not comply, they would be penalized, even terminated.

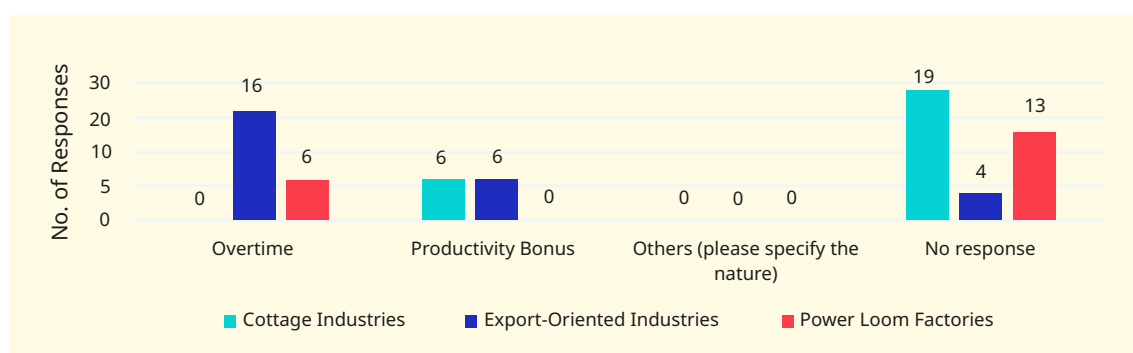
As a result of the long working hours and forced overtime, women found it difficult to get reliable and secure transport to return home late at night. There was no provision of medical leave for the workers.

► **Figure 14: Overtime frequency among surveyed textiles and garment unit workers, by unit type, Sindh and Punjab, 2020 (number of workers)**



Source: Mapping data.

► **Figure 15: Other remuneration of surveyed textiles and garment unit workers, by unit type, Sindh and Punjab, 2020 (number of workers)**



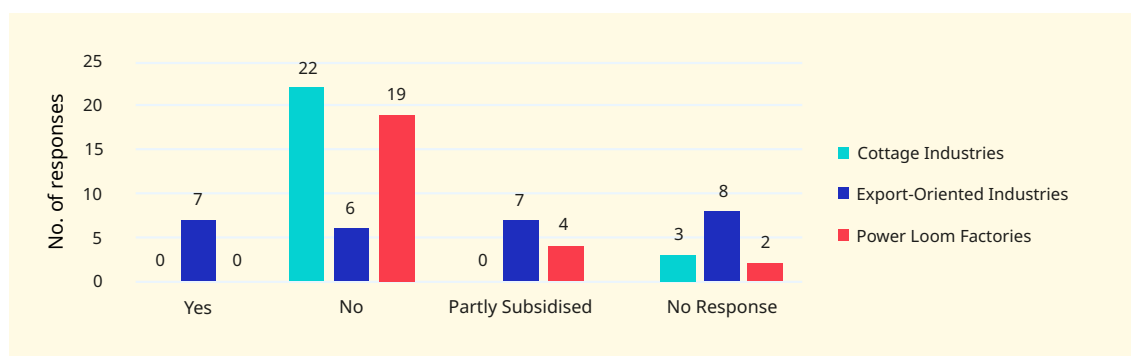
Source: Mapping data.

Provident fund and housing

Finding 31

- **Export-oriented industry units:** Employers contributed to the provident fund for the workers in the export-oriented units. Some of the respondents had taken advantage of the housing facilities provided by the employer.
- **Power looms and cottage industry units:** There was no coverage of provident fund in the power loom factories or in cottage industries. Only a handful respondents in power loom had partially subsidized housing facilities.

► **Figure 16: Employers' benefits for surveyed textiles and garment workers, by unit type, Sindh and Punjab, 2020 (number of workers)**



Source: Mapping data.

► **Table 9: Minimum wage between 2020 and 2021, Punjab and Sindh Provinces**

Workers	Minimum wage/month (in PKR)							
	Punjab				Sindh			
	2020		2021		2020		2021	
	Month	Day	Month	Day	Month	Day	Month	Day
All workers (male, female, juvenile)	16 500	635	20 000	769	17 500	673	25 000	961

Source: Mansoor, M, "Punjab raises workers' minimum wage to Rs20,000", Dawn 2 May 2021; Minimum Wage Notification 2021, Government of Sindh

In Sindh, the 2021 Minimum Wage notification was issued by the Labour and Human Resources Department under Section 4 of the Sindh Minimum Wages Act, 2015, covering all industrial and commercial establishments, whether registered or not.⁵⁵ Full-time employees (working 8 hours a day) are entitled to weekly holidays, employees old-age benefits, social security and overtime pay. Contractors, temporary and part-time workers are not entitled to these benefits. This leaves out of coverage many

55 EFP, "Minimum Wage Notification 2021 – Labour Dept, GOS", Blog, Employers' Federation of Pakistan. Available at: <https://efp.org.pk/minimum-wage-notification-2021-labour-dept-gos/>.

workers in the informal economy's micro and small enterprises – including those in the cottage and power loom units, and even some in export-oriented units. The Act specifies that piece-rate workers must be ensured of an earning of not less than 120 rupees per hour. The Act stipulates equal remuneration for female and male workers.

In the 2021 notification, the term 'wage' has been used in the sense of remuneration to include both salary and wage. Generally, 'wage' is the variable amount of compensation which is paid on the basis of a unit cost – such as per day, per hour, or per item produced or completed. In this report and in the case of cotton workers, the basis of unit for the rate includes quantity or weight per unit of seed cotton picked, or the number of tanks of pesticide sprayed or amount of land that has been sowed with seed, or where other tasks, such as thinning, hoeing or weeding have been performed.

Calculating estimated overtime in the cottage industry is challenging since time needed to complete different items vary. Hence, determining wage rate against the statutory minimum hourly, daily or monthly wage is difficult (Zhou 2017). The average time for which the workers worked was 12 to 13 hours a day in most cases.

► 5. Conclusions and recommendations

5.1. Conclusions

The results and analyses of the findings of the mapping survey and review of available studies and information, show decent work deficits compounded by a general lack of knowledge and limited, though increasing incrementally, the application of fundamental principles and rights at work in the cotton growing communities in the five districts surveyed in Punjab and Sindh Provinces.

The report has described how the cotton supply chain at the community level is under strain. The cotton farms in Pakistan contribute significantly to the GDP, generate critical foreign exchange for the country, feed raw material to the critical textile and garment industry, provide livelihoods to millions in the cotton supply chain and enable it to be the fifth largest producer in the world. Yet, cotton production is under multi-dimensional stress brought about collectively by factors ranging from low yields to low government support, to outdated farming methods on many of the farms, to constant fragmentation of farm land from one generation to the next, to the palpable impact of climatic change. By and large, the cotton smallholder farmers, farmworkers, and workers and businesses along much of the supply chain do not partake of the success and abundance of the cotton success story.

Pakistan's agriculture sector is critical to the country's economic and human resource growth. Policies and initiatives, including social protection measures, are being introduced to support the sector further, but implementation is weak. As the report shows, policies and initiatives do not always benefit the most marginalized farmers and farmworkers. They continue to rely on dependency-inducing informal credit providers as they ponder and then switch from cotton to other more profitable crops, such as wheat and sugarcane – following their peers. These are the **farmers** who are not aware of or cannot afford farm inputs that would be best for their needs, who cannot afford to buy or rent farm machinery, whose young children are working on the farms, likely involved in child labour, and missing out on education. These are the **workers** with inadequate income, no job security or work quality, no social protection, few skills and no capacity to protect themselves and their children from hazards at work. These are the **female workers** with triple responsibilities (of motherhood, houseworker and farmworker) who yearn to be trained and to be able to use farm machinery, to have equality and equity at work and in their lives. These are the human resources on the farm that carry much of the burden in traditional farming, as the report shows.

The pursuit of solutions for remediation for the impacts of cotton stress intensified over the past several years. Numerous evidence-supported biological, technological, and environmental causes have been put forth. These have inspired policies and action in attempts to halt and reverse the steep decline in yield and the shuttering of businesses along the supply chain. The wide-ranging policies on input subsidies, quality seeds, certification, farm machinery, cess and subsidies, import and export of cotton, and crop support, among others, are contributing in varying degrees to improving cotton production.

However, the **human infrastructure**, which is a critical set of **people factors** that contribute to the yield, is often overlooked in research and discussions and so subsequently in policies and responses. Supportive and adequate human infrastructure enables farmers and farmworkers to advance from deficit riddled work to decent work and to maximize their productivity. Among the key **people factors** creating the elements of human infrastructure is decent work, including fundamental principles and rights at work, employment and income opportunities, social protection, and tripartism and social dialogue. The report shows inadequacies in important specific supportive factors, such as literacy, education, skills,

knowledge and information, affordable access to health services, and targeted community support particularly in the case of migrants and minorities. The report shows a varying degree of awareness among the key stakeholders of the cotton production community of the lack of important factors, including working conditions, fair and timely wages, support for childcare. Even so, there is either a lack of capability or negligence that leads to inadequacies in the human infrastructure. **Just as research, discussions, policies and action are needed for the biological, technological, and environmental factors, so they are needed for the provision of an enabling human infrastructure.**

The mapping data does not provide evidence of the promotion of fundamental principles or rights at work or their application at the community level in the cotton growing communities that were surveyed. This also is the case for the textile and garment units surveyed in the cottage industry and the power loom industry units. Those in the export-oriented industry units that were surveyed demonstrated somewhat greater respect for and application of fundamental principles and rights at work. More specific description of the situation of these rights is given below.

Freedom of association and the right to collective bargaining, in the form of freedom to form unions, have representation, be a part of them, or have a recognized and authoritative body negotiate on behalf of workers, were not observed on the farms. Workers in industrial units, other than export-oriented industry units, were afraid of reprisals and pushed back on conversation or input into the matter. This was to be expected in the informal economy, given the general level of low unionization in the country, even in the formal sector. The legislative provision in Sindh through the Sindh Industrial Act, 2013 can have an impact if implemented vigorously. Punjab is yet to make the provision for industrial action in the agriculture sector.

Forced labour or compulsory labour includes all work or services (with specified potential exclusions) that are extorted from persons under the threat of penalty and for which the persons have not offered themselves voluntarily. The report highlights certain prevailing practices that make the workers vulnerable to conditions of forced labour. The practice of providing advance payments (*peshgi*) from informal sources to workers and farmers is reported as a common practice. The report shows that at times the borrowers are not able to pay back in time, which rolls over their loan payment to the next cycle, and increases the commission they end up paying. In the case of farmers, the rollover binds them to the lender (usually the village or market traders, the *beoparis* and *arthis*), taking away their freedom to seek a better sale option for their crop. The practice of *peshgi* is associated with bonded labour in Pakistan.

The findings confirm that workers in the cottage industry and power loom industry units witnessed and experienced forced overtime and threats in case of non-compliance. This practice points to coercion, intimidation and threats and excessive or mandatory overtime, which are indicators of forced labour. The findings do not address whether the element of “involuntariness” in undertaking the work was involved on part of the workers. It is possible that this could be the case for the children who accompanied their family and continued to work.

Child labour: Children’s work on cotton farms or in off-farm activities, to attend to other chores while parents work, can be illegal and harmful to them. Although children were not among the key informants in the survey, there is evidence of their involvement in child labour. According to ILO Conventions, when children’s work interferes with their schooling in that they do not attend school, or drop out of school without completion, or some combine school with excessively long and heavy work, then that constitutes child labour. The report findings make it clear that this is the case when school dropout rates increase during harvesting because girls are involved in harvesting, or according to the NRSP report (NRSP n.d.) when boys are involved in sowing the seeds.

Application of chemical sprays by children below 18 years of age is prohibited by provincial laws in Punjab and Sindh Provinces and according to ILO Conventions on child labour. Children’s mere presence in the area will cause them harm. In the absence of targeted and immersive campaigns to educate and raise awareness about the harms and protection against them, children and communities will continue to be damaged for life, with impacts on their productivity and the yield.

It is also likely that children are involved in hazardous work in the industrial units: Adult key respondents in the cottage industry and power loom units surveyed confirmed the presence of children 15 to 17 years of age. Although it could be legal for this group of children to be working because of their age, it is also possible that their work was hazardous and illegal. The work they did, the number of hours they put in each day, the breaks from work that they took, off-days that they were provided, their working environment, the nature of their interaction with their supervisors and other workers, and their education status would have helped to determine whether or not they were in hazardous work as defined in the provincial laws or international Conventions. These details were not provided through the mapping exercise.

Most units in the informal economy seldom come under the purview of the inspection systems, and even if they do, inspections are rare, follow-up is even more rare, and compliance almost non-existent. Given this context, it is likely that the adolescents are involved in what is defined as hazardous work, which is a worst form of child labour – “work which, by its nature or the circumstances in which it is carried out, is likely to harm the health, safety or morals of children” (Article 3 (d) of ILO Convention No. 182 on the Worst Forms of Child Labour).

Discrimination: The findings show discrimination against girls in respect to their education. Elements included not sending them to school, downplaying the importance of girls’ education, and stopping the education of girls rather than of boys in case of resource constraints. The mapping data points to some religious, social and wage discrimination, but the limited information does not allow for a clear picture to emerge. Male and female workers in the cottage and power loom industry units said they had experienced discrimination.

The cottage and power loom industry units surveyed were in the informal and unregulated section of the economy, where most workers are vulnerable. The report notes intra-provincial and non-citizen migrants as among the workers. It may be noted that non-citizen workers are not entitled to the national identity cards, compounding their vulnerability, and leading them to work without written contracts and with low pay.

Safe and healthy working environment: The mapping shows that for most of the surveyed farmers their perception of safety did not sync with their reality and neither with the fact that the risks of today will likely become serious illnesses, or even incapacity, in the years to come. The use of personal protective equipment, if any, was very limited. Impact on workers’ productivity and quality of life can be impacted by issues with access to health care services and the quality of the services that were reported as well as neglecting to use protective equipment and taking other safety and health measures. More direct data and information on children’s work and child labour through direct interviews with the children and disaggregation of the data by age would have helped to shed more light on the hazardous nature of child labour on the farms and in the textile and garment units.

The report brings to the forefront the need for an integrated and mutually reinforcing approach to fundamental principles and rights at work in the cotton producing communities and along the supply chain. When smallholder farmers and workers have a voice and representation to help improve their work, lives and the policies that impact them, their social and economic situation is improved. With no further interventions to improve respect for fundamental principles and rights at work, child labour, forced or compulsory labour, and discrimination will continue. Without the necessary human infrastructure, the workers’ productivity will be minimized, many will continue to seek alternative sources of livelihood, impacting the yield and the economic situation of the families, and the communities along the supply chain.

The report shows that export-oriented units with connection to the multi-national or national export-oriented companies demonstrate better application of fundamental principles and rights at work than units in the informal economy that do not have these connections. Many of their workers are aware about the issues and they can enjoy these rights, at least to some degree. The broader supply chain and communities must benefit from these actions, which becomes difficult when multi-nationals have very narrow and exclusive supply chains.

Exclusive vertical supply chains may benefit business in the short term, providing swift opportunities for tracking and tracing, verification and transparency. However, they are not fool proof and sustainable if the entire supply chain (and corresponding supply chains) and the environment in which they are embedded, continue to be incognizant of the standards and where fundamental principles and rights at work are not respected.

The ILO experience in Pakistan and worldwide has shown that there are actions that can help improve fundamental principles and rights at work within specific supply chains, particularly if they are on sectoral basis. Such action to uplift communities in a sustainable manner requires commitment and action, including a coordinated community-based sectoral approach, by government, employers' and workers' organizations and multi-layered partnerships. For the cotton supply chain, it involves national, international and multi-national business stakeholders that source cotton and cotton products from Pakistan.

The recommendations that follow provide suggestions as to what needs to be done for a human-centred approach to the cotton supply chain.

5.2. Recommendations

The report enhances understanding of the context in which a typical cotton supply chain at the community level operates, and its key actors and dynamics, in selected communities in Punjab and Sindh Provinces of Pakistan. It does so by describing the pertinent national socio-economic and legal contexts. It also looks into the policy and programme framework of cotton production from the decent work perspective with a focus on fundamental principles and rights at work (which are: freedom of association and collective bargaining, elimination of all forms of forced labour and compulsory labour, effective abolition of child labour, elimination of discrimination in respect of employment and occupation, and a safe and healthy working environment). The report breaks down the various processes of cotton production and explores how they relate to one another and impact the workers and smallholder farmers. It examines the role of the workers and other key players and their intermingled relationships at the farm and market levels. It further examines working conditions in a number of diverse textile and garment units. Besides findings and analysis of the mapping survey, the report examines other current and recent studies and literature on the subject. These further validate the mapping survey findings.

The report draws on these findings and discussions to make the following evidence-based recommendations for impactful action so workers in the cotton fields and those in textile and garment units and farmers are provided access to and avail their fundamental rights at work and have their socio-economic conditions improved. The recommendations also draw on the ILO's vast experience providing policy advice and supporting the planning and implementation of technical cooperation interventions focused on agriculture, including cotton, workers and smallholder farmers and their local communities. As the recommendations are interconnected their impact, particularly in the long-term, will be maximized when implemented in entirety and cohesively. However, they can be phased in incrementally.

The recommendations are directed at the ILO, its constituents, and partners (including sectoral and development partners), who can take them forward, leveraging national and international partnerships.

Recommendation 1

ILO should support legal and policy reform and advocacy for a more equitable and enabling environment for cotton production that fosters respect for fundamental principles and rights at work.

More specifically, the recommendation includes the following sets of action:

- **Advocacy** at different levels to promote the review of laws and policies related to fundamental principles and rights at work as they impact the agriculture/cotton growing communities;
- Research, in collaboration with **competent research bodies**, on specific issues in the laws and policies that need to be changed. (Some research topics listed below would also apply.);
- Support to **employers' and workers' organizations** in their advocacy for legal reform impacting the agriculture/cotton workers and their coverage under the laws, particularly the industrial relations laws;
- Technical support to the **Federal and Provincial Governments** to review and revise laws and policies that restrict freedom of association and collective bargaining, do not adequately address conditions of forced or compulsory labour and child labour, that hinder gender equality, are discriminatory, and do not ensure safe and healthy working environments.

For effective implementation of this recommendation, its advocacy must be evidence-based – supported by relevant, updated, high-quality research and evidence from real life experiences, including from project implementation.

The report has noted in section 3.3. universities and research institutes in Pakistan that are actively involved in research on cotton growing. Partnerships with these and other institutions should be explored for undertaking the recommended research.

The report highlights legal obstacles to cotton farm workers to freely form and be part of trade unions that can enter into collective bargaining. If these laws do not cover agriculture workers, then the organizations cannot be registered and consequently cannot enter into collective bargaining on their behalf. With some apprehension about trade unions for fear of retaliation, 72 per cent of cotton workers surveyed in Punjab and Sindh said they wanted an organization to look after their rights.

Further supporting efforts of Pakistan Workers' Federation, Employers' Federation of Pakistan and other stakeholders to amend the Punjab Industrial Act will enable coverage of agriculture workers in Punjab under the Act – as was done through amendment the Acts in Sindh and Balochistan.

Supporting Employers' Federation of Pakistan and Pakistan Workers' Federation to continue to form local community-based employers'/farmers' and workers' organizations and developing their capacity for sustained operation will enable the cotton producers to have a voice in their work and to be served and protected by collective bargaining.

In addition to the Industrial Relations Acts mentioned above, other laws and policies that can improve the socio-economic and rights at work for workers are occupational safety and health, labour and other inspection systems, applicable taxation/cess collection, farm inputs that affect health, social protection, transportation, housing, and children's access to education.

Women in cotton production, as in agriculture in general, are disadvantaged, in part, because of laws that do not extend adequate coverage for their rights, protection and growth. The findings in this report are clear on the lack of support to women in connection with training, transfer of information and technology, and access to credit, women's access to ownership and assets (including land ownership), gender-based violence and discrimination, and education and training opportunities.

Potential outcomes

The intended outcome of the various actions under this recommendation will be legal protection and greater opportunities for cotton farmworkers and farmers to avail of their rights at work, which, which among other impacts, will ultimately enable children to pursue schooling without the burden of child labour, and secure greater gender equality and equity, and social protection. More immediately, the planning and implementation of evidence-based advocacy for legal reform will result in deepened understanding of the legal issues that are creating obstacles and conduct.

Recommendation 2

ILO should undertake analytical research and strengthen information systems on fundamental principles and rights at work in the cotton supply chain in Pakistan.

The report illustrates the need for research that enhances understanding of the interlinkages between different facets of cotton production (such as human infrastructure, technological, biological and environmental) and their mutual impacts to enable integrated policy responses and action.

- The role of **human infrastructure**, particularly for women and children, in achieving cotton yield targets;
- The role of the intermediaries, particularly the **arthis** and **beoparis**, in the cotton growing community and the trading market;
- The expected and actual impact of existing **cotton taxation (including cess)** on the cotton growing communities and the revenue gaps created by barriers to implementation of the policies;
- Access to formal and informal **credit and loan facilities** by farmworkers and smallholder farmers;
- Involvement of children (below 18 years of age) in cotton production and support activities, including domestic and other non-farm/caring responsibilities ensuing from their parents' work in cotton production;
- Safe and permissible tasks that **adolescents** can perform; and
- **Gender-responsive** situation analysis.

The role of human infrastructure in achieving cotton yield targets

The report argues that discussions and policies around farm yield often exclude consideration of the numerous people factors that form the human infrastructure, which contributes to the success or failure of the cotton yield. It is recommended to undertake research to investigate in a comprehensive manner how and why the presence or absence or the equality of these people factors impact yield positively or negatively. Among the factors would be the health and wellness of the people, literacy and education for them and their children, decent work including freedom to form and join trade unions that are able to enter into collective bargaining, access grievance mechanisms, gender equality and equity, no child labour for the children in their community, no forced labour or bonded labour conditions, non-discrimination at work, ensuring a safe and healthy work environment, among other factors. (See section 2.2.1. for a brief discussion on the factors affecting yield.

Potential outcomes

The research will deepen understanding and the recommendations will enhance the profile of the human factors as contributors to the success or failure of the yield. The recommendations will provide guidance for policy and action of different stakeholders, including the government, private and non-profit sector. It will help farmers organize better and support the human infrastructure, resulting in improved yield. It will highlight the need for cotton research and support to cover these issues as well.

The role of the *arthi* and *beopari* in the cotton growing community and the trading market

The research should be geared towards knowledge about the critical power points that have notable influence on the routine processes (including pricing) involved in cotton production and supply chain, and which in turn impact the cotton growing communities. For instance, the report highlights the vital role of the local traders (the *beoparis* and the *arthis*) and the constant interdependence between the local traders and the farmers. It emerges from the report that the *arthis* must be part of the solution for any meaningful and sustainable changes to the market practices that presently are stacked against the smallholder farmer and thereby the farmworkers.

Potential outcomes

The role of the *arthis* has a historical perspective that must be understood for any positive and sustainable changes. This research will help develop that understanding. The recommendations will contribute to improved processes for cotton pricing, evaluation and sale, and partnerships and for policy recommendations on the same.

The expected and actual impact of cotton taxation, including cess, on the cotton growing communities

Similarly, a better understanding of some of the measures that trade associations put their weight behind and support. While beneficial to their members, some of these measures may be having negative impact on the cotton growing community and enterprises in the informal economy along the supply chain, as well as institutions whose functionality and productivity depend on revenue from the cess.

Potential outcomes

The research will increase understanding of the various taxes collected and lead to policy recommendations that would benefit the cotton communities and involved institutions.

Access to formal and informal credit and loan facilities

Advance payments provided by local traders to smallholder farmers and workers create a relationship of dependency that makes the borrowers vulnerable to conditions of forced labour/ bonded labour. Yet, the farmers and workers prefer these providers to the formal ones because of the 'ease of business' and no direct markups – in most cases. Easily accessible credit and loan facilities are needed for the cotton farmers and workers.

Potential outcomes

Recommendations of this research will create a better understanding of the dynamics of the informal loan market and help to design formal market loans with widespread and easy access, with more reliable and transparent credit and loan facilities for the cotton growing communities. They will also help improve the conditions of the informal loan services through competition and an informed customer base.

Safe and permissible tasks that adolescents can perform

Certain occupations and tasks in cotton production communities and processing units in the supply chain can be appropriate permissible work for children of 15 to 17 years, if done under the right conditions, environment and inspection/monitoring. The finding highlighted a lack of understanding about the minimum age for employment and of work that is hazardous for children. Prohibiting 'all' work by children 15 to 17 years in cotton production and processing units is stipulated neither by national law nor international Conventions. More research is needed on the use of pesticides and harmful substances that make apparently permissible work to be hazardous for children below 18 years of age. Determination of safe weights for loads that children and adults can carry safely is also important in the context of cotton production and transportation.

The children's employment laws in Punjab and Sindh identify some tasks that need to be looked into and guidance provided to the stakeholders.

Potential outcomes

The recommendations of this research will contribute to better understanding and help develop codes of practice and policies that protect children from hazardous work and enable those at the legally permissible age to work (and earn) while doing age-appropriate work in safe and healthy environments.

Gender-responsive situation analysis

The analysis should investigate not only the existing roles and responsibilities of women and men (off-farm and on-farm) and the division of labour, but the policy, cultural and policy frameworks that impact power relationships within and outside the family; access to and control over resources and benefits; and their practical and strategic needs, among other aspects. Land ownership, access to training, use of farm machinery, and involvement in market/trade decisions are some specific aspects where the report has highlighted information gaps. The report's limited findings on the involvement of children in cotton production and farm and non-farm support activities (including domestic and caring) ensuing from their parents' work in cotton production underscore the need for further in-depth and research into the subject.

Potential outcomes

Recommendations of this research will facilitate addressing challenges, building capacities and providing opportunities to promote gender equality and equity. The research findings and recommendations will inform design and implementation of gender- and child-responsive policies and programming in the cotton producing communities and enterprises in the informal economy along the supply chain.

Recommendation 3

ILO to convene a regional meeting of experts on fundamental principles and rights at work in the cotton supply chain, followed by a national meeting for policymakers and implementing stakeholders.

The regional meeting of experts will gather researchers and scholars on the subjects of cotton growing/agriculture and fundamental principles and rights at work, together with programme implementers with first-hand knowledge and experience of the issues, opportunities and risks on ground. The meeting's recommended objectives would include: developing an in-depth understanding of the issues and available research, forming multidisciplinary alliances and a network to highlight the role of the human infrastructure that contributes to the yield, discussing and advocating policy options for governments, employers' and workers' organizations, businesses, farm input providers, microfinance institutions, among others.

The second meeting(national) for policymakers will examine the recommendations from the experts meeting with a view to improving policies and the legal framework, programmes, action and multidisciplinary collaboration on the subject.

Potential outcomes

The experts' meeting will provide an opportunity to present, discuss and debate the diverse issues related to the cotton supply chain. It will bring together different stakeholders who develop better appreciation for and understanding of the different facets of cotton production. It will help create a core group of experts to act in advisory capacity for work on the cotton supply chain in Pakistan and the region.

Potential outcomes

The national meeting will put the recommendations of the experts' meeting before policy makers and stakeholders to encourage action for improving policies and programmes. It will give them a greater understanding of the supply chain and the issues involved, particularly as related to fundamental principles and rights at work.

Recommendation 4

ILO should implement a capacity development and capability building programme on fundamental principles and rights at work in the cotton supply chain, at the institutional and individual levels, through traditional and innovative approaches.

Gender analysis and training needs analysis should be undertaken prior to developing a programme to meet the needs of the target groups and to integrate gender equality in the programmes.

The target groups for the programme include:

- Concerned government departments at the federal, provincial and district level, including extension service workers, employers' and workers' organizations, trade associations, non-governmental organizations;
- At farm and market levels for workers, farmers, input providers, intermediaries (*beoparis* and *arthis*), members of the market committee, other trade committees, community level non-governmental organizations and other bodies; and
- Agriculture universities and research institutes, particularly those with existing extension agriculture programmes or with a potential to develop programmes that can connect university students and faculty with cotton growers for mutual learning.

In addition to capacity building on fundamental principles and rights at work, the ILO should, in collaboration with the social partners, provide business training to farmers, including workforce management, negotiations, marketing, planning, and business expansion.

ILO's tripartite constituents and key partners at the national and provincial levels in Pakistan are likely to have adequate, or even deep, understanding of fundamental principles and rights at work, the report shows that key players involved in cotton production and supporting organizations at the farm, community and district level do not. For the realization of these rights, that understanding needs to be embodied in their daily lives in their local contexts. For this reason, the capacity development programme must be accompanied by capability building activities to develop skills in applying fundamental principles and rights in their contexts.

Learning-by-doing is a highly effective approach. It is therefore recommended the programme include, in addition to group training, real-life opportunities for sharing, doing and learning through practical programmes in real life settings, field trips and exchange visits from community to community, from district to district and province to province.

Programmes that enable placement of students, faculty or staff in the communities and give opportunities to farm level stakeholders to participate in district, provincial or national initiatives are also among real-life experiences that go a long way in developing understanding, innovation, capability building, capacity development and creating links.

The report indicates gaps in outreach, particularly by the extension services agents to female farm workers. It is recommended to strengthen the public and private sector extension services to the farm workers, particularly the women, and involve them in the capability building programmes. These services must also use and promote amongst the cotton communities the use of appropriate, cost-effective technologies for farm and non-farm use.

For those not familiar with fundamental principles and rights at work, the conceptual grasp of what these rights entail can be daunting. It is, therefore, recommended to pair training and other activities on fundamental principles and rights at work with other everyday themes that the beneficiaries easily see as relevant, such as healthcare, livelihoods enhancement, childcare and education, creating links with local opportunities and benefits.

The report shows the need for farmers to be trained and be knowledgeable about issues related to their farms and workforce and improving the business aspects of farming. Farmers can also help with reducing hazards and hardships at work for the farmworkers. For instance, the recycling of the empty pesticide containers that pollute the environment affect the health of the farmworkers or providing clean drinking water and sanitation on the farms.

Potential outcomes

This recommendation will help the farmers see the importance of fundamental principles and rights at work as part of an integrated approach to their business, increasing its chances of being applied.

This recommendation intends to deal with the fundamental issue of individual and institutional capacities and capabilities, without which the other actions and input will not be effective. As fundamental principles and rights at work can be vague as an issue for many people not familiar with the rights-based framework, the recommendation provides for concrete ways of making the issues more concrete. Having core capacity at the local levels and in different institutions will help generate greater respect for fundamental principles and rights at work.

Recommendation 5

The ILO to develop localized training programmes, particularly for smallholder farmers, to support them in improving the business of farming.

It is recommended that the farmers' training programmes should be needs-based and multidisciplinary to the extent possible but focused on farming from a business perspective. Social partners and local trade and business associations must be trained to deliver the programme and support the growth of the farm business.

The report shows the need for farmers to be trained and be knowledgeable about issues related to their farms and workforce and improving the business aspects of farming. Farmers can reduce hazards and hardships at work for the farmworkers. For instance, the recycling of the empty pesticide containers that pollute the environment affect the health of the farmworkers or providing clean drinking water and sanitation on the farms.

Potential outcomes

This recommendation will help the farmers see the importance of fundamental principles and rights at work as part of an integrated approach to their business, increasing its chances of being applied.

Recommendation 6

Support the local communities to launch and implement locally developed multi-media community-based campaigns to promote decent work practices, respect for fundamental principles and rights at work.

It is recommended that the project's successes with awareness raising be further reinforced with lessons learned in the areas covered by the project and expanded to new areas through existing and new community-based partnerships.

The campaigns should seek to raise awareness of the forms of forced or compulsory labour, child labour, discrimination and safe and healthy work environments in their context; and what in concrete terms would be freedom of association and collective bargaining.

The campaigns should be developed by professional agencies but locally and based on focus group discussions and delivered using multi-media including radio, religious and social centres, street theatre, local artists, among others. Extension workers should be involved in developing and delivering the messages. Pre-, mid- and post-campaign assessments should be put in place to see the impact of the campaigns and make any changes as needed.

Potential outcomes

This recommendation will create awareness where it can have most impact in people's life – in the villages and communities, with the view to transferring knowledge and information directly to the people concerned so they make bring positive changes to their lives.

Recommendation 7

Support the mandated local government agency and partners in the development and implementation of a simple and do-able integrated inspection and community-based monitoring of fundamental principles and rights at work.

As one of the three most dangerous sectors to work in, the agriculture sector needs a particular focus on inspections, monitoring and awareness raising on occupational safety and health issues as well as child labour and hazardous work for older children, use of chemicals and machinery safely.

The report findings show that there is little, if any, awareness of the hazards that the workers and their families face and that preventative measures are missing.

It is recommended to promote the development and implementation of an integrated programme, involving government inspectors with expertise and mandates from the various agencies and on the issues involved. Simple, cost-effective and do-able community-based monitoring systems that are developed as support systems can go a long way in creating awareness of the issues involved and changing mindsets on forced labour, child labour and other issues. When linked with labour inspection systems, they are able to make valuable contribution.

Potential outcomes

The inspections and community-based monitoring will protect people from hazards at work and other harmful working conditions, create awareness of issues, create a greater sense among the community for keeping themselves and each other protected. It will give the people a better sense of respect for fundamental principles and rights at work and how that respect can be enhanced in their context.

Recommendation 8

Based on evidence and experience, continue to support partners in creating and strengthening linkages of farmworkers and farmers to social protection programmes, credit providers and other opportunities that will help improve prospects for securing their fundamental rights at work and supporting them and their families with the social protection needed.

Different ILO projects in Pakistan, as many other initiatives, have successfully linked various target groups to the social protection programmes available to them. A range of Federal and Provincial Government and other programmes are available and need to be explored for linkages.

Recommendation 9

Support, through a multi-stakeholder sectoral project, national partners and domestic and international businesses that are sourcing from Pakistan to improve the cotton supply chain through greater formality with respect for fundamental principles and rights at work in the cotton growing communities and businesses along the supply chain.

The scope of the mapping exercise is limited to the cotton growing community and a few textile and garment units, which exists in the informal economy, removed from the tier-1 and tier-2 businesses that operate in the formal economy. It is recommended that the ILO continue to develop collaborative partnerships with tier-1 and tier-2 to leverage their support in promoting fundamental principles and rights, decent work in the cotton growing communities and along the supply chain.

The report shows the export-oriented textile and garment units tend to show more respect for fundamental principles and rights at work than the other units which have no direct links with the export market. While the national and international brands maintain their exclusive vertical supply chains, this cannot be sustainable unless standards in the rest of the supply chain are improved.

It is recommended the ILO take a sectoral approach, bringing onboard, for a project partnership, the private players, companies and brands that have significant sourcing share or influence over the cotton supply chain to address major fundamental principles and rights issues in the cotton supply chain. The role of employers' and workers' organizations will be critical in forming this partnership.

In addition to the ILO's current worldwide experience with public-private-partnership initiatives, ILO Office in Pakistan has historical and successful sectoral partnership experience to guide the initiative proposed in the recommendation.

This will, in the longer-term, make the supply chains more sustainable with greater transparency and accountability.

Potential outcomes

Sectoral collaboration in the form of a project can have systemic and lasting impact in improving the diverse aspects of the cotton supply chain in the producing communities and the businesses along the supply chain.

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► Glossary and definitions

Glossary

Arthis: commission agents; oversee/conduct auctions in the market; powerful figure in the market and cotton trading; licensed by the food department and represented on the Cotton Market Committees through the *arthi* associations, such as Punjab *Arthis* Association Punjab president (PAAP). The *arthi* are also referred to as **katcha arthis** (commission agents) and **pukka arthis** (wholesale dealers). In this report, the term *arthi* is used for both types.

Beoparis: intermediaries who buy seed cotton from the farmer (usually at their doorsteps) and bring it to the market for auction.

Biri/bidi: they are locally made rolled cigarettes.

Cotton bolls: these are the rounded mature fruit, that contain the seed and the fluffy cotton fibre/lint. From harvest to the ginnery, cotton remains in the boll form, which open at harvest time. At the ginnery, the seeds and fibre are separated, and they follow different paths of the supply chains from there on. The fibre is spun into cotton yarn, woven into fabric, and then manufactured into garments and other consumer products. The seeds are made into oil and other products.

Seed cotton: This term is used for raw unginned cotton that consists of the seed attached to the white lint as picked from the boll. The ginning process separates the seeds from the lint, the latter being used to produce cotton lint and then fibre, among other products. Some cottonseed crop is taken back for planting, but most is used as raw material and processed to make products, such as oil and feeds.

Cottonseed: these are seeds that grow in the bolls, along with the cotton fibre/lint and are separated from the cotton fibre during the ginning process

Palledars: help with loading and unloading in the market; not licensed.

Pharias – intermediaries, working collaboratively with the *arthi*; buy in large bulk in the auction and then either divide these into smaller lots and sell to retailers and individual customers, or resell in large bulk to business.

Definitions of concepts

Decent Work: Decent Work is productive employment that fetches a fair income; and is performed at a safe and secure place, where workers have a voice, where they can organize themselves to have representation, and where they can participate in the decisions that affect their work and lives. Underlying Decent Work is the principle of equality of opportunity and treatment for all, irrespective of gender.

Family labour: Family labour is work done by contributing family members/the household unit. Generally, the contributing family members are unpaid or at least not paid directly by the employer, but they share (without having to pay for) the food and lodging provided by the family. The head of the family/unit receives the remuneration. Persons in family labour may be considered as self-employed.

Self-employment: Self-employed (or own account) workers are persons whose remuneration from work is dependent directly upon the profits (or the potential for profits, that include own consumption) from their work/job. Moreover, a self-employed person makes the operational decisions or delegates such decision while retaining responsibility. (In this context “enterprise” includes one-person operations and can include the family/economic unit.) – Current guidelines - The 14th International Conference of

Labour Statisticians (ICLS) adopted the Resolution concerning the International Classification of Status in Employment (ICSE), known as ICSE-93.

Intermediaries: In the context of this Study, intermediaries refer to business entities or individuals who play a business or trade role in the movement of the raw material from the source (producers/growers/workers) to the manufacturer (companies). They include the suppliers, sub-suppliers, vendors, street hawkers, and entrepreneurs. They do not include workers in the fields or the waste material collectors unless these workers also purchase the material to sell it further.

Salary: Salary is the fixed amount of compensation which is paid for the performance of an employee at a regular interval – for instance, weekly or monthly. It often comes with other benefits, such as contribution to social security or pension, entitlement to weekly days off and holidays. Such workers often also have the option of being member of a union that is able to enter into collective bargaining for members.

Wage is the variable amount of compensation which is paid on the basis of a unit cost – such as per day, or per hour, or per item produced or completed. Although in some countries, in the case of piece-rate, employers must still comply with minimum wage, overtime and record keeping requirements, in many instances, workers receiving wages do not get additional benefits and their work is of a temporary nature.

Supply chain: For purpose of this Formative Study, a supply chain is defined as a web of multiple and varied stakeholders who participate in, and across, different segments of the processes involved in developing or producing a product. The processes span from segment where the raw material is sourced, through various other segments where intermediary players add value to the material or do transactions, all the way to the manufacturer of the final product. Manufacturing, distribution and consumption are also segments of the supply chain, but they are beyond the scope of this Formative Study.

Supply streams: The paper and paperboard supply chain as discussed in this Study consists of two distinct 'supply channels' which the Study refers to as supply streams. This distinction makes clear the highly informal, unorganized and at times blurry nature of the paths that the raw materials travel and those who work on making that material available as it is transacted and as it moves. Introducing the concept of the 'stream' helps to portray the complexity and the different nature of the different strands of processes and actors involved. Together these form the supply chain.

Tiers: For purposes of this Report, tier-4 is the community level, where seed cotton, the raw material for the cotton supply chain is cultivated and harvested (produced). It is carried through several trading and processing levels through what here is referred to as tier-2 and tier-4. The final level, tier-1, is where the final produced is manufactured, in this case garments or textile fabrics. Tier-4 operates in the informal economy, focusing on livelihood issues. As the tiers step up, there is increasing compliance and enforcement. There is no uniformity from tier to tier in the cotton supply chain and most others in the developing world.

Tripartism: A process whereby governments consult and involve representatives of employers and workers in social and economic matters, social dialogue refers to exchange of information, consultations or negotiations between or among representatives of governments, employers' and workers' organizations.

Wages: See above under Salary.

► Annexes

Annex 1: Villages – data collection

Sr#	District	Respondents	Village
1	Vehari	5 farmers. 5 male workers	Chak 32 WB
2	Vehari	5 farmers. 5 male workers	Chak 82 WB
3	Vehari	5 farmers. 5 male workers	Chak 47 WB
4	Vehari	5 farmers. 5 male workers	Chak 50 WB
5	Vehari	5 farmers. 5 male workers	Chak 157 WB
6	Vehari	5 farmers. 5 female workers	Chak 72 WB
7	Vehari	5 farmers. 5 female workers	Chak 228 EB
8	Vehari	5 farmers. 5 female workers	Chak 192 WB
9	Vehari	5 farmers. 5 female workers	Chak 204 EB
10	Vehari	5 farmers. 5 female workers	Chak 95 WB
11	Bahawalpur	5 farmers. 5 workers	Chak 18 BC Bahawalpur
12	Bahawalpur	5 farmers. 5 workers	Chak 115 DB
13	Bahawalpur	5 farmers. 5 workers	Rangpur
14	Bahawalpur	5 farmers. 5 workers	Chak 39 DB
15	Bahawalpur	5 farmers. 5 workers	Chak 46 DB
16	Bahawalpur	5 farmers. 5 workers	Chak 91/DB
17	Bahawalpur	5 farmers. 5 workers	Chak 47 DB Gharbi
18	Bahawalpur	5 farmers. 5 workers	Chak 69 DB
19	Bahawalpur	5 farmers. 5 workers	Chak 72 DB
20	Bahawalpur	5 farmers. 5 workers	Kirani Village
21	Rahim Yar Khan	5 farmers. 5 workers	Chak 103 P
22	Rahim Yar Khan	5 farmers. 5 workers	Chak 87/P Old 26
23	Rahim Yar Khan	5 farmers. 5 workers	Chak 80/P
24	Rahim Yar Khan	5 farmers. 5 workers	Chak 156/P
25	Rahim Yar Khan	5 farmers. 5 workers	Chak 181 Np
26	Rahim Yar Khan	5 farmers. 5 workers	Chak 173 P
27	Rahim Yar Khan	5 farmers. 5 workers	Chak 120 P

Sr#	District	Respondents	Village
28	Rahim Yar Khan	5 farmers. 5 workers	Chak 155 P
29	Rahim Yar Khan	5 farmers. 5 workers	Chak 35/A
30	Rahim Yar Khan	5 farmers. 5 workers	Chak 222/P
31	Mirpur Khas	5 farmers. 5 workers	Goth Kamal khan Kapri
32	Mirpur Khas	5 farmers. 5 workers	Goth Ghulam Muhammad
33	Mirpur Khas	5 farmers. 5 workers	Village Digri
34	Mirpur Khas	5 farmers. 5 workers	Goth Ahmed Khan
35	Mirpur Khas	5 farmers. 5 workers	Goth Ch. Ismail
36	Mirpur Khas	5 farmers. 5 workers	Goth Mahi Khaskeli
37	Mirpur Khas	5 farmers. 5 workers	Goth Baghaan wali
38	Mirpur Khas	5 farmers. 5 workers	Goth Mola Bux
39	Mirpur Khas	5 farmers. 5 workers	Village Mehdiabad
40	Mirpur Khas	5 farmers. 5 workers	Goth Gidoo
41	Sanghar	5 farmers. 5 workers	Village Landi
42	Sanghar	5 farmers. 5 workers	Village Dost Muhammad Khoso
43	Sanghar	5 farmers. 5 workers	Shahdadpur
44	Sanghar	5 farmers. 5 workers	Village Mosa Khan Umrani
45	Sanghar	5 farmers. 5 workers	Village Qazi Juman Dahri
46	Sanghar	5 farmers. 5 workers	Village Karim Dino Wasan
47	Sanghar	5 farmers. 5 workers	Village Baloch khan Laghari
48	Sanghar	5 farmers. 5 workers	Sinjhoro
49	Sanghar	5 farmers. 5 workers	Village Muhammad Uris Brohi
50	Sanghar	5 farmers. 5 workers	Goth Shamas din Arain
51	Bahawalpur	4 farmers 5 workers	Kirani Village Bahawalpur
52	Bahawalpur	6 farmers	Rangpur
53	Bahawalpur	5 workers	Jageer Bhatti Dahri
54	Bahawalpur	1 farmer	Goth Lal
55	Rahim Yar Khan	5 farmers	Mud Hassan
56	Rahim Yar Khan	3 farmers	Chak 106/P
57	Rahim Yar Khan	2 farmers 5 workers	Chak 139/P
58	Rahim Yar Khan	5 workers	Basti Mirmohammad
59	Mirpur Khas	6 farmer 5 workers	Haji Muhammad Khan Bangalani

Sr#	District	Respondents	Village
60	Mirpur Khas	2 farmers	Purani Landi
61	Mirpur Khas	2 farmers 4 workers	Kot Ghulam Muhammad
62	Sanghar	5 farmers 6 workers	Goth Shamas din Arain
63	Sanghar	6 farmers 5 workers	Goth Muhammad Saeed Arain
64	Vehari	6 farmers 8 workers	Chak 82 WB Vehari
65	Vehari	4 farmers 2 workers	Chak 50 WB Vehari
Total		302 farmers, 300 workers	

Annex 2: Villages – focus group discussions

Sr#	District	Activity	Village
1	Bahawalpur	FGD with farmers	Kirani Village Bahawalpur
2	Vehari	FGD with female workers	Chak 82 WB Vehari
3	Vehari	FGD with farmers	Chak 82 WB Vehari
4	Vehari	FGD with male workers	Chak 82 WB Vehari
5	Sanghar	FGD with male workers	Goth Muhammad Saeed Arain
6	Sanghar	FGD with Female workers	Goth Muhammad Saeed Arain
7	Sanghar	FGD with farmers	Goth Muhammad Urs Brohi
8	Sanghar	Sami structured interview	Ginner Sultan Cotton
9	Mirpur Khas	FGD with Female workers	Kot Ghulam Muhammad
10	Rahim Yar Khan	FGD with female workers	Mud Hassan
11	Mirpur Khas	FGD with male workers	Kot Ghulam Muhammad
12	Rahim Yar Khan	FGD with male workers	Chak 222/P
13	Bahawalpur	FGD with male workers	Jageer Bhatti Dahri
14	Bahawalpur	FGD with female workers	Chak 18 BC Bahawalpur
15	Rahim Yar Khan	FGD with farmers	Chak 139/P
16	Mirpur Khas	FGD with farmers	Haji Muhammad Khan Bangalani

Annex 3: Stakeholders interviewed

Sr.#	District	Department/Organization	Designation
1	Multan	CCRI Multan	Head, Fiber Technology
2	Multan	PCSI Multan	Head of Department
3	Multan	PCGA Multan	Ex- Chairman
4	Vehari	Alwasey Ginners	Cotton Ginner
5	Vehari	Pakistan Kissan Ittehad	President PKI
6	Vehari	Baidari Vehari Women Organization	District Manager
7	Bahawalpur	Agriculture Department	Agriculture Officer
8	Bahawalpur	Cotton Research Station Bahawalpur	Head, Cotton Research Station
9	Bahawalpur	Cotton Inspector	Cotton Inspector Bahawalpur
10	Bahawalpur	Agronomic Research Station	Head of Department
11	Bahawalpur	WWF	Project Coordinator
12	Bahawalpur	Quality International, Pesticide Co.	Business Manager
13	Bahawalpur	Farmer Organization, Kissan Welfare Association	KWA
14	Bahawalpur	Nazim Industries (Ginner)	Owner
15	Bahawalpur	Askari Bank	GM Agri loaning
16	Bahawalpur	Child Protection Bureau	DO
17	Bahawalpur	Labour Department	Labour officer
18	Bahawalpur	Cheema Corporation	Commission Agent
19	Rahim Yar Khan	Seed Company (Jhallander)	General Manager
20	Rahim Yar Khan	Seed Company (Sinker seed)	General Manager
21	Rahim Yar Khan	Cotton Inspector	Cotton Inspector
22	Rahim Yar Khan	Pesticide Company (Agros)	Owner
23	Rahim Yar Khan	Fatima Fertilizer	Technical Sale Officer
24	Rahim Yar Khan	Ginner Zafer Cotton	General Manager
25	Rahim Yar Khan	Farmer Associates	Executive Director
26	Rahim Yar Khan	Reeds-Pakistan, BCI Implementing Partner	Executive Director
27	Rahim Yar Khan	PCSI Lab	Assistant Director
28	Rahim Yar Khan	Bank ZTBL	Mobile Credit Officer
29	Rahim Yar Khan	Cotton Connect	Assistant Manager

Sr.#	District	Department/Organization	Designation
30	Rahim Yar Khan	BCI	Programme Coordinator
31	Rahim Yar Khan	Labour Department	Labour Officer
32	Rahim Yar Khan	Grain Markert	Commision Agent
33	Mirpur Khas	Agriculture Extension Department	Deputy Director
34	Mirpur Khas	Worker Organization	Manager
35	Mirpur Khas	Cotton Research Station	Senior Scientific Officer
36	Mirpur Khas	Ginner Lal Cotton	Owner

Annex 4: Cotton producing districts in Punjab and Sindh Provinces

PUNJAB		SINDH	
District	Cotton Area 000 Hec.	District	Cotton Area 000 Hec.
Bahawal pur	279.2	Sanghar	122.9
Rahimyar Khan	254.1	Khairpur	78.9
Bahawalnagar	211.2	Ghotki	76.9
Lodhran	197.5	Nawabshah	46.3
Vehari	182.1	N. Feroze	36.9
M. Garh	182.1	Mirpur khas	35.3
Multan	167.1	Matiali	33.1
Khanewal	165.5	Sukkur	30.2
Rajanpur	126.7	Umarkot	23.2
D.G. Khan	95.1	Tando Allahyar	20.8
Sahiwal	71.2	Jamshoro	17.5
Jhang	55.4	Badin	13.6
Pakpattan	46.1	Dadu	9.2
Toba Tek Singh	43.3		
Faisalabad	39.3		
Layyah	38.4		
Okara	25.1		

PUNJAB		SINDH	
District	Cotton Area 000 Hec.	District	Cotton Area 000 Hec.
Mianwali	13.4		
Kasur	13		
Bhakkar	8.1		
Sargodha	6.9		

Annex 5: Key constituents consulted on the draft and publication of report

	Name of Department/Organization	Remarks
1	Ministry of Overseas Pakistanis and Human Resource Development (MOPHRD)	The report was shared with the MOPHRD on 23 December 2021 and 4 January 2024. No comments were received.
2	Employers' Federation of Pakistan (EFP)	The EFP reviewed the report and consented to publish this study on 3 February 2022.
3	Pakistan Workers' Federation (PWF)	The PWF reviewed the report and consented to publish this study on 25 January 2022.

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